INTRODUCTION TO GENERAL RELATIVITY

TOBIAS J. OSBORNE

1. Prerelativity gravitation

1.1. Course overview. General relativity is, by wide consensus among physicists, one of the most beautiful and elegant physical theories we possess. The connection it draws between the curvature of spacetime and the motion of freely falling bodies has an almost miraculous quality: there is something deeply compelling, perhaps even inevitable, about the way the theory holds together.

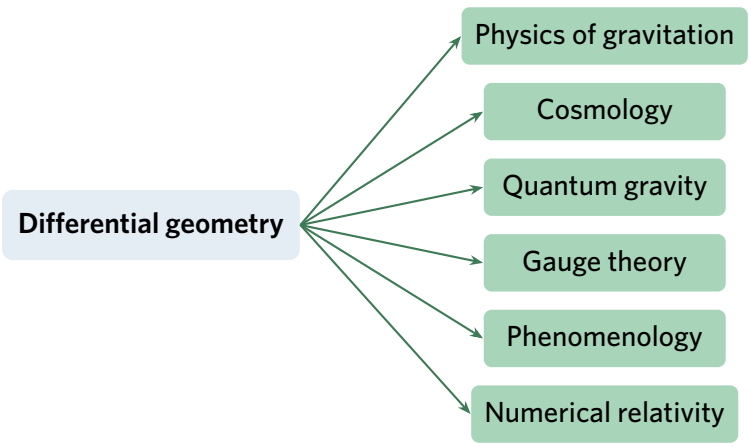
That said, general relativity is also *wrong*, or at least incomplete: as far as we know, it is not compatible with quantum mechanics in any fully satisfactory way. A deeper, more fundamental theory must underlie both. Nonetheless, the predictions of general relativity have been confirmed to extraordinary accuracy across an enormous range of cosmological and astrophysical phenomena, and it remains the indispensable framework for understanding gravity.

Text. Wald, *General Relativity* [Wal84]. We follow Wald closely, though we deviate where other sources offer clearer treatments.

Other sources:

- Misner, Thorne & Wheeler [MTW73]: the "telephone book," a comprehensive and physically rich discussion of general relativity.
- Weinberg [Wei72]: an excellent resource for explicit calculations.

Structure. The course proceeds from mathematical methods to the physics of gravitation. The emphasis falls deliberately on the mathematical side: in particular, on differential geometry. While it is perfectly possible to do general relativity concretely, working in a single coordinate patch and never invoking coordinate-free notation, a command of the underlying differential geometry is indispensable for further studies in theoretical physics. Anytime one wishes to do serious work in quantum gravity or string theory, fluency in differential geometry is a prerequisite; the investment made here will pay dividends well beyond this course.



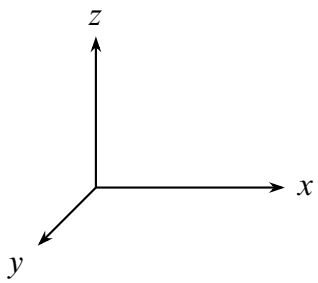
Videos. Lenny Susskind's lecture series on general relativity offers an absolutely different, and completely complementary, perspective to the one presented here.

Prerequisites:

Date: May 6, 2026.

- Special relativity
- Calculus
- Linear algebra
- Combinatorics

1.2. Newtonian mechanics and inertial frames. **Assumption:** Newtonian physics presupposes the existence of a preferred reference system and clocks. This is the core assumption that will be broken in the transition from pre-relativity to relativity physics: a common thread in the development of physics is that one makes (often implicit) assumptions in a physical theory, and as one learns more about the universe one is forced to re-evaluate and potentially discard them.



A free test mass (i.e. no force acts) moves along a straight trajectory at a constant rate, uniformly in time: constant displacement per unit time. This defines an *inertial reference frame*. There is one preferred reference frame, but one can obtain others by performing Galilean boosts: an observer moving at constant velocity with respect to the preferred frame will also see free test masses moving along straight trajectories at constant rates.

Newton's second law is written with respect to an inertial reference system:

$$\boldsymbol{F}(\boldsymbol{x}(t),\, t) \;=\; m_I\, \ddot{\boldsymbol{x}}(t)\,, \tag{1}$$

where $\boldsymbol{F}$ is the force, m_I is the *inertial mass*, and $\ddot{\boldsymbol{x}}$ is the acceleration.

1.2.1. *Non-inertial reference frames.* With respect to a non-inertial reference frame, define three (time-dependent) unit vectors $\hat{\boldsymbol{x}}'(t)$, $\hat{\boldsymbol{y}}'(t)$, $\hat{\boldsymbol{z}}'(t)$. A trajectory can be expressed in either basis:

$$\boldsymbol{w}(t) = w_x(t)\,\hat{\boldsymbol{x}} + w_y(t)\,\hat{\boldsymbol{y}} + w_z(t)\,\hat{\boldsymbol{z}} \tag{2}$$

$$= w_{x'}(t)\,\hat{\boldsymbol{x}}'(t) + w_{y'}(t)\,\hat{\boldsymbol{y}}'(t) + w_{z'}(t)\,\hat{\boldsymbol{z}}'(t). \tag{3}$$

Index notation. For a vector $\boldsymbol{a}$ and a matrix M:

$$[\boldsymbol{a}]_x = a_x\,, \qquad [M]_{xy} = M_{xy}\,.$$

Physics in non-inertial frames (as captured by Newton's second law) gives rise to additional *inertial force* terms, e.g. centrifugal, Coriolis, Euler forces, etc.

Exercise 1.1. Distinguish "true forces" from apparent (inertial) forces.

Why inertial forces matter

In a rotating reference frame, a ball thrown in a straight line *appears* to curve (Coriolis effect). The "force" causing this curve is not due to any physical interaction: it is an artefact of the non-inertial frame. In Newtonian physics, gravity is classified as a *true* force. A central insight of general relativity is that gravity, too, can be understood as an artefact of geometry, effectively reclassifying it alongside the inertial forces.

1.3. Mass: three conceptually distinct quantities.

(i) **Inertial mass** m_I: the constant of proportionality in Newton's second law,

$$F = m_I\, a\,. \tag{4}$$

(ii) **Active gravitational mass** $m_g^{(a)}$: the mass that *produces* the gravitational field that other masses respond to. The gravitational field of a point source at position $\boldsymbol{x}'$ is

$$\boldsymbol{g}(\boldsymbol{x}) = -G\, m_g^{(a)}\, \frac{\boldsymbol{x}-\boldsymbol{x}'}{\|\boldsymbol{x}-\boldsymbol{x}'\|^3}\,, \tag{5}$$

where G is the gravitational constant,

$$G = 6.674\,30(15) \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}\,. \tag{6}$$

(iii) **Passive gravitational mass** $m_g^{(p)}$: the mass that *responds* to an external gravitational field by accelerating,

$$\boldsymbol{F}_g = m_g^{(p)}\, \boldsymbol{g}\,. \tag{7}$$

It is assumed that m_I, $m_g^{(a)}$, $m_g^{(p)} \geq 0$. This non-negativity is an assumption, not a theorem: one *cannot* deduce it from the other laws of Newtonian physics.

Remark. It is possible to build a consistent Newtonian physics with negative masses. The consequences are exotic: negative masses behave somewhat like negative charges, and all manner of counterintuitive phenomena arise, but the theory does not immediately break down. The non-negativity of mass is supported by extensive experimental evidence, not by logical necessity.

Exercise 1.2. Using Newton's third law, argue that $m_g^{(a)}/m_g^{(p)}$ is a universal constant.

Choosing units: $m_g^{(a)} = m_g^{(p)} \equiv m_g$.

Three masses, one body

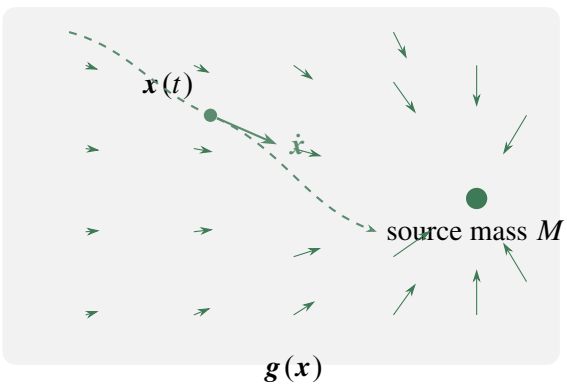
Every massive body carries *three* logically independent numbers: how much it resists acceleration (m_I), how strongly it sources a gravitational field ($m_g^{(a)}$), and how strongly it responds to one ($m_g^{(p)}$). That nature chooses to make all three equal is not obvious: it is a deep empirical fact that general relativity elevates to a geometric principle.

1.4. The universality of free fall.

Extensive experimental evidence suggests $m_g = m_I$.

Definition 1.1 (Universality of Free Fall)**.** In a gravitational field $\boldsymbol{g}$, the orbit $\boldsymbol{x}(t)$ of a test mass depends *only* on the initial conditions. That is, identical initial conditions produce identical trajectories regardless of the composition or internal structure of the test mass.

Consider a test mass in a gravitational field:



The equation of motion reads

$$m_g\, \boldsymbol{g}(\boldsymbol{x}(t)) = m_I\, \ddot{\boldsymbol{x}}(t)\,, \tag{8}$$

or equivalently,

$$\boxed{\ddot{\boldsymbol{x}}(t) = \frac{m_g}{m_I}\, \boldsymbol{g}(\boldsymbol{x}(t))\,.} \tag{9}$$

Since m_g/m_I is a universal constant, we may choose units such that $m_g = m_I$.

Remark. (1) The universality of free fall is *not* a consequence of Newtonian theory: it is a falsifiable hypothesis. That $m_g = m_I$ cannot be proved from Newton's laws; it is an empirical statement that could, in principle, be refuted by experiment. Experiments testing this relationship continue to this day.

(2) UFF applies specifically to *test masses*. A test mass is an idealised notion: one should visualise it as a "point" particle. The concept is already subtle in Newtonian physics: a finite mass concentrated into an arbitrarily small volume would, in general relativity, form a black hole, and it becomes more so when test masses are numerous enough to generate their own gravitational fields. These subtleties will be resolved once we have the full apparatus of general relativity.

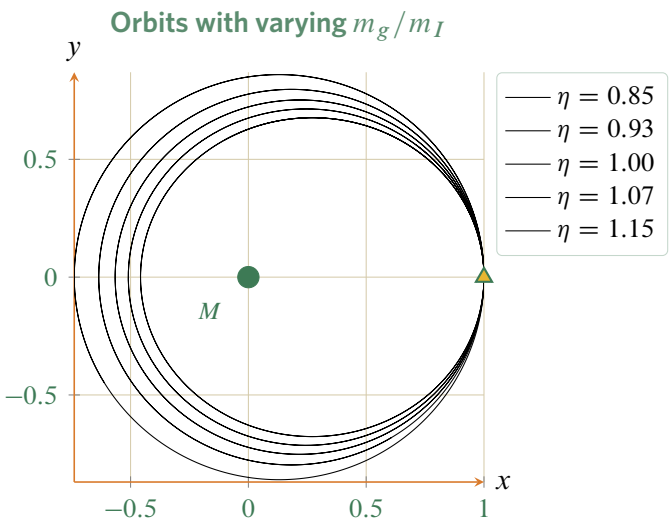


FIGURE 1. Orbits of five test particles launched from the same initial conditions ($r_0 = 1$, tangential velocity $v_0 = 0.85\, v_{\text{circ}}$) in a Kepler field, with varying mass ratio $\eta = m_g/m_I$. If $\eta = 1$ (universality of free fall), all particles follow the same orbit; different η values produce visibly different trajectories. No such deviation has ever been observed.

Experimental tests of the equivalence principle

The equality $m_g = m_I$ has been tested with extraordinary precision:

Newton (1687)	Eötvös (1922)	Lunar laser (1969)	MICROSCOPE (2022)
$\sim 10^{-3}$	$\sim 10^{-9}$	$\sim 10^{-13}$	$\sim 10^{-15}$

The Eötvös experiment [vEPF22] uses a *torsion balance*: two masses of different composition are suspended at opposite ends of a bar. If m_g/m_I differed between materials, the bar would rotate. No such rotation has ever been observed [RKD64, AGH+09]. Lunar laser ranging [WTB12] exploits the Apollo retroreflectors to compare the free-fall accelerations of the Earth and Moon toward the Sun. The MICROSCOPE satellite (2016–2018; final results published 2022) [TMR+22] achieved the most stringent test to date, constraining the Eötvös parameter to $|\eta| \equiv |(m_g/m_I)_A - (m_g/m_I)_B| < 10^{-15}$.

1.5. From Newtonian gravity to general relativity. We now reformulate Newton's gravity as a *field theory*, drawing on the analogy between Newtonian gravity and electrostatics. The motivation is direct: electrodynamics is compatible with special relativity: Lorentz transformations leave Maxwell's equations invariant and information propagates at the speed of light. If Newtonian gravity looks structurally like electrostatics (and it does), then perhaps we can add dynamics in the same way and arrive at a relativistic theory of gravity. As we shall see, this hope is partially fulfilled: the approach does lead somewhere, but the destination is more radical than a simple gravitational analog of Maxwell's equations.

1.5.1. *Coulomb vs. Newton: a detailed comparison.*

Electrostatics **Newtonian gravity**

Attractive *and* repulsive Attractive *only*

	Electrostatics	**Newtonian gravity**
Force law	$\boldsymbol{F} = k_e \frac{q_1 q_2}{r^2}\hat{\boldsymbol{r}}$	$\boldsymbol{F} = -G\frac{m_1 m_2}{r^2}\hat{\boldsymbol{r}}$
Field	$\boldsymbol{E}$	$\boldsymbol{g}$
Source	charge density ρ_e	mass density ρ
Source sign	both signs	$\rho \geq 0$ only
Potential	V	ϕ
Field from potential	$\boldsymbol{E} = -\nabla V$	$\boldsymbol{g} = -\nabla\phi$
Gauss's law	$\nabla\cdot\boldsymbol{E} = \rho_e/\varepsilon_0$	$\nabla\cdot\boldsymbol{g} = -4\pi G\rho$
Poisson equation	$\Delta V = -\rho_e/\varepsilon_0$	$\Delta\phi = 4\pi G\rho$

The gravity–electrostatics analogy

The structural parallel is almost perfect: both are $1/r^2$ force laws sourced by a density and governed by a Poisson equation. The crucial difference is that electric charge comes in two signs (allowing shielding and cancellation), while mass is always positive. This means gravity cannot be screened, every mass in the universe contributes, which is ultimately why a geometric description (curved spacetime) becomes necessary.

1.5.2. *Gravitational field.* The gravitational field is a map

$$\boldsymbol{g}\colon \mathbb{R}\times\mathbb{R}^3 \to \mathbb{R}^3\,, \qquad (t,\,\boldsymbol{x}) \mapsto \boldsymbol{g}(t,\,\boldsymbol{x})\,.$$

1.5.3. *Sources.* The (gravitational) mass density is

$$\rho\colon \mathbb{R} \times \mathbb{R}^3 \to \mathbb{R}^+ ,$$

and the field equation reads

$$\boxed{\nabla \cdot \boldsymbol{g} = -4\pi G\, \rho\,.} \tag{10}$$

1.5.4. *Gravitational potential.* Since $\nabla \times \boldsymbol{g} = 0$, the field $\boldsymbol{g}$ is conservative. By Poincaré's lemma there exists a scalar potential $\phi\colon \mathbb{R} \times \mathbb{R}^3 \to \mathbb{R}$ such that

$$\boxed{\boldsymbol{g} = -\nabla\phi} \qquad \text{(sign convention).} \tag{11}$$

Substituting into (10):

$$\boxed{\Delta\phi = 4\pi G\, \rho} \qquad \text{(Poisson's equation).} \tag{12}$$

1.5.5. *Equation of motion for test masses.*

$$\boxed{\ddot{\boldsymbol{x}}(t) = -\nabla\phi\big(t,\, \boldsymbol{x}(t)\big)\,.} \tag{13}$$

1.5.6. *Boundary conditions and uniqueness.* Assume $\operatorname{supp}(\rho)$ is compact (closed and bounded by the Heine–Borel theorem), with

$$\phi(t,\, \boldsymbol{x}) \to 0 \quad \text{as} \quad \|\boldsymbol{x}\| \to \infty\,.$$

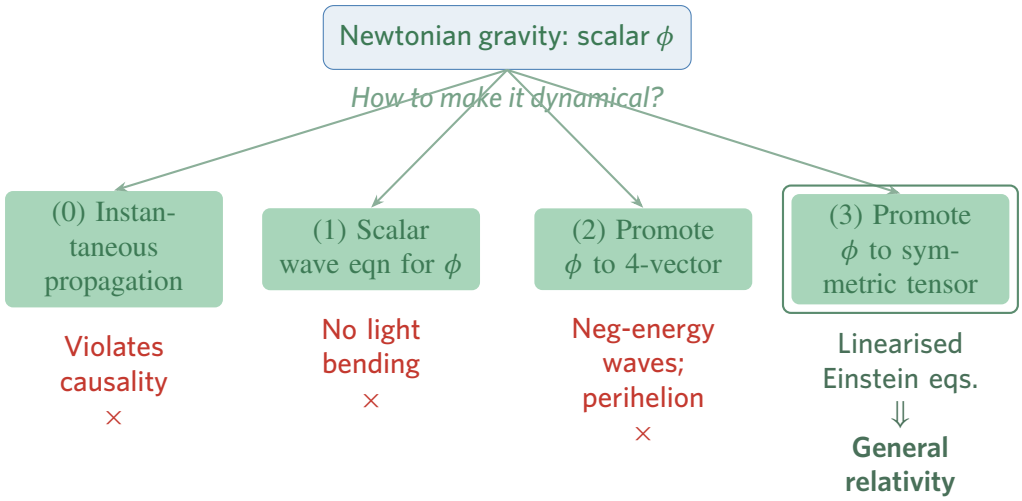
Then the unique solution to Poisson's equation is

$$\boxed{\phi(t,\, \boldsymbol{x}) = -G \int_{\mathbb{R}^3} \mathrm{d}^3 x'\, \frac{\rho(t,\, \boldsymbol{x}')}{\|\boldsymbol{x} - \boldsymbol{x}'\|}\,.} \tag{14}$$

This gives the "gravitostatic" formulation of Newtonian gravity: symbol for symbol, it mirrors electrostatics. Just as in electrostatics, the field equation, potential, and integral solution form a self-contained package. The key difference, as always, is that $\rho \geq 0$: gravity has only one sign of "charge."

1.6. Adding dynamics: routes to general relativity.

How do we add dynamics to the gravitational field? At present, the gravitational field in our formulation is static: the Poisson equation determines ϕ from the instantaneous mass distribution. But what happens when the sources move? Several approaches suggest themselves, with very different outcomes:



See Misner, Thorne & Wheeler [MTW73], Chapter 7, for a detailed and beautiful discussion of all four routes.

Route (0) has obvious causality problems: by rearranging masses one could signal faster than light, violating special relativity. Route (1), promoting the Poisson equation to a scalar wave equation, is mathematically consistent but predicts no bending of light by gravity, which has been observed. Route (2), completing ϕ to a 4-vector, as one does for the electromagnetic potential, suffers from negative-energy wave solutions and fails to reproduce the observed perihelion precession of Mercury. Only route (3), completing ϕ to a symmetric rank-2 tensor, survives all experimental tests. Remarkably, the dynamics one is forced to write down turn out to be precisely the linearised form of Einstein's field equations, from which the full nonlinear theory of general relativity can be recovered.

We will *not* follow this route to discover general relativity; it is not the path Einstein took (at least not as far as the historical record reveals). Instead, we will arrive at the theory from a completely different direction, starting from the equivalence principle in the next lecture.

Why a tensor?

A scalar field ϕ has one component: not enough to encode the geometry of a 4-dimensional spacetime. A 4-vector has four components, but leads to unphysical negative-energy solutions. A symmetric rank-2 tensor $g_{\mu\nu}$ has ten independent components in four dimensions: exactly the right number to describe a metric on spacetime. This is the gravitational field of general relativity.

1.7. Example: Lagrange points in a two-body system. The Newtonian gravitational potential for a system of two bodies in the *co-rotating frame* illustrates the interplay between gravity and inertial (centrifugal/Coriolis) forces. The **effective potential** is

$$\Phi_{\text{eff}}(x, y) = -\frac{1-\mu}{r_1} - \frac{\mu}{r_2} - \tfrac{1}{2}(x^2 + y^2)\,, \tag{15}$$

where $\mu = M_2/(M_1 + M_2)$ is the mass ratio, and $r_{1,2}$ are the distances to the two masses. This potential has five critical points: the **Lagrange points** L_1–L_5. The collinear points L_1, L_2, L_3 are saddle points (unstable), while L_4 and L_5 form equilateral triangles with the two masses and are linearly stable for $\mu < \mu_{\text{crit}} \approx 0.0385$.

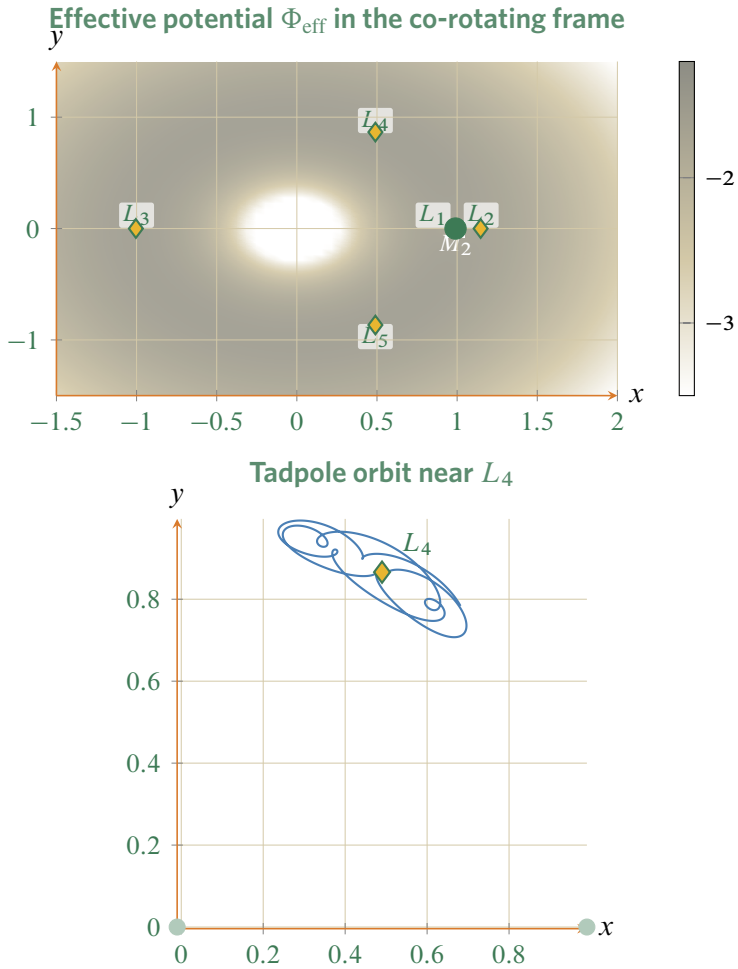


FIGURE 2. Top: pseudo-colour plot of the effective potential Φ_{eff} in the co-rotating frame ($\mu = 0.01$), with the five Lagrange points marked. Bottom: tadpole orbit of a test particle released near L_4 with a small perturbation, exhibiting the characteristic epicyclic motion.

2. The equivalence principle and Mach's principle

In the previous lecture we reviewed the fundamental assumptions of Newtonian physics—in particular, the preferred reference frame and the three conceptually distinct notions of mass. We now turn to the two big principles that motivated Einstein to formulate general relativity: the *equivalence principle* and *Mach's principle*.

2.1. Einstein's equivalence principle (EEP). The first, and arguably most important, principle underlying the formulation of general relativity is the Einstein equivalence principle. In its modern form, EEP consists of three parts:

(1) **Universality of free fall (UFF)** (also known as the *weak equivalence principle*, WEP). Not rejected (a key distinction: in physics we cannot *confirm* a hypothesis, only fail to reject it) e.g. in torsion balance experiments (cf. §1.4).

(2) **Local Lorentz invariance (LLI).**
In a freely falling reference frame, the laws of non-gravitational physics are those of special relativity—there is no preferred reference frame. Recall that the existence of a preferred frame was a fundamental assumption of Newtonian physics (§1); this is one of

the assumptions one must already abandon in the transition to special relativity. In particular, the laws of physics do not depend on the orientation or velocity of the observer.

Best bounds from Michelson–Morley–type experiments:

$$\frac{\Delta c}{c} < 5 \times 10^{-17} . \tag{16}$$

(3) **Local position invariance (LPI).**
Special relativity holds independent of spacetime position (no "preferred" location).

The three pillars of EEP

UFF says *all bodies fall the same way*. LLI says *physics looks the same regardless of your velocity*. LPI says *physics looks the same regardless of where you are*. Together, they force gravity to be a property of spacetime geometry rather than an ordinary force.

2.1.1. *LPI and gravitational redshift.* LPI is equivalent (exercise) to the *universality of gravitational redshift* (UGR): clock rates are universally affected by the gravitational field.

Exercise 2.1. Show that for small velocities and weak gravitational fields, the fractional change in clock rate between two positions $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ is

$$\boxed{\frac{\Delta \nu}{\nu} = \frac{\Delta \phi}{c^2}} , \tag{17}$$

where $\nu(\boldsymbol{x}_1)$, $\nu(\boldsymbol{x}_2)$ are the clock rates at positions 1, 2 and $\Delta\phi = \phi(\boldsymbol{x}_1) - \phi(\boldsymbol{x}_2)$. *Hint:* There are (at least) two approaches—a Doppler-shift argument, or an argument using the quantum expression $E = h\nu$ for the energy of a photon and tracking how it changes at different points in the gravitational field.

Estimate: On Earth's surface, for a height difference $\|\boldsymbol{x}_1 - \boldsymbol{x}_2\| \sim 30$ cm, this effect is $\Delta\phi/c^2 \sim 10^{-17}$—and has been measured! Modern atomic clocks are sensitive enough that gravitational redshift is now routinely measurable over laboratory-scale height differences.

Gravity slows clocks

A clock at the bottom of a gravitational potential well runs slower than a clock at the top. GPS satellites must correct for this: their clocks run faster (by $\sim 45\ \mu$s/day) than ground-based clocks. Without this correction, GPS positions would drift by ~ 10 km/day.

2.2. From EEP to geometry. **Observation:** Using EEP, it is possible to "derive" (or at least argue convincingly) that gravity should be described *geometrically* (K. Thorne, D. Lee, & A. Lightman, *Phys. Rev. D*, **7**, 3563–3578, 1973).

One should be careful here: formulating a physical theory in geometric language is, by itself, no particular magic. Newton's mechanics can be formulated geometrically (via symplectic geometry), as can thermodynamics, statistical mechanics, and quantum mechanics (via Kähler manifolds). If you are willing to work hard enough, just about *any* physical theory admits a geometric formulation.

What makes general relativity special is not merely that it *has* a geometric formulation, but what that geometry *means* physically:

The geometric consequence of EEP

All matter—electrons, protons, gluons, the Higgs field—couples to the *same geometry*, namely that of spacetime. A single metric suffices to describe all of these couplings simultaneously. This universal coupling is what distinguishes general relativity from any other theory that happens to admit a geometric formulation.

2.3. Calculations with EEP. **Statement:** No external, static, homogeneous gravitational field can be detected in a freely falling elevator.

This is one of Einstein's classic thought experiments—"Einstein's elevator." Note the qualifiers *static* and *homogeneous*: if the gravitational field varies in space (is not translation-invariant), then tidal forces arise, and these *can* be detected even in a freely falling frame. This distinction will become important when we discuss curvature.

Suppose we have N (non-relativistic) particles moving in a pair force field $\boldsymbol{F}(\boldsymbol{x}_j - \boldsymbol{x}_k)$ and an external gravitational field $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{g}$ (constant), with $\boldsymbol{F}(\boldsymbol{0}) = \boldsymbol{0}$.
Equations of motion:

$$m_j \frac{d^2 \boldsymbol{x}_j}{dt^2} = m_j \boldsymbol{g} + \sum_{k=1}^{N} \boldsymbol{F}(\boldsymbol{x}_j - \boldsymbol{x}_k) . \tag{18}$$

We now make a non-Galilean change to an *accelerating (freely falling) frame*:

$$\boldsymbol{x}' = \boldsymbol{x} - \tfrac{1}{2} \boldsymbol{g} t^2 , \qquad t' = t . \tag{19}$$

Then the equations of motion become (exercise):

$$\boxed{m_j \frac{d^2 \boldsymbol{x}'_j}{dt'^2} = \sum_k \boldsymbol{F}(\boldsymbol{x}'_j - \boldsymbol{x}'_k)} . \tag{20}$$

The gravitational field has been *eliminated*—the freely falling frame sees only the inter-particle forces. This is a first concrete demonstration of EEP within Newtonian physics.

2.4. Gravitational forces and freely falling coordinates. We now tell essentially the same story, but within special relativity—a more demanding calculation, and a good check on one's special-relativistic mechanics. These calculations are meant to give facility with non-Galilean, non-Lorentzian coordinate changes; along the way, many of the symbols of differential geometry will already start to appear in this quite elementary setting.

According to EEP, for *any* particle moving purely under the influence of a gravitational field, there exists a freely falling coordinate system $(\xi^0, \xi^1, \xi^2, \xi^3)$ such that

$$\boxed{\frac{d^2 \xi^\alpha}{d\tau^2} = 0} \qquad (\equiv \text{ force-free motion}) , \tag{21}$$

where τ is the *proper time*, defined by

$$d\tau^2 = -\eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta , \tag{22}$$

or equivalently,

$$-1 = \eta_{\alpha\beta} \frac{d\xi^\alpha}{d\tau} \frac{d\xi^\beta}{d\tau} . \tag{23}$$

Here $\eta_{\alpha\beta}$ is the *Minkowski metric*:

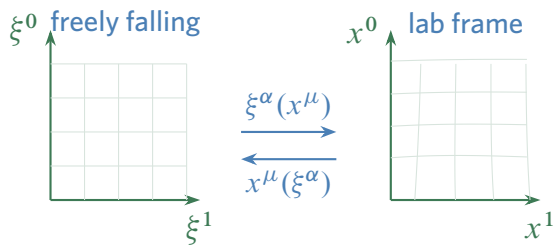
$$\eta_{\alpha\beta} = \mathrm{diag}(-1, +1, +1, +1) . \tag{24}$$

Einstein's summation convention:

$$u_\alpha \, \xi^\alpha \equiv \sum_{\alpha=0}^{3} u_\alpha \, \xi^\alpha \,. \tag{25}$$

Repeated indices (one "upstairs," one "downstairs") are summed over.

2.5. The geodesic equation. Suppose x^μ denotes some other—completely general—set of coordinates: a lab frame, a rotating frame, a frame "jumping up and down on a trampoline"—anything at all:



The freely falling coordinates are functions of the lab coordinates:

$$\xi^\alpha = \xi^\alpha(x^0, x^1, x^2, x^3) \,.$$

The force-free equation (21) becomes

$$0 = \frac{d^2\xi^\alpha}{d\tau^2} = \frac{d}{d\tau}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{dx^\mu}{d\tau}\right) = \frac{\partial \xi^\alpha}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \, \partial x^\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} \,. \tag{26}$$

Multiplying by $\frac{\partial x^\lambda}{\partial \xi^\alpha}$ and using $\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{\partial x^\lambda}{\partial \xi^\alpha} = \delta^\lambda{}_\mu$, we obtain the **geodesic equation**:

$$\boxed{\frac{d^2x^\lambda}{d\tau^2} + \Gamma^\lambda{}_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0} \,, \tag{27}$$

where the *affine connection* (Christoffel symbol) is

$$\Gamma^\lambda{}_{\mu\nu} = \frac{\partial x^\lambda}{\partial \xi^\alpha}\frac{\partial^2 \xi^\alpha}{\partial x^\mu \, \partial x^\nu} \,. \tag{28}$$

The affine connection encodes how the freely falling coordinates depend on our (possibly wild) lab coordinates, together with a Jacobian factor. When we come to formulate differential geometry properly (starting in lecture 3), we will encounter the geodesic equation and affine connections in much greater generality—but they already appear here, in this entirely elementary calculation.

Geodesics on a curved surface

To build intuition for the geodesic equation, consider a 2D surface embedded in $\mathbb{R}^3$. The shortest paths on the surface—the geodesics—are *not* straight lines in the ambient space; they bend to stay on the surface. The deviation from a straight line is governed by the Christoffel symbols, which encode the surface's curvature.

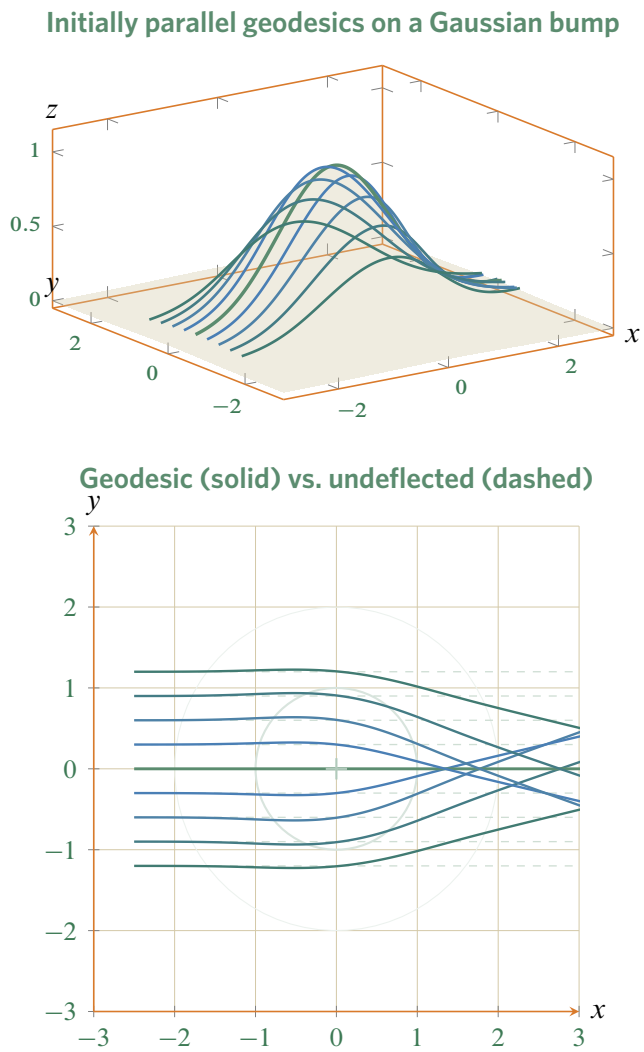


FIGURE 3. Initially parallel geodesics on a Gaussian-bump surface $z = e^{-(x^2+y^2)/2}$. Nine geodesics all start from $x = -2.5$ moving in the $+x$ direction at equally spaced y offsets. Top: 3D view showing the geodesics bending over the bump. Bottom: top-down view comparing geodesics (solid) with undeflected straight paths (dashed). The bump acts as a "gravitational lens": curvature causes the initially parallel paths to converge, cross, and then diverge.

2.6. The metric. The proper time (22) expressed in the lab coordinates becomes

$$\begin{aligned} d\tau^2 &= -\eta_{\alpha\beta}\, d\xi^\alpha\, d\xi^\beta = -\eta_{\alpha\beta}\, \frac{\partial \xi^\alpha}{\partial x^\mu}\, \frac{\partial \xi^\beta}{\partial x^\nu}\, dx^\mu\, dx^\nu \\ &= -g_{\mu\nu}\, dx^\mu\, dx^\nu\,, \end{aligned} \tag{29}$$

where

$$g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu}\, \frac{\partial \xi^\beta}{\partial x^\nu}\, \eta_{\alpha\beta} \tag{30}$$

is the **metric**. The metric collects all the information that appeared when we changed to the new coordinate system; it is the object that allows us to compute proper time (and hence distances and clock rates) without having to transform back to freely falling coordinates every time.

What the metric encodes

The metric $g_{\mu\nu}$ tells you how to measure distances and time intervals in arbitrary coordinates. In freely falling coordinates it reduces to the flat Minkowski metric $\eta_{\alpha\beta}$. The deviation of $g_{\mu\nu}$ from $\eta_{\alpha\beta}$ encodes the gravitational field. Ultimately, $g_{\mu\nu}$ is all one needs to describe the geometry of spacetime.

So far we have seen only one half of the story: the geometry of spacetime tells freely falling particles how to move (via the geodesic equation). The other half—that matter tells spacetime how to curve—requires Mach's principle and, eventually, Einstein's field equations.

2.7. Mach's principle. Mach's principle is the second key set of ideas—really a *circle* of ideas, less precise than EEP and harder to make mathematically sharp, but profoundly influential on Einstein's thinking.

Mach's principle (summary)

All matter in the universe should contribute to the local definition of "non-accelerating" and "non-rotating." In a universe devoid of matter, these concepts should have no meaning.

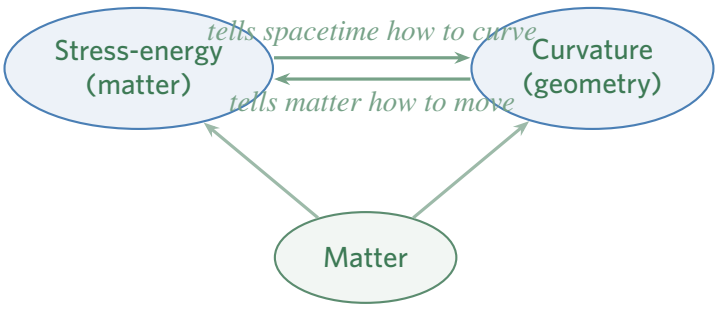
This is a deep statement: the very notions of acceleration and rotation are not absolute but are determined by the matter content of the universe. Without matter, there is nothing with respect to which one could be "accelerating" or "rotating."

2.8. General relativity: the synthesis. General relativity is the following statement:

General relativity

The observer-independent properties of spacetime are described by a *spacetime metric* $g_{\mu\nu}$—a symmetric rank-2 tensor field assigning four numbers to each spacetime event—which need not have the flat form $\eta_{\mu\nu}$ of special relativity. *Curvature* is that property of the metric that accounts for the physical effects of gravitation. Furthermore, curvature is determined by the distribution of stress-energy and momentum, and vice versa.

The "vice versa" is crucial: stress-energy tells spacetime how to curve, and curvature tells matter how to move. This self-referential, nonlinear interplay between geometry and matter is what makes general relativity so remarkable. The extraordinary—one might say almost miraculous—thing is that this all holds together: the theory does not collapse into inconsistency, but instead forms a beautifully self-contained whole.



Actually implementing the physical content of this formulation—computing curvature, writing down Einstein's field equations, extracting predictions—requires the mathematics of spacetime manifolds that are not flat. That is the work we commence in the next lecture, beginning with the theory of differentiable manifolds.

3. Manifolds

It is a core underlying assumption of both special and general relativity—indeed of much of physics—that spacetime is a continuum: to characterise an event we need four numbers (one temporal, three spatial). In special relativity this continuum is the familiar flat $\mathbb{R}^4$, but in general relativity the geometry itself is dynamical—it changes in response to the distribution of matter, energy, and stress throughout the universe.

Because we now allow the geometry to be curved and dynamical, we need a mathematical apparatus to describe and quantify the situation precisely; without it we would be limited to qualitative remarks, whereas in physics we want to make sharp, falsifiable predictions. One approach would be to model curved continua as surfaces embedded in a higher-dimensional ambient space $\mathbb{R}^m$. This works, but it raises an uncomfortable question: do these extra dimensions exist? We sidestep the issue entirely by studying curved continua *intrinsically*, without reference to any ambient space—and this leads us to the theory of *manifolds*.

Why manifolds?

In general relativity, spacetime is a manifold $\mathcal{M}$. Although we may picture $\mathcal{M}$ as sitting inside some higher-dimensional $\mathbb{R}^m$, all physically meaningful statements should be intrinsic: they must not depend on the embedding. The intrinsic approach requires more formalism upfront, but it is ultimately more natural and more transferable to other areas of theoretical physics.

3.1. Notation and prerequisites.

We work in n dimensions throughout. One might ask: why bother with arbitrary n when spacetime is four-dimensional? There are several good reasons. First, it costs almost no extra effort and yields considerable generality for free. Second, we often wish to restrict attention to dynamics on a lower-dimensional hypersurface (a particle constrained to the surface of a sphere, for instance). Third—and more speculatively—there are indications from quantum gravity (the holographic principle, for example) that the effective dimension of spacetime may itself be emergent. Finally, in a quantum theory of gravity one might need to superpose spacetimes of different dimensions, so it is wise not to hard-code $n = 4$.

- $\mathbb{R}^n = \{(x^1, \dots, x^n) \mid x^j \in \mathbb{R}\}$.
- For $\boldsymbol{x} = (x^1, \dots, x^n)$, the Euclidean norm is

$$\|\boldsymbol{x} - \boldsymbol{y}\| = \Big(\sum_{j=1}^{n} (x^j - y^j)^2\Big)^{1/2}.$$

- The *open ball* of radius r about $\boldsymbol{y}$ is

$$B_r(\boldsymbol{y}) = \{\boldsymbol{x} \in \mathbb{R}^n \mid \|\boldsymbol{x} - \boldsymbol{y}\| < r\}.$$

- A subset $U \subseteq \mathbb{R}^n$ is *open* if for every $\boldsymbol{x} \in U$ there exists $\varepsilon > 0$ such that $B_\varepsilon(\boldsymbol{x}) \subset U$.
- C^k denotes the set of k-times continuously differentiable functions (from $\mathbb{R}^n$ to $\mathbb{R}$, or between open subsets thereof): all partial derivatives of order $\leq k$ exist and are continuous.
- C^0 = continuous functions, C^∞ = smooth (infinitely differentiable) functions.

3.2. Definition of a manifold.

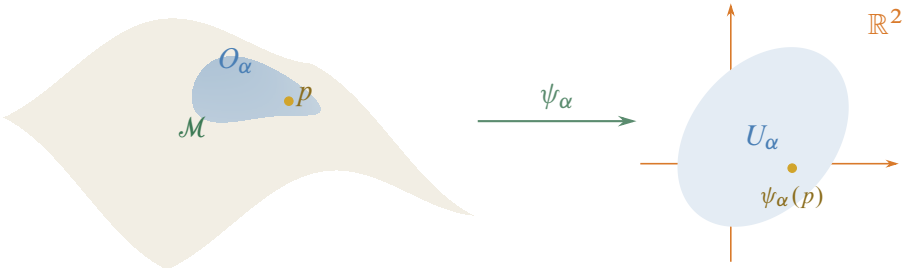
Definition 3.1 (Manifold)**.** An n-dimensional, C^∞, real **manifold** $\mathcal{M}$ is a set together with a collection $\{O_\alpha\}$ of subsets satisfying:

(0) $\mathcal{M}$ is a topological space (Hausdorff and second-countable).[1]

(1) The subsets $\{O_\alpha\}$ **cover** $\mathcal{M}$, i.e. $\bigcup_\alpha O_\alpha = \mathcal{M}$. Equivalently: for every $p \in \mathcal{M}$ there exists at least one α with $p \in O_\alpha$.

(2) For each α there is a **homeomorphism**

$$\psi_\alpha \colon O_\alpha \to U_\alpha\,, \qquad U_\alpha \subset \mathbb{R}^n \text{ (open)}\,,$$

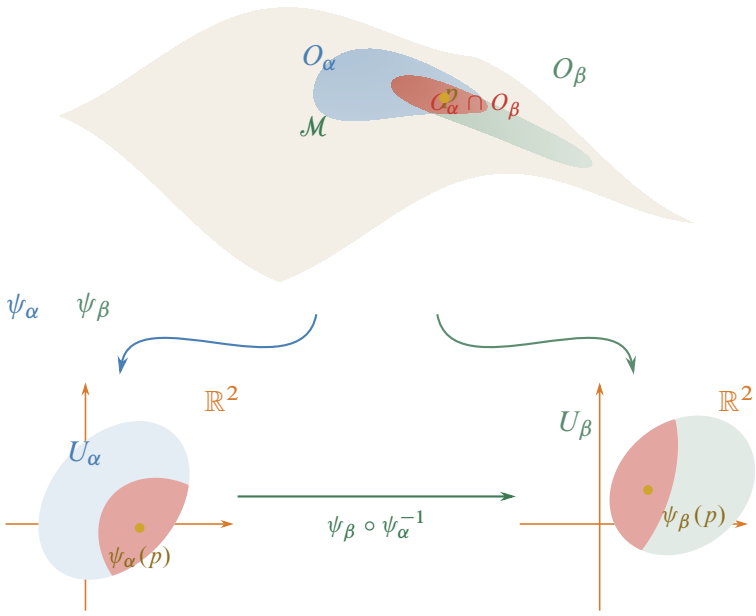


One should think of each pair (O_α, ψ_α) as identifying a piece of the manifold with a piece of Euclidean space. A manifold is thus *built up—sewn together—from patches of* $\mathbb{R}^n$.

(3) If any two sets O_α and O_β **overlap**, i.e. $O_\alpha \cap O_\beta \neq \varnothing$, then the *transition map*

$$\psi_\beta \circ \psi_\alpha^{-1} \colon \psi_\alpha[O_\alpha \cap O_\beta] \;\to\; \psi_\beta[O_\alpha \cap O_\beta]$$

(which is a map between open subsets of $\mathbb{R}^n$) is C^∞. This condition governs the *gluing*: wherever two patches overlap, the two coordinate descriptions must be smoothly related.



3.3. Charts and atlases. The maps ψ_α are called **charts** (in mathematics) or **coordinate systems** (in physics).

The definition as stated depends on the choice of cover $\{O_\alpha\}$ and charts $\{\psi_\alpha\}$. Adding a new chart ψ' compatible with the existing ones would formally give a "new" manifold, even though it carries no new information. To eliminate this ambiguity we require the atlas $\{\psi_\alpha\}$ to be **maximal**: *all* coordinate systems compatible with conditions (2) and (3) are included.

[1] The Hausdorff and second-countability conditions exclude pathological examples; together they imply paracompactness and hence the existence of partitions of unity.

Maximal atlas

Requiring maximality is a bookkeeping device. In practice one specifies a convenient atlas and declares that every compatible chart is implicitly included.

3.4. Examples.

3.4.1. *$\mathbb{R}^n$ as a manifold.* $\mathbb{R}^n$ is the trivial example: it can be covered by a *single* chart $O_1 = \mathbb{R}^n$, $\psi_1 = \mathrm{id}$. As a manifold, $\mathbb{R}^n$ has uncountably many compatible covers (any collection of open sets that covers $\mathbb{R}^n$, together with the identity or other smooth maps, will do).

3.4.2. *Spacetime.* Minkowski spacetime is $\mathbb{R}^1 \times \mathbb{R}^3 \cong \mathbb{R}^{1,3} \cong \mathbb{R}^4$ as a manifold. (The Lorentzian structure—the metric signature—comes later.)

3.4.3. *The sphere S^n.* The n-sphere is

$$S^n = \left\{(x^1, \dots, x^{n+1}) \in \mathbb{R}^{n+1} \;\middle|\; \textstyle\sum_{j=1}^{n+1} (x^j)^2 = 1\right\}. \tag{31}$$

S^n cannot be covered by a single chart (it is compact, while any chart image is an open subset of $\mathbb{R}^n$). Define the $2(n+1)$ open sets

$$O_k^+ = \{(x^1, \dots, x^{n+1}) \in S^n \mid x^k > 0\}, \qquad O_k^- = \{(x^1, \dots, x^{n+1}) \in S^n \mid x^k < 0\}, \tag{32}$$

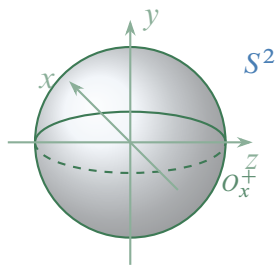
for $k = 1, \dots, n+1$. These cover S^n. The corresponding charts are

$$\psi_k^\pm \colon O_k^\pm \to \mathbb{R}^n, \qquad \psi_k^\pm(x^1, \dots, x^{n+1}) = (x^1, \dots, x^{k-1}, x^{k+1}, \dots, x^{n+1}), \tag{33}$$

i.e. one simply drops the k-th coordinate (whose sign is determined by the choice of $O_k^\pm$).

Special case: S^2. Consider the charts ψ_x^+ (drop x, restricted to $x > 0$) and ψ_y^- (drop y, restricted to $y < 0$). On the overlap $\{(y, z) \in \mathbb{R}^2 \mid y < 0,\ y^2 + z^2 < 1\}$, the transition function is

$$(\psi_y^- \circ (\psi_x^+)^{-1})(y, z) = \left(\sqrt{1 - y^2 - z^2},\ z\right).$$



Exercise 3.1. Find the remaining transition functions for the atlas $\{\psi_k^\pm\}$ of S^n and show that they are all C^∞.

Do we need manifolds?

For many basic solutions of Einstein's field equations the underlying manifold is simply $\mathbb{R}^4$—only the geometry (the metric) varies. So one might wonder whether the full manifold machinery is really necessary. The answer is *yes*: as soon as one considers cosmological spacetimes (which may be compact, like $S^3 \times \mathbb{R}$) or black hole spacetimes (which have non-trivial causal structure and topology), more general manifolds become unavoidable. In quantum gravity one may even wish to sum over manifolds of different topologies, making this generality essential.

3.5. New manifolds from old: product manifolds. Suppose $\mathcal{M}$ and $\mathcal{M}'$ are manifolds of dimension n and n', respectively. We can form the **product manifold** $\mathcal{M} \times \mathcal{M}'$.

Let $\psi_\alpha \colon O_\alpha \to U_\alpha$ and $\psi'_\beta \colon O'_\beta \to U'_\beta$ be charts for $\mathcal{M}$ and $\mathcal{M}'$. Define charts for $\mathcal{M} \times \mathcal{M}'$ via

$$\psi_{\alpha\beta} \colon O_\alpha \times O'_\beta \;\to\; U_\alpha \times U'_\beta \;\subset\; \mathbb{R}^{n+n'} ,$$

with

$$\psi_{\alpha\beta}(p, p') = \big(\psi_\alpha(p),\ \psi'_\beta(p')\big) \;\in\; \mathbb{R}^{n+n'} . \tag{34}$$

Exercise 3.2. Verify that $\mathcal{M} \times \mathcal{M}'$ with the charts $\{\psi_{\alpha\beta}\}$ satisfies the manifold axioms.

Example 3.1 (Products build familiar spaces)**.**

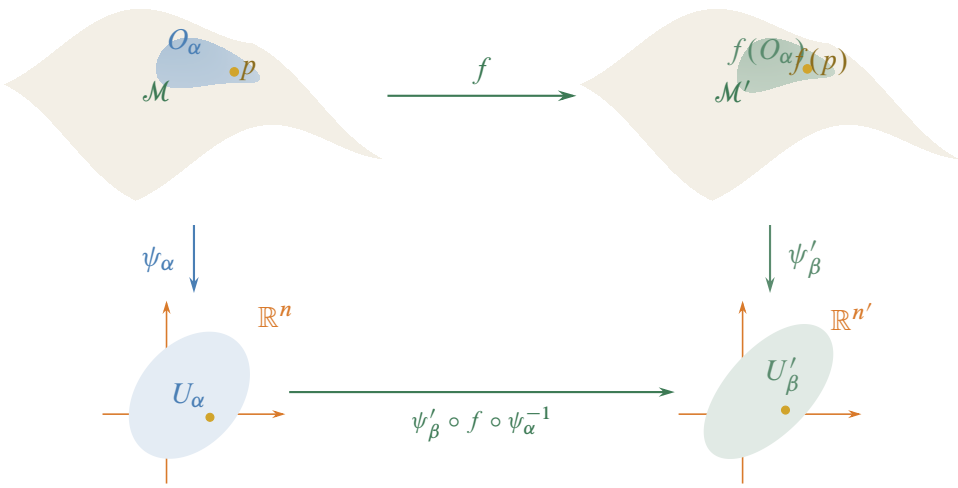
- $\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{n}$.
- The *torus*: $T^2 = S^1 \times S^1$.
- With just $\mathbb{R}$ and S^n and their products, one can construct many of the manifolds relevant to general relativity.

3.6. Smooth maps and diffeomorphisms. Let $\mathcal{M}$ and $\mathcal{M}'$ be manifolds with charts $\{\psi_\alpha\}$ and $\{\psi'_\beta\}$.

Definition 3.2 (Smooth map)**.** A map $f \colon \mathcal{M} \to \mathcal{M}'$ is said to be C^∞ (*smooth*) if for all α, β the composite

$$\psi'_\beta \circ f \circ \psi_\alpha^{-1} \colon U_\alpha \to U'_\beta$$

is C^∞ (as a map between open subsets of Euclidean spaces).



Definition 3.3 (Diffeomorphism)**.** A map $f \colon \mathcal{M} \to \mathcal{M}'$ is a **diffeomorphism** if it is C^∞, one-to-one, onto, and its inverse f^{-1} is also C^∞. If such a map exists, $\mathcal{M}$ and $\mathcal{M}'$ are said to be **diffeomorphic**: they have identical manifold structure.

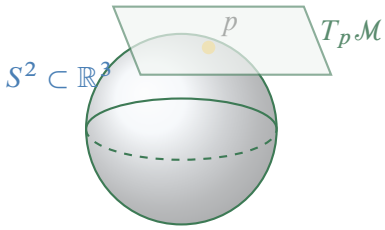
3.7. Vectors on manifolds. Euclidean space $\mathbb{R}^n$ (and Minkowski space) carries a natural *global* vector space structure: $V = \mathbb{R}^n$ is simultaneously a manifold and a vector space.

For a general manifold this is **no longer true**—there is no natural additive structure. This is an instance of a general phenomenon: when one generalises a definition so that it applies to more examples, some of the structure enjoyed by the original example is inevitably lost. Euclidean space is simultaneously a manifold, a vector space, and a group; a generic manifold is none of the latter two.

Example 3.2 (S^2 is not a vector space)**.** The 2-sphere S^2 is a manifold, but it is *not* a vector space: there is no natural way to "add" two points on S^2 and obtain another point on S^2.

We want to preserve as much of this vector-space structure as possible, because without vectors we cannot write down equations of motion for test particles. The key observation is that even though S^2 is not globally a vector space, at each point p there is still a natural *local* vector space—the tangent plane—and this idea generalises to arbitrary manifolds.

It is, however, still possible to associate a vector space to each point of a manifold. For a manifold $\mathcal{M}$ embedded in $\mathbb{R}^m$ (such as $S^2 \subset \mathbb{R}^3$), one can visualise the **tangent plane** at a point p:



But we want an *intrinsic* definition, independent of any embedding.

Intrinsic tangent vectors

The picture above relies on the embedding $S^2 \subset \mathbb{R}^3$, which we have declared off-limits. The solution is a pleasant roundabout: one first observes that there is a one-to-one correspondence between tangent vectors and *directional derivatives* of smooth functions, and then notes that directional derivatives can be defined using only the manifold structure—no embedding required. We therefore *define* tangent vectors as directional derivatives. This is developed in the next lecture.

3.8. Formal definitions (summary). We collect the precise versions of the definitions introduced above for reference.

Definition 3.4 (Locally Euclidean space)**.** A **locally Euclidean space** of dimension d is a Hausdorff topological space $\mathcal{M}$ such that every point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^d$.

Definition 3.5 (Differentiable structure)**.** A **differentiable structure** $\mathcal{F}$ of class C^k ($1 \leq k \leq \infty$) on a locally Euclidean space $\mathcal{M}$ is a collection of coordinate systems $\{(O_\alpha, \psi_\alpha) \mid \alpha \in A\}$ such that:

(a) $\bigcup_{\alpha \in A} O_\alpha = \mathcal{M}$.
(b) $\psi_\alpha \circ \psi_\beta^{-1}$ is C^k for all $\alpha, \beta \in A$.
(c) The collection $\mathcal{F}$ is maximal with respect to (b).

A d-dimensional **differentiable manifold of class** C^k is a pair $(\mathcal{M}, \mathcal{F})$ consisting of a second-countable, locally Euclidean space $\mathcal{M}$ together with a differentiable structure $\mathcal{F}$ of class C^k. (The C^k condition can be generalised to C^∞, complex analytic, etc.)

4. Tangent space

In the previous lecture we saw that a general manifold $\mathcal{M}$ does not carry a natural vector space structure. Nevertheless, we showed that one can *visualise* a tangent plane at each point when the manifold is embedded in some ambient $\mathbb{R}^m$, and we identified a strategy: replace tangent vectors by directional derivatives, which *can* be defined intrinsically. The goal of this lecture is to carry out that programme—giving a purely *intrinsic* definition of the tangent space that makes no reference to any embedding.

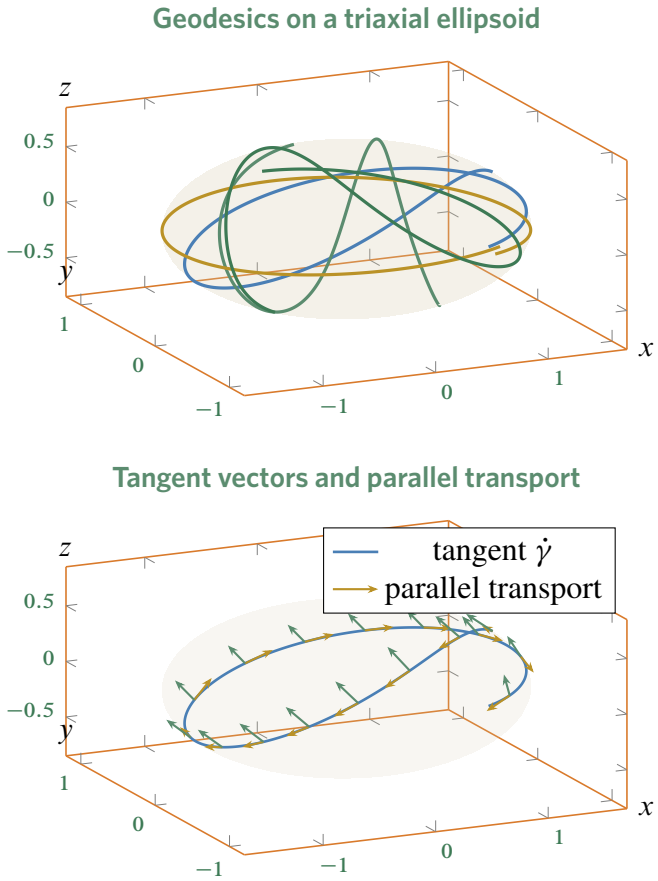


FIGURE 4. Top: geodesics on a triaxial ellipsoid ($a = 1.5$, $b = 1.0$, $c = 0.7$). The curves are shortest paths intrinsic to the surface. Bottom: tangent vectors (gold) and a parallel-transported vector (teal) along a geodesic, illustrating how parallel transport preserves inner products but the transported vector rotates relative to the tangent direction.

4.1. Directional derivatives as vectors. The key idea is to **identify tangent vectors with directional derivatives**.

Let $\boldsymbol{v} = (v^1, \ldots, v^n) \in \mathbb{R}^n$. At a point $p \in \mathbb{R}^n$, define the *directional derivative* along $\boldsymbol{v}$ as follows. Suppose f is a C^1 function from $\mathbb{R}^n \to \mathbb{R}$, and define

$$\boldsymbol{v} \cdot \nabla = \sum_{\mu=1}^{n} v^\mu \frac{\partial}{\partial x^\mu}, \qquad v(f) \equiv (\boldsymbol{v} \cdot \nabla) f(\boldsymbol{x})\Big|_{\boldsymbol{x}=p} = \sum_{\mu=1}^{n} v^\mu \frac{\partial f}{\partial x^\mu}(p). \tag{35}$$

This is the *directional derivative of f at p* along $\boldsymbol{v}$.

Conversely, given the directional derivative operator $(\boldsymbol{v}\cdot\nabla)_p$, we can recover the vector $\boldsymbol{v} \in \mathbb{R}^n$ (apply it to each coordinate function). Furthermore, directional derivatives at p form a **vector space**:

Vectors as derivatives

In $\mathbb{R}^n$, there is a one-to-one correspondence between vectors and directional derivative operators. On a general manifold, where there is no ambient space, we *define* tangent vectors to be directional derivative operators.

4.2. The tangent space V_p.

Notation. Write $\mathscr{F}(\mathcal{M})$ for the set of all C^∞ functions $f\colon \mathcal{M} \to \mathbb{R}$.

Note that $(\boldsymbol{v}\cdot\nabla)_p$ satisfies two properties:

(1) **Linearity:** $v(af+bg) = a\,v(f) + b\,v(g)$ for all $f, g \in \mathscr{F}(\mathcal{M})$ and all $a, b \in \mathbb{R}$.
(2) **Leibniz rule:** $v(fg) = f(p)\,v(g) + g(p)\,v(f)$ (the product rule).

Definition 4.1 (Tangent space)**.** Let $\mathcal{M}$ be a smooth manifold and $p \in \mathcal{M}$. The **tangent space** V_p at p is the set of all maps $v\colon \mathscr{F}(\mathcal{M}) \to \mathbb{R}$ satisfying (1) and (2) above.

Exercise 4.1. Using only properties (1) and (2), show that if $h \in \mathscr{F}(\mathcal{M})$ is constant then $v(h) = 0$ for every $v \in V_p$.

Exercise 4.2. Prove that V_p is a vector space (under pointwise addition and scalar multiplication of the maps v).

Lemma 4.1 (Locality of tangent vectors)**.** *If $v \in V_p$ and $f, g \in \mathscr{F}(\mathcal{M})$ agree on some neighbourhood of p, then $v(f) = v(g)$.*

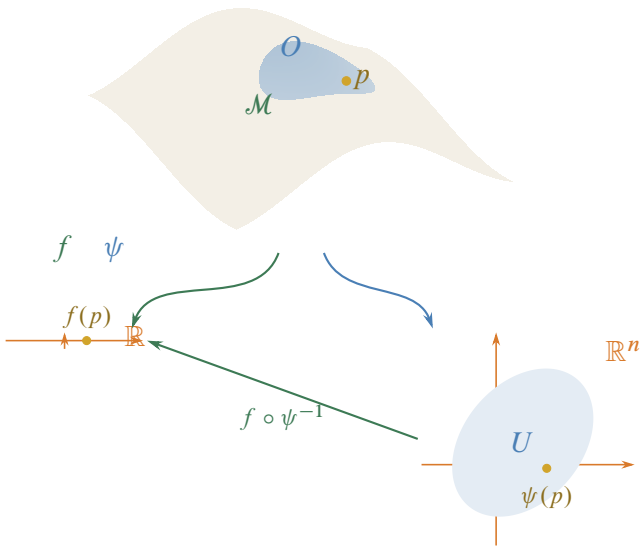
Proof. Let $h = f - g$; by assumption h vanishes on some neighbourhood U of p. By paracompactness (axiom (0) of Def. 3.1) there exists a bump function $\phi \in \mathscr{F}(\mathcal{M})$ with $\phi(p) = 0$ and $\phi \equiv 1$ on $\mathcal{M} \setminus U$. Then $h = \phi h$ pointwise, so by the Leibniz rule $v(h) = \phi(p)\,v(h) + h(p)\,v(\phi) = 0$. □

In particular, a tangent vector depends only on the *germ* of its argument at p, so we may freely apply v to a function defined only on a neighbourhood of p by smoothly extending it to all of $\mathcal{M}$ (any two such extensions give the same value).

Have we created a monster? By demanding only linearity and the Leibniz rule, we might inadvertently have produced a huge—even infinite-dimensional—vector space that has nothing to do with tangent vectors. Fortunately, this is not the case:

Theorem 4.2. *Let $\mathcal{M}$ be an n-dimensional smooth manifold and let $p \in \mathcal{M}$. Then* $\dim(V_p) = n$.

4.3. Coordinate basis. Let $\psi\colon O \to U \subset \mathbb{R}^n$ be a chart with $p \in O$. If $f \in \mathscr{F}(\mathcal{M})$, then the composite $f \circ \psi^{-1}\colon U \to \mathbb{R}$ is C^∞.



Definition 4.2 (Coordinate basis vectors)**.** For $\mu = 1, \ldots, n$, define

$$X_\mu(f) = \left.\frac{\partial(f\circ\psi^{-1})}{\partial x^\mu}\right|_{\psi(p)}, \tag{36}$$

where $(x^1, \ldots, x^n)$ are the coordinates of $\mathbb{R}^n$.

Claim. The X_μ so defined are tangent vectors (i.e. elements of V_p).

Verification. Linearity of X_μ is inherited from that of $\partial/\partial x^\mu$. For the Leibniz rule, use $(fg)\circ\psi^{-1} = (f\circ\psi^{-1})\,(g\circ\psi^{-1})$ together with the ordinary product rule in $\mathbb{R}^n$:

$$X_\mu(fg) = \left.\frac{\partial\big((f\circ\psi^{-1})(g\circ\psi^{-1})\big)}{\partial x^\mu}\right|_{\psi(p)} = f(p)\,X_\mu(g) + g(p)\,X_\mu(f)\,.$$

Proof that $\{X_\mu\}$ is a basis for V_p. We use the following lemma.

Lemma. Let $V\subset\mathbb{R}^n$ be a *convex* open neighbourhood of $\boldsymbol{a} = (a^1,\dots,a^n)$, and let $F\colon V\to\mathbb{R}$ be C^∞. Then there exist C^∞ functions $H_\mu\colon V\to\mathbb{R}$ such that

$$F(\boldsymbol{x}) = F(\boldsymbol{a}) + \sum_{\mu=1}^{n}(x^\mu - a^\mu)\,H_\mu(\boldsymbol{x})\,, \tag{37}$$

with $H_\mu(\boldsymbol{a}) = \left.\dfrac{\partial F}{\partial x^\mu}\right|_{\boldsymbol{x}=\boldsymbol{a}}$. (Convexity ensures $H_\mu(\boldsymbol{x}) = \int_0^1(\partial F/\partial x^\mu)(\boldsymbol{a}+t(\boldsymbol{x}-\boldsymbol{a}))\,dt$ is well-defined; see the homework.)

Let $F = f\circ\psi^{-1}$ and $\boldsymbol{a} = \psi(p)$, and fix a convex open neighbourhood $V\subset U_\alpha$ of $\boldsymbol{a}$ (e.g. an open ball). Then by (37), for all $q\in\psi^{-1}(V)\subset O$,

$$f(q) = f(p) + \sum_{\mu=1}^{n}\big((x^\mu\circ\psi)(q) - (x^\mu\circ\psi)(p)\big)\,H_\mu(\psi(q))\,. \tag{38}$$

The functions $x^\mu\circ\psi$ and $H_\mu\circ\psi$ on $\psi^{-1}(V)$ are extended smoothly to all of $\mathcal{M}$ via a bump function; by Lemma 4.1, v gives the same value on any such extension. Now suppose $v\in V_p$. Apply v to f:

$$\begin{aligned} v(f) &\overset{(38)}{=} v\Big(f(p) + \textstyle\sum_\mu(\cdots)\,H_\mu(\psi(\cdot))\Big) \\ &= \underbrace{v(f(p))}_{=0} + \sum_{\mu=1}^{n}\Big[\underbrace{(H_\mu\circ\psi)(p)}_{\partial F/\partial x^\mu|_{\psi(p)}}\cdot v(x^\mu\circ\psi) + \underbrace{\big((x^\mu\circ\psi)(p) - (x^\mu\circ\psi)(p)\big)}_{=0}\cdot v(H_\mu\circ\psi)\Big] \\ &= \sum_{\mu=1}^{n}\underbrace{v(x^\mu\circ\psi)}_{=:\,v^\mu}\;\left.\frac{\partial(f\circ\psi^{-1})}{\partial x^\mu}\right|_{\psi(p)}\,. \end{aligned} \tag{39}$$

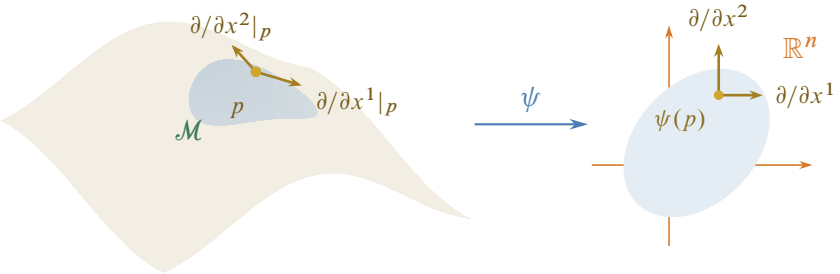
Therefore

$$\boxed{v(f) = \sum_{\mu=1}^{n} v^\mu\,X_\mu(f)}\,, \tag{40}$$

and the $\{X_\mu \mid \mu = 1,\dots,n\}$ form a basis for V_p. □

The basis $\{X_\mu\}$ is called the **coordinate basis**. It is often denoted

$$X_\mu \equiv \left.\frac{\partial}{\partial x^\mu}\right|_p\,, \qquad \text{or} \qquad \partial_\mu\big|_p\,, \qquad \text{or} \qquad e_\mu\,. \tag{41}$$



4.4. Change of basis. Suppose we choose a different chart ψ', giving a coordinate basis $\{X'_\nu\}$. By the chain rule:

$$X_\mu = \sum_{\nu=1}^{n} \frac{\partial x'^\nu}{\partial x^\mu}\bigg|_{\psi(p)} X'_\nu \,. \tag{42}$$

Given a tangent vector $v = \sum_\mu v^\mu X_\mu$, the components in the new basis are

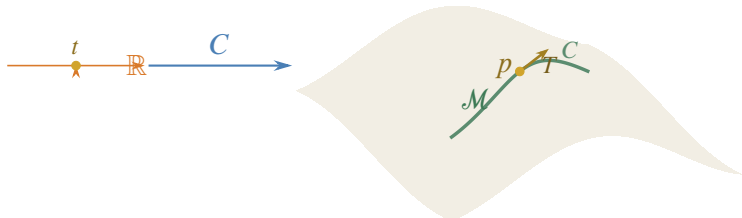
$$\boxed{v'^\nu = \sum_{\mu=1}^{n} v^\mu \frac{\partial x'^\nu}{\partial x^\mu}\bigg|_{\psi(p)}} \tag{43}$$

This is the **(contravariant) vector transformation law**.

4.5. Curves on manifolds. To do mechanics on a manifold we need the notion of a trajectory—a smooth path that a test particle might follow.

Definition 4.3 (Smooth curve)**.** A **smooth curve** C on a manifold $\mathcal{M}$ is a C^∞ map

$$C\colon \mathbb{R} \to \mathcal{M} \qquad \text{(or from an interval } I \subset \mathbb{R}\text{)}\,, \qquad t \mapsto C(t)\,.$$



To each point $p \in \mathcal{M}$ on the curve C (say $p = C(t_0)$), we associate a tangent vector $T \in V_p$ as follows. For $f \in \mathscr{F}(\mathcal{M})$:

$$T(f) = \frac{d}{dt}(f \circ C)\bigg|_p = \sum_\mu \frac{\partial(f \circ \psi^{-1})}{\partial x^\mu}\bigg|_{\psi(p)} \frac{dx^\mu}{dt} = \sum_\mu \frac{dx^\mu}{dt} X_\mu(f)\,, \tag{44}$$

where $x^\mu(t) \equiv (x^\mu \circ \psi \circ C)(t)$ denotes the μ-th component of $\psi(C(t)) \in \mathbb{R}^n$.

This expansion works for *any* coordinate basis. The components of T in the basis $\{X_\mu\}$ are

$$T^\mu = \frac{dx^\mu}{dt}\,. \tag{45}$$

4.6. The tangent bundle. We call V_p the **tangent space at** p. The **tangent bundle** is the disjoint union of all tangent spaces:

$$T\mathcal{M} = \bigcup_{p \in \mathcal{M}} V_p\,. \tag{46}$$

Remark. Although $\dim(V_p) = \dim(V_q) = n$ for all $p, q \in \mathcal{M}$, and thus $V_p \cong V_q$ as vector spaces, the isomorphisms are **not natural**. There is no canonical way to identify a tangent vector at p with one at q; the identification can even vary wildly as one moves around $\mathcal{M}$. To obtain a "good" (smooth, geometrically meaningful) identification one needs extra data—a *connection*, which we introduce in Lecture 7.

4.7. Tangent vector fields.

Definition 4.4 (Tangent vector field)**.** A **tangent vector field** v on a manifold $\mathcal{M}$ is a smooth assignment of a tangent vector $v|_p \in V_p$ to each point $p \in \mathcal{M}$. We say v is *smooth* if for every $f \in \mathscr{F}(\mathcal{M})$, the function $v(f)\colon \mathcal{M} \to \mathbb{R}$ defined by $p \mapsto v|_p(f)$ is C^∞.

Note that smoothness of a vector field is defined purely in terms of the manifold structure—no metric is required. This is perhaps surprising: we cannot yet compare tangent vectors at different points, yet we can still say whether a vector field varies smoothly.

Lemma 4.3. *The coordinate basis fields X_μ are smooth: $X_\mu(f)(p) = \partial(f \circ \psi^{-1})/\partial x^\mu|_{\psi(p)}$ is a C^∞ function of p.*

Since an arbitrary tangent vector v is a linear combination of the X_μ with components v^μ, smoothness of v is equivalent to smoothness of its component functions $v^\mu \in \mathscr{F}(\mathcal{M})$.

Velocity fields

A *velocity field* v is a tangent vector field. If C is a smooth curve solving the equation of motion, then $T(f) = v(f)$ along C—i.e. the tangent to the solution curve equals the velocity field.

5. Flows and tensors

In the previous lecture we defined the tangent space V_p at a point $p \in \mathcal{M}$, coordinate bases, curves, and tangent vector fields. We now study how vector fields generate *flows* (one-parameter groups of diffeomorphisms), how to form new vector fields from old ones via the *commutator*, and the algebraic machinery of *tensors* that lies at the heart of general relativity.

5.1. Flows (one-parameter groups of diffeomorphisms).

Definition 5.1 (One-parameter group of diffeomorphisms)**.** Let $\mathcal{M}$ be a smooth manifold. A **one-parameter group of diffeomorphisms** is a C^∞ map

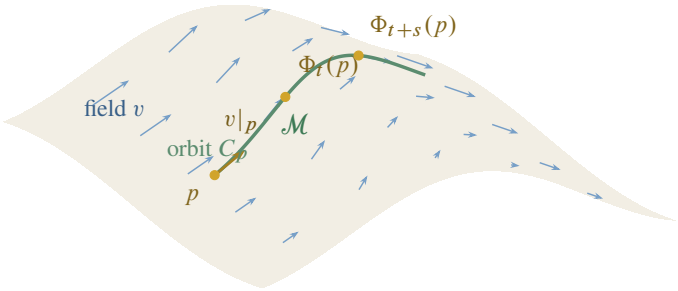
$$\Phi\colon \mathbb{R} \times \mathcal{M} \to \mathcal{M}\,, \qquad (t, p) \mapsto \Phi_t(p)\,,$$

such that

(1) for each fixed $t \in \mathbb{R}$, the map $\Phi_t\colon \mathcal{M} \to \mathcal{M}$ is a diffeomorphism, and
(2) for all $s, t \in \mathbb{R}$,

$$\boxed{\Phi_s \circ \Phi_t = \Phi_{t+s}}\,. \tag{47}$$

Choosing $s = -t$ in (47) immediately gives $\Phi_0 = \mathrm{id}$ (the identity map on $\mathcal{M}$).



5.1.1. *From flows to vector fields.* Given a one-parameter group Φ_t, we obtain a tangent vector field v on $\mathcal{M}$ as follows. For a fixed $p \in \mathcal{M}$, the map

$$C_p \colon \mathbb{R} \to \mathcal{M}\,, \qquad t \mapsto \Phi_t(p)\,,$$

is a smooth curve through p (with $C_p(0) = p$), called the **orbit** of p under Φ_t. Define $v|_p$ to be the tangent vector to C_p at $t = 0$:

$$v|_p(f) = \frac{d}{dt}(f \circ \Phi_t(p))\bigg|_{t=0}\,, \qquad f \in \mathscr{F}(\mathcal{M})\,. \tag{48}$$

Repeating this for every $p \in \mathcal{M}$ yields a smooth vector field v.

5.1.2. *From vector fields to flows (integral curves).* Conversely, given a smooth vector field v on $\mathcal{M}$, we seek **integral curves** of v: a family of curves in $\mathcal{M}$, one through each point p, such that the tangent vector to the curve at p equals $v|_p$.

Let C be such an integral curve through p, and let $(x^1, \dots, x^n)$ be local coordinates. Then the tangent vector to C acts on $f \in \mathscr{F}(\mathcal{M})$ by

$$T(f) = \frac{d}{dt}(f \circ C) = \sum_{\mu=1}^{n} \frac{dx^\mu}{dt}\, X_\mu(f)\,, \tag{49}$$

where $X_\mu = \partial/\partial x^\mu|_p$ is the coordinate basis. Meanwhile, the vector field gives

$$v(f) = \sum_{\mu=1}^{n} v^\mu\, X_\mu(f)\,. \tag{50}$$

Setting (49) equal to (50), we must solve the system of coupled ordinary differential equations

$$\boxed{\frac{dx^\mu}{dt} = v^\mu(x^1, \dots, x^n)} \qquad \mu = 1, \dots, n\,. \tag{51}$$

By the standard existence and uniqueness theorem for ODEs (proved via fixed-point arguments; see, e.g., Coddington and Levinson [CL55]), there exists $\varepsilon > 0$ such that a unique solution exists for all $t \in (-\varepsilon, \varepsilon)$.

Remark. If $\mathcal{M}$ is compact, the integral curves exist for all $t \in \mathbb{R}$, and the family of integral curves defines a genuine one-parameter group of diffeomorphisms.

Flows as fluid motion

Think of v as a velocity field describing steady fluid flow on $\mathcal{M}$. The integral curves are the streamlines: each fluid particle follows the integral curve through its initial position. The map Φ_t advances every particle by time t.

With flows and integral curves in hand, we have assembled the basic toolkit for describing "mechanics on manifolds"—how points move under smooth evolution. Notice, however, that nothing in our discussion so far has involved special or general relativity: we are still working with bare manifolds, without any notion of distance, angle, or causal structure. The machinery we develop next—the commutator of vector fields, and especially the algebraic theory of tensors—will supply precisely the additional structure needed to formulate physics on curved spacetimes.

5.2. New vector fields from old: the commutator.

Definition 5.2 (Commutator / Lie bracket)**.** Let v and w be smooth vector fields on a manifold $\mathcal{M}$. Define a new map $u\colon \mathscr{F}(\mathcal{M}) \to \mathscr{F}(\mathcal{M})$ by

$$u(f) = v\big(w(f)\big) - w\big(v(f)\big) \qquad \text{for all } f \in \mathscr{F}(\mathcal{M})\,. \tag{52}$$

This map u is called the **commutator** (or **Lie bracket**) of v and w, and is written $u = [v, w]$.

Exercise 5.1. Show that $[v, w]$ as defined in (52) is indeed a vector field, i.e. it satisfies linearity and the Leibniz rule at every point.

Remark. Although v and w each involve first-order derivatives, the second-order terms cancel in the combination $v(w(f)) - w(v(f))$, leaving $[v, w]$ a first-order operator—hence a genuine vector field. The commutator plays a central role in the theory of Lie groups and will reappear when we study Lie derivatives and the curvature of connections.

5.3. New vector spaces from old. Before defining tensors on manifolds, we review the key algebraic constructions that build new vector spaces from given ones. The guiding observation is this: our manifold $\mathcal{M}$ canonically provides a single "atomic" vector space at each point—the tangent space V_p. Every other geometric object we need (the metric, the curvature tensor, differential forms, …) lives not in V_p itself but in some vector space *built from* V_p via elementary algebraic operations.

One should assess the value of each construction by whether it produces quantities with a clear physical interpretation. Just because an operation is available does not mean physics requires it.

Let V and W be finite-dimensional real vector spaces. The principal operations are:

(i) **Taking duals:** $V \mapsto V^*$.
(ii) **Taking direct sums:** $V,\ W \mapsto V \oplus W$.
(iii) **Taking tensor products:** $V,\ W \mapsto V \otimes W$.
(iv) **Taking subspaces:** $U \subset V$.
(v) **Taking exterior (wedge) products:** $V,\ W \mapsto V \wedge W$.
(vi) **Taking joins:** $V,\ W \mapsto V \vee W$.

In general relativity, *duals* and *tensor products* are by far the most important: the metric, for instance, naturally lives in a tensor product of two copies of V_p^*. Direct sums play a role in fiber-bundle theory but rarely appear in our treatment; wedge products feature in the theory of differential forms; joins and intersections are more prominent in quantum information theory. We develop the first three in detail below.

5.4. The dual space V^*.

Definition 5.3 (Dual space)**.** Let V be a real vector space. The **dual space** V^* is the vector space of all linear functions $f\colon V \to \mathbb{R}$. Elements of V^* are called **dual vectors** (or **covectors**, or **one-forms**).

Exercise 5.2. Show that V^* is a real vector space under pointwise addition and scalar multiplication.

Remark. Let $\{v_1, \dots, v_n\}$ be a basis for V. We obtain a **dual basis** $\{(v^\mu)^* \mid \mu = 1, \dots, n\}$ for V^* defined by

$$(v^\mu)^*(v_\nu) = \delta^\mu{}_\nu\,, \tag{53}$$

where $\delta^\mu{}_\nu$ is the Kronecker delta.

Notice the placement of indices: elements of V carry a subscript (lower index), while elements of V^* carry a superscript (upper index) in their labelling. This convention will become essential when we work with tensor components on manifolds.

Exercise 5.3. Show that $(V^*)^* \cong V$ (the double dual is canonically isomorphic to V). This result will be used frequently and without further comment throughout the course.

Exercise 5.4. Show that $\dim(V^*) = \dim(V)$.

Remark. The map $v_\mu \mapsto (v^\mu)^*$ gives an isomorphism between V and V^*, but this isomorphism **depends on the choice of basis**—it is *non-canonical*. Had we chosen a different basis for V, we would obtain a different identification between V and V^*. In physics we seek statements that are independent of any particular coordinate system chosen by the observer. A *canonical* (basis-free) isomorphism $V_p \to V_p^*$ arises only once we equip the manifold with a metric; this is the operation of "lowering an index" that we shall meet shortly.

5.5. The tensor product.

The tensor product is one of the most versatile constructions in all of mathematical physics. Beyond its central role in general relativity, it appears whenever one deals with multi-particle systems: the state space of N quantum-mechanical particles is the tensor product of the individual Hilbert spaces, and tensor products of qubit spaces underpin quantum information theory and quantum computing. We give three perspectives on the construction—an informal definition, a formal quotient-space definition, and the dual (multilinear-map) characterisation used in most relativity texts—so that you can make reference to whichever viewpoint is most convenient in your future studies.

5.5.1. *Informal definition.*

Definition 5.4 (Tensor product (informal))**.** Let V and W be finite-dimensional real vector spaces. The **tensor product** $V \otimes W$ consists of elements of the form $v \otimes w$ (for $v \in V$, $w \in W$) together with all their linear combinations, subject to the **bilinearity condition**:

(54)
$$(a_1 v_1 + a_2 v_2) \otimes (b_1 w_1 + b_2 w_2) = a_1 b_1\, v_1 \otimes w_1 + a_1 b_2\, v_1 \otimes w_2 + a_2 b_1\, v_2 \otimes w_1 + a_2 b_2\, v_2 \otimes w_2\,.$$

Example 5.1. Let $V = \mathbb{R}^2$, $W = \mathbb{R}^2$. Then $V \otimes W = \mathbb{R}^2 \otimes \mathbb{R}^2 \cong \mathbb{R}^4$.

5.5.2. *Formal construction.* *(This subsection may be skipped on a first reading without loss of continuity.)*

More precisely, one defines

$$V \otimes W \;\equiv\; F(V \times W)\big/\!\sim\,, \tag{55}$$

where $F(V \times W)$ is the free vector space on the set $V \times W$ (formal linear combinations of ordered pairs (v, w)). Since $V \times W$ is an infinite set, $F(V \times W)$ is infinite-dimensional and far too large; the equivalence relation $\sim$ collapses it to the correct finite-dimensional space. The relation $\sim$ is generated by:

(1) identity, symmetry, and transitivity (equivalence relation axioms),
(2) distributivity: $(v_1 + v_2,\, w) \sim (v_1, w) + (v_2, w)$ and $(v,\, w_1 + w_2) \sim (v, w_1) + (v, w_2)$,
(3) scalar multiples: $c(v, w) \sim (cv,\, w) \sim (v,\, cw)$ for all $c \in \mathbb{R}$.

The equivalence class of (v, w) is written $v \otimes w$. Elements of $V \otimes W$ are linear combinations of **simple tensors** $v_j \otimes w_k$.

Not every tensor is simple

Warning: given $u \in V \otimes W$, it is *not* always true that $u = v \otimes w$ for some $v \in V$, $w \in W$. Generically, u is a sum of simple tensors that cannot be factored.

Exercise 5.5. Show that $\dim(V \otimes W) = \dim(V) \cdot \dim(W)$. *Hint:* If $\{v_\mu\}$ is a basis for V and $\{w_\nu\}$ is a basis for W, show that $\{v_\mu \otimes w_\nu\}$ is a basis for $V \otimes W$.

The tensor product is associative:

$$U \otimes (V \otimes W) \;\cong\; (U \otimes V) \otimes W\,, \tag{56}$$

so we may write $U \otimes V \otimes W$ without ambiguity.

5.5.3. *The dual characterisation of tensors.*

Definition 5.5 (Tensor as multilinear map)**.** Let V and W be real vector spaces. A **tensor** (in the dual sense) is a multilinear map $T\colon V \times W \to \mathbb{R}$, i.e.

$$T(a_1v_1+a_2v_2,\; b_1w_1+b_2w_2) = a_1b_1\,T(v_1,w_1)+a_1b_2\,T(v_1,w_2)+a_2b_1\,T(v_2,w_1)+a_2b_2\,T(v_2,w_2)\,. \tag{57}$$

This is the "classical" approach to tensor products, and the one most commonly seen in general relativity textbooks (including Wald [Wal84]). It is related to the informal definition by

$$(V \otimes W)^* \;\cong\; \{T\colon V \times W \to \mathbb{R} \mid T \text{ multilinear}\}\,. \tag{58}$$

Exercise 5.6. Show that $(V \otimes W)^* \cong V^* \otimes W^*$.

5.6. Tensors on manifolds.

We now apply these algebraic constructions to the tangent space V_p at a point $p \in \mathcal{M}$.

From V_p we build:

- the dual space V_p^* (cotangent vectors),
- tensor products such as $V_p \otimes V_p$, $V_p^* \otimes V_p$, etc.,
- and more generally, the space $\underbrace{V_p \otimes \cdots \otimes V_p}_{k} \otimes \underbrace{V_p^* \otimes \cdots \otimes V_p^*}_{l} \;\cong\; V_p^{\otimes k} \otimes (V_p^*)^{\otimes l}$.

Since $(V_p^*)^* \cong V_p$, an element $t \in V_p$ can be viewed as a linear function on V_p^* as well.

One could also form "mixed" tensor products such as $V_p \otimes V_p^* \otimes V_p$, where the factors are interleaved. In general relativity, however, the order of the individual V_p and V_p^* factors rarely matters (many of the tensors we encounter have symmetry properties that make the ordering irrelevant), so it is conventional to cluster all V_p factors to the left and all V_p^* factors to the right. In quantum information theory, by contrast, the ordering of tensor-product factors carries physical significance (each factor corresponds to a distinct subsystem), and one must keep careful track of it.

5.6.1. *Tensors of type (k,l).*

Definition 5.6 (Tensor of type (k,l))**.** A **tensor of type** (k,l) at p is an element

$$T \;\in\; V_p^{\otimes k} \otimes (V_p^*)^{\otimes l}\,.$$

Equivalently, T is a multilinear map

$$T\colon \underbrace{V_p^* \times \cdots \times V_p^*}_{k} \times \underbrace{V_p \times \cdots \times V_p}_{l} \longrightarrow \mathbb{R}\,.$$

The vector space of all tensors of type (k,l) at p is denoted $\mathcal{T}(k,l)$.

$$\dim\big(\mathcal{T}(k,l)\big) = n^{k+l}\,, \qquad \text{where } n = \dim(V_p)\,. \tag{59}$$

Remark. We adopt the convention $V^{\otimes 0} \cong \mathbb{R}$, so that $\mathcal{T}(0,0) \cong \mathbb{R}$ (scalars).

5.7. Examples of tensors.

Example 5.2 (Type $(0,1)$: dual vectors)**.** A tensor of type $(0,1)$ is an element of $V_p^{\otimes 0} \otimes (V_p^*)^{\otimes 1} \cong \mathbb{R} \otimes V_p^* \cong V_p^*$. It is simply a **dual vector** (covector).

Example 5.3 (Type $(1,0)$: vectors)**.** A tensor of type $(1,0)$ is an element of $V_p^{\otimes 1} \otimes (V_p^*)^{\otimes 0} \cong V_p \otimes \mathbb{R} \cong V_p$. It is simply a **vector**.

Example 5.4 (Type $(1,1)$: linear maps)**.** A tensor of type $(1,1)$ is an element of $V_p \otimes V_p^*$. Such a tensor T can be interpreted in three equivalent ways:

(i) as an element of $V_p \otimes V_p^*$,
(ii) as a multilinear map $V_p^* \times V_p \to \mathbb{R}$ (equivalently, a dual vector in $(V_p^* \otimes V_p)^*$),
(iii) as a **linear transformation** $T\colon V_p \to V_p$, i.e. a matrix.

5.8. Tensor components and the action of tensors.

Choose a basis $\{v_\mu\}$ of V_p with dual basis $\{(v^\nu)^*\}$ for V_p^*. A type-$(1,1)$ tensor T can be expanded as

$$T = \sum_{\mu,\nu} t^\mu{}_\nu \, v_\mu \otimes (v^\nu)^* \;\in\; V_p \otimes V_p^* \,. \tag{60}$$

The scalars $[T]^\mu{}_\nu = t^\mu{}_\nu$ are called the **components** (or **matrix elements**) of T in the basis $\{v_\mu\}$.

Action on a vector. Let $w \in V_p$, with $w = \sum_\nu w^\nu \, v_\nu$. Then

$$\begin{aligned} Tw &= \left(\sum_{\mu,\nu} t^\mu{}_\nu \, v_\mu \otimes (v^\nu)^*\right)(w) = \sum_{\mu,\nu} t^\mu{}_\nu \, (v^\nu)^*(w) \, v_\mu \\ &= \sum_{\mu,\nu} t^\mu{}_\nu \, w^\nu \, v_\mu \,. \end{aligned} \tag{61}$$

Therefore the components of Tw are

$$\boxed{[Tw]^\mu = \sum_{\nu=1}^{n} t^\mu{}_\nu \, w^\nu} \,. \tag{62}$$

This is an example of **tensor contraction**.

5.9. Tensor contraction.

Definition 5.7 (Contraction)**.** Let $T \in \mathcal{T}(k,l)$ with $k \geq 1$ and $l \geq 1$. The **contraction** over the j-th contravariant and j'-th covariant index is the map

$$C_{j,j'}\colon \mathcal{T}(k,l) \to \mathcal{T}(k-1,l-1)$$

defined (interpreting T as a multilinear map) by

$$(C_{j,j'}\,T)(\dots) = \sum_{\sigma=1}^{n} T\big(\dots, \underbrace{(v^\sigma)^*}_{j\text{-th slot}}, \dots; \underbrace{v_\sigma}_{j'\text{-th slot}}, \dots\big)\,, \tag{63}$$

where $\{v_\sigma\}$ is any basis for V_p and $\{(v^\sigma)^*\}$ is the dual basis.

In terms of components:

$$\boxed{(C_{j,j'}\,T)^{\mu_1\cdots\mu_{k-1}}{}_{\nu_1\cdots\nu_{l-1}} = \sum_{\sigma=1}^{n} T^{\mu_1\cdots\sigma\cdots\mu_{k-1}}{}_{\nu_1\cdots\sigma\cdots\nu_{l-1}}} \tag{64}$$

where σ appears in the j-th upper position and the j'-th lower position and is summed over.

Exercise 5.7. Show that the contraction is independent of the choice of basis.

5.10. The outer product.

Definition 5.8 (Outer product)**.** Let $S \in \mathcal{T}(k,l)$ and $T \in \mathcal{T}(k',l')$. Their **outer product** (or **tensor product**) $S \otimes T \in \mathcal{T}(k+k', l+l')$ is defined in components by

$$(S \otimes T)^{\mu_1 \cdots \mu_k \, \mu_{k+1} \cdots \mu_{k+k'}}{}_{\nu_1 \cdots \nu_l \, \nu_{l+1} \cdots \nu_{l+l'}} = S^{\mu_1 \cdots \mu_k}{}_{\nu_1 \cdots \nu_l} \cdot T^{\mu_{k+1} \cdots \mu_{k+k'}}{}_{\nu_{l+1} \cdots \nu_{l+l'}} \,. \tag{65}$$

Intrinsically, $S \otimes T$ lives in

$$\left(V_p^{\otimes k} \otimes (V_p^*)^{\otimes l}\right) \otimes \left(V_p^{\otimes k'} \otimes (V_p^*)^{\otimes l'}\right),$$

which is rearranged (collecting all V_p factors to the left and all V_p^* factors to the right) to give an element of $V_p^{\otimes(k+k')} \otimes (V_p^*)^{\otimes(l+l')}$.

Building tensors

Every tensor can be built from vectors and dual vectors using the outer product, and then contracted. The outer product increases the type, while contraction decreases it. Together, these two operations generate the full algebra of tensor manipulations used throughout general relativity.

6. Tensors continued

Let $p \in \mathcal{M}$ be a point in a manifold $\mathcal{M}$. We write V_p for the tangent space at p. In the previous lecture we constructed tensor products of V_p purely algebraically, ignoring the manifold structure entirely. But a manifold provides *extra data*: not just one vector space but a whole field of vector spaces, one at each point, linked by coordinate charts. Our goal in this lecture is to study how vectors $v \in V_p$, covectors $\omega \in V_p^*$, and more general tensors transform under a change of coordinates on $\mathcal{M}$.

6.1. The cotangent space (dual of V_p).

Definition 6.1 (Cotangent space)**.** The **cotangent space** at p is the dual vector space

$$V_p^* \equiv T_p^* \mathcal{M} \,.$$

Elements of V_p^* are called **covectors** (or *one-forms*).

Recall that V_p has a coordinate basis $\{e_\mu\} = \{\partial/\partial x^\mu|_p\}$. We formally define the **dual basis** $\{e^\mu\} = \{\mathrm{d}x^\mu\}$ by the requirement

$$e^\mu(e_\nu) = \delta^\mu{}_\nu \,. \tag{66}$$

In coordinate notation this reads

$$\boxed{\mathrm{d}x^\mu\left(\left.\frac{\partial}{\partial x^\nu}\right|_p\right) = \delta^\mu{}_\nu} \,. \tag{67}$$

Thus $\mathrm{d}x^\mu$ is a *linear function of tangent vectors*, fully determined by (67).

Remark. At this stage, $\mathrm{d}x^\mu$ is **just a symbol** for a particular covector—a linear function on tangent vectors defined by the Kronecker-delta condition (67). It has nothing (yet) to do with derivatives or integration measures. Later, $\mathrm{d}x^\mu$ will acquire an interpretation as a differential form and integration measure, but for now resist the temptation to read more into the notation than is given by the definition.

Covectors as measuring devices

A tangent vector v specifies a direction and magnitude at p. A covector ω is a linear machine that "measures" tangent vectors: it eats a vector and returns a real number. The dual basis element $\mathrm{d}x^\mu$ measures the μ-th component of a vector.

6.2. Transformation law for covectors. Recall the *contravariant* vector transformation law (Lecture 4):

$$v'^\mu = \sum_\nu v^\nu \frac{\partial x'^\mu}{\partial x^\nu} . \tag{68}$$

Now let $\omega \in V_p^*$. In the coordinate basis,

$$\omega = \sum_\mu \omega_\mu \, \mathrm{d}x^\mu .$$

We ask: how are the components $\omega'_{\mu'}$ in the new coordinate system x'^μ related to the old components ω_μ?

Apply ω to an arbitrary vector $v \in V_p$:

$$\omega(v) = \omega\left(\sum_\mu v^\mu \left.\frac{\partial}{\partial x^\mu}\right|_p\right) = \sum_\mu v^\mu \, \omega\left(\left.\frac{\partial}{\partial x^\mu}\right|_p\right) = \sum_\mu \omega_\mu \, v^\mu . \tag{69}$$

In the primed coordinates the same contraction gives

$$\omega(v) = \sum_{\mu'} \omega'_{\mu'} \, v'^{\mu'} = \sum_{\mu',\mu} \omega'_{\mu'} \frac{\partial x'^{\mu'}}{\partial x^\mu} \, v^\mu . \tag{70}$$

Since (69) and (70) must agree for *all* v^μ, we read off

$$\boxed{\omega_\mu = \sum_{\mu'} \omega'_{\mu'} \frac{\partial x'^{\mu'}}{\partial x^\mu}} \quad \Longleftrightarrow \quad \boxed{\omega'_{\mu'} = \sum_\mu \frac{\partial x^\mu}{\partial x'^{\mu'}} \, \omega_\mu} . \tag{71}$$

This is the **covariant vector transformation law**.

Remark. Compare the two transformation laws:

- **Contravariant** (vectors): $v'^{\mu'} = (\partial x'^{\mu'}/\partial x^\mu)\, v^\mu$.
- **Covariant** (covectors): $\omega'_{\mu'} = (\partial x^\mu/\partial x'^{\mu'})\, \omega_\mu$.

The Jacobian matrices that appear are *inverse transposes* of each other.

6.3. General tensor transformation law. Let $T \in \mathcal{T}(k,l)$ be a tensor of type (k,l). In a coordinate basis, T is expanded as

$$T = \sum T^{\mu_1 \cdots \mu_k}{}_{\nu_1 \cdots \nu_l} \, e_{\mu_1} \otimes \cdots \otimes e_{\mu_k} \otimes e^{\nu_1} \otimes \cdots \otimes e^{\nu_l} , \tag{72}$$

where $e_\mu = \partial/\partial x^\mu|_p$ and $e^\nu = \mathrm{d}x^\nu$.

Under a change of coordinates $x^\mu \to x'^{\mu'}$, the components transform as

$$\boxed{T'^{\mu'_1 \cdots \mu'_k}{}_{\nu'_1 \cdots \nu'_l} = \sum T^{\mu_1 \cdots \mu_k}{}_{\nu_1 \cdots \nu_l} \frac{\partial x'^{\mu'_1}}{\partial x^{\mu_1}} \cdots \frac{\partial x'^{\mu'_k}}{\partial x^{\mu_k}} \frac{\partial x^{\nu_1}}{\partial x'^{\nu'_1}} \cdots \frac{\partial x^{\nu_l}}{\partial x'^{\nu'_l}}} \tag{73}$$

This is the **tensor transformation law**.

Classical definition of a tensor

Classically, a *tensor of type* (k,l) at a point $p \in \mathcal{M}$ is a collection of numbers $T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\nu_l}(p)$ that transforms under coordinate changes via (73).

Remark. Not every collection of numbers indexed by coordinate labels is a tensor. One can associate n^{k+l} numbers to each point $p \in \mathcal{M}$ that do *not* obey (73) under coordinate changes (the Christoffel symbols, which we shall meet in Lecture 7, are a prominent example). Tensor fields are therefore quite special: they are the coordinate-indexed quantities whose transformation law is purely algebraic and involves only Jacobian factors.

6.4. Smooth tensor fields.

Definition 6.2 (Smooth tensor field)**.** A **smooth tensor field** T of type (k,l) on $\mathcal{M}$ is an assignment of a tensor $T|_p \in \mathcal{T}(k,l)(V_p)$ to each point $p \in \mathcal{M}$, such that for all smooth covector fields $\omega^1,\dots,\omega^k$ and all smooth vector fields $v_1,\dots,v_l$, the function

$$p \longmapsto T|_p(\omega^1|_p,\dots,\omega^k|_p;\ v_1|_p,\dots,v_l|_p)$$

is C^∞ on $\mathcal{M}$.

Remark. A covector field ω is *smooth* if and only if $\omega(v)$ is a C^∞ function on $\mathcal{M}$ for every smooth vector field v.

6.5. Examples from special relativity. The preceding lectures have built up an abstract formalism for multilinear objects on manifolds. We now put it to work by re-examining several familiar physical quantities—the four-current, the energy-momentum tensor, and the metric—through the lens of tensor analysis, all set in Minkowski spacetime $\mathcal{M} = \mathbb{R}^{1,3}$.

6.5.1. *Currents and charge densities.* Consider N particles with worldlines $x_j(t)$ and charges q_j, $j = 1,\dots,N$.

Definition 6.3 (Charge density and current)**.** The **charge density** is

$$\varepsilon(\boldsymbol{x},t) = \sum_{j=1}^{N} q_j\,\delta^{(3)}\big(\boldsymbol{x}-\boldsymbol{x}_j(t)\big)\,, \tag{74}$$

and the **current density** is

$$\boldsymbol{J}(\boldsymbol{x},t) = \sum_{j=1}^{N} q_j\,\delta^{(3)}\big(\boldsymbol{x}-\boldsymbol{x}_j(t)\big)\,\frac{d\boldsymbol{x}_j(t)}{dt}\,. \tag{75}$$

Define a **four-current** J^μ by setting

$$J^\mu = (\varepsilon,\ \boldsymbol{J})\,. \tag{76}$$

Exercise 6.1. Argue that J^μ is a vector (i.e. an element of V_p) field under Lorentz transformations $x'^\nu = \Lambda^\nu{}_\mu\, x^\mu$.

6.5.2. *The energy-momentum tensor.* Consider N particles with energy-momentum four-vectors p_j^μ, $j = 1,\dots,N$.

The density of the μ-th component of four-momentum is

$$T^{\mu 0}(\boldsymbol{x},t) = \sum_{j=1}^{N} p_j^\mu(t)\,\delta^{(3)}\big(\boldsymbol{x}-\boldsymbol{x}_j(t)\big)\,. \tag{77}$$

The corresponding current (the flux of p^μ in the x^k-direction) is

$$T^{\mu k}(\boldsymbol{x}, t) = \sum_{j=1}^{N} p_j^\mu(t) \frac{dx_j^k(t)}{dt} \delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{x}_j(t)\big) . \tag{78}$$

Combining into a single formula (with $x^0(t) = t$):

$$T^{\mu\nu}(x) = \sum_{j=1}^{N} p_j^\mu \frac{dx_j^\nu(t)}{dt} \delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{x}_j(t)\big) . \tag{79}$$

Since $p_j^\nu = E_j \, dx_j^\nu/dt$, this may be rewritten as

$$\boxed{T^{\mu\nu}(x) = \sum_{j=1}^{N} \frac{p_j^\mu \, p_j^\nu}{E_j} \delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{x}_j(t)\big)} \quad . \tag{80}$$

Hence $T^{\mu\nu}$ is **symmetric**: $T^{\mu\nu} = T^{\nu\mu}$.

In covariant (proper-time) form:

$$T^{\mu\nu}(x) = \sum_{j=1}^{N} \int \mathrm{d}\tau \; p_j^\mu \frac{dx_j^\nu}{d\tau} \delta^{(4)}\big(x - x_j(\tau)\big) . \tag{81}$$

Exercise 6.2. Under Lorentz transformations $x'^\nu = \Lambda^\nu{}_\mu \, x^\mu$ (with $\Lambda^{\mu'}{}_\mu = \partial x'^{\mu'}/\partial x^\mu$), show that

$$T'^{\mu'\nu'} = \Lambda^{\mu'}{}_\mu \, \Lambda^{\nu'}{}_\nu \, T^{\mu\nu} . \tag{82}$$

Conclude that $T^{\mu\nu}$ is a tensor of type $(2, 0)$. *Hint:* the covariant form (81) makes the argument more transparent, since the four-dimensional delta function and proper-time integral are manifestly Lorentz-invariant.

The energy-momentum tensor plays an extraordinarily important role in general relativity: it is the object that encodes the complete stress-energy content of matter and radiation, and will appear as the source term on the right-hand side of Einstein's field equations.

6.5.3. *The metric tensor.*

Definition 6.4 (Metric tensor)**.** A **metric tensor** g on a manifold $\mathcal{M}$ is a smooth tensor field of type $(0, 2)$ that is

(1) **symmetric:** $g(v_1, v_2) = g(v_2, v_1)$ for all $v_1, v_2 \in V_p$, and
(2) **nondegenerate:** $g(v, w) = 0$ for all $w \in V_p$ implies $v = 0$.

Why we need a metric

A manifold by itself has no notion of distance or angle. A metric is the extra datum that supplies an *infinitesimal length*: an infinitesimal displacement is modelled by a tangent vector, and the "infinitesimal squared distance" is a quadratic function of that tangent vector—precisely what $g(v, v)$ provides.

In a coordinate basis $\{\partial/\partial x^\mu|_p\}$, expand the metric as

$$g = \sum_{\mu,\nu} g_{\mu\nu} \, \mathrm{d}x^\mu \otimes \mathrm{d}x^\nu . \tag{83}$$

This is often written (omitting the tensor product sign) as the **line element**

$$\boxed{\mathrm{d}s^2 = \sum_{\mu,\nu} g_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu} \quad . \tag{84}$$

The metric defines an **inner product** on V_p for each $p \in \mathcal{M}$:

$$\begin{aligned}(v,w)_p &= \sum_{\mu,\nu} g_{\mu\nu}\,(\mathrm{d}x^\mu \otimes \mathrm{d}x^\nu)(v,w) = \sum_{\mu,\nu} g_{\mu\nu}\,\mathrm{d}x^\mu(v)\,\mathrm{d}x^\nu(w) \\ &= \sum_{\mu,\nu} g_{\mu\nu}\,v^\mu\,w^\nu\,. \end{aligned} \tag{85}$$

6.5.4. *Signature.* By the Gram–Schmidt procedure, there exists an orthonormal basis $\{v_{\hat\mu}\}$ for V_p such that

$$g(v_{\hat\mu},\,v_{\hat\nu}) = s_\mu\,\delta_{\mu\nu}\,, \qquad s_\mu \in \{+1,\,-1\}\,. \tag{86}$$

Exercise 6.3. Prove that an orthonormal basis satisfying (86) exists.

The number of $+1$'s and -1's is independent of the choice of orthonormal basis. This invariant is the **signature** of g.

Definition 6.5 (Riemannian and Lorentzian metrics)**.**

- A metric g with $s_\mu = +1$ for all μ is called **Riemannian** (positive definite).
- The metric of spacetime has signature $(-1,+1,+1,+1)$—this is called a **Lorentzian** metric.

6.6. The metric as a map $V_p \to V_p^*$. A metric tensor g can be viewed simultaneously as:

(1) a $(0,2)$ tensor,
(2) a bilinear map $V_p \times V_p \to \mathbb{R}$, and
(3) a linear map $V_p \to V_p^*$.

The third interpretation is induced by the map

$$v \longmapsto g(\cdot\,,v) \;=:\; \omega\,. \tag{87}$$

Here $g(\cdot\,,v)\colon V_p \to \mathbb{R}$, so indeed $g(\cdot\,,v) \in V_p^*$.

Theorem 6.1. *If g is nondegenerate, then the map (87) is a linear isomorphism $V_p \xrightarrow{\sim} V_p^*$.*

Proof. The map $v \mapsto g(\cdot\,,v)$ is linear by bilinearity of g. Suppose $g(\cdot\,,v) = 0$, i.e. $g(w,v) = 0$ for all $w \in V_p$. By nondegeneracy, $v = 0$. Hence the kernel is trivial and the map is injective. Since $\dim V_p = \dim V_p^*$, it is also surjective. □

Raising and lowering indices

The metric provides a canonical, basis-independent correspondence between vectors and covectors. In components:

$$v_\mu = \sum_\nu g_{\mu\nu}\,v^\nu \qquad \text{(lowering)}\,, \qquad \omega^\mu = \sum_\nu g^{\mu\nu}\,\omega_\nu \qquad \text{(raising)}\,,$$

where $g^{\mu\nu}$ is the matrix inverse of $g_{\mu\nu}$.

Remark. This canonical isomorphism is so convenient that many authors identify vectors with covectors from the outset, never introducing the cotangent space as a separate object. We prefer to keep the distinction explicit: the metric is extra data that may not always be present (e.g. in

symplectic geometry or contact geometry), and understanding the difference between V_p and V_p^* clarifies which structures depend on that extra data and which do not.

6.7. Abstract index notation. Suppose $T \in \mathcal{T}(k,l)$. Think of T as a multilinear map

$$T\colon \underbrace{V_p^* \otimes \cdots \otimes V_p^*}_{k} \otimes \underbrace{V_p \otimes \cdots \otimes V_p}_{l} \longrightarrow \mathbb{R}\,.$$

One can specify T by its components $T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\nu_l}$ in some basis. However, it is often sufficient to know merely *which arguments of T take vectors and which take covectors*, without committing to a particular basis.

Definition 6.6 (Abstract index notation)**.** In the **abstract index notation**, each argument of a tensor is labelled by a lower-case Latin letter.

- A **superscript** (upper) index labels a *contravariant* slot (takes a covector).
- A **subscript** (lower) index labels a *covariant* slot (takes a vector).

Example 6.1.

- A vector is written v^a—one contravariant slot.
- A covector is written ω_a—one covariant slot.
- $T^{ab}{}_{cd}$ denotes a tensor of type $(2,2)$.

Here the lower-case Latin letters $a, b, c, d, \ldots$ label *arguments and their type*, not components with respect to a basis (for which we use Greek indices $\mu, \nu, \rho, \ldots$).

Remark. The abstract index notation (due to R. Penrose) is a compromise between the coordinate-free and component-based approaches. It carries the structural information of index placement (contravariant vs. covariant) without fixing a basis, yet retains the computational convenience of the index calculus. In practice, any physical calculation eventually requires choosing a basis—there is no such thing as doing physics entirely without coordinates. But one can, and should, carry out as many manipulations as possible in a basis-free manner; the abstract index notation is designed to facilitate exactly this.

7. Derivative operators and parallel transport (I)

In the previous lectures we developed the algebra of tensors on a manifold $\mathcal{M}$. We can add tensors, take tensor products, and contract indices—but we cannot yet *differentiate* tensor fields. In $\mathbb{R}^n$ one differentiates by comparing values of a tensor at nearby points; on a general manifold this comparison requires extra structure. That structure is a **derivative operator** (or **affine connection**).

Recall our earlier discussion of parallel transport: given a curve on $\mathcal{M}$, we want a rule that "carries" a tensor from one point to a nearby point along the curve. One should expect a close correspondence between such parallel-transport operations and derivative-type operators. Indeed, an infinitesimal parallel transporter is precisely a covariant derivative: the derivative $\nabla_X Y$ measures how a vector field Y fails to be "parallel" as one moves in the direction X. Our strategy in this lecture is to abstractly and intrinsically characterise those properties that any reasonable covariant derivative operator should possess, and then to explore what freedom remains in choosing such an operator.

Notation. We write $\mathscr{X}(\mathcal{M})$ for the space of smooth vector fields on $\mathcal{M}$, and $\mathscr{F}(\mathcal{M})$ for the space of smooth functions $f\colon \mathcal{M} \to \mathbb{R}$.

7.1. Affine connections: definition.

Definition 7.1 (Affine connection / covariant derivative)**.** An **affine connection** (or **covariant derivative operator**) on a manifold $\mathcal{M}$ is a map

$$\nabla\colon \mathscr{X}(\mathcal{M}) \times \mathscr{X}(\mathcal{M}) \to \mathscr{X}(\mathcal{M}), \qquad (X, Y) \mapsto \nabla_X Y,$$

which we immediately extend to a map $\nabla\colon \mathscr{X}(\mathcal{M}) \times \mathcal{T}(k,l) \to \mathcal{T}(k,l)$, satisfying axioms (0)–(5) below.

7.2. The axioms.

Axiom (0): Additivity in the first argument. For all $X, Y \in \mathscr{X}(\mathcal{M})$ and $Z \in \mathcal{T}(k,l)$:

$$\nabla_{X+Y} Z = \nabla_X Z + \nabla_Y Z. \tag{88}$$

Axiom (0′): $\mathscr{F}(\mathcal{M})$-linearity in the first argument. For all $f \in \mathscr{F}(\mathcal{M})$, $X \in \mathscr{X}(\mathcal{M})$, and $Y \in \mathcal{T}(k,l)$:

$$\nabla_{fX} Y = f\, \nabla_X Y. \tag{89}$$

Remark. Axioms (0) and (0′) together say that ∇_X depends $\mathscr{F}(\mathcal{M})$-linearly on the "direction" X. In abstract index notation we write $\nabla_X \equiv X^a \nabla_a$, so that the derivative in direction X is determined by the components of X and the operator ∇_a.

Axiom (1): Linearity. For all $A, B \in \mathcal{T}(k,l)$:

$$\nabla_X (A + B) = \nabla_X A + \nabla_X B. \tag{90}$$

In abstract index notation:

$$\nabla_c \big(A^{a_1 \cdots a_k}{}_{b_1 \cdots b_l} + B^{a_1 \cdots a_k}{}_{b_1 \cdots b_l} \big) = \nabla_c A^{a_1 \cdots a_k}{}_{b_1 \cdots b_l} + \nabla_c B^{a_1 \cdots a_k}{}_{b_1 \cdots b_l}.$$

Axiom (2): Leibniz rule. For all $A \in \mathcal{T}(k,l)$ and $B \in \mathcal{T}(k',l')$:

$$\nabla_X (A \otimes B) = (\nabla_X A) \otimes B + A \otimes (\nabla_X B). \tag{91}$$

In abstract index notation:

$$\begin{aligned} \nabla_e \big(A^{a_1 \cdots a_k}{}_{b_1 \cdots b_l}\, B^{c_1 \cdots c_{k'}}{}_{d_1 \cdots d_{l'}} \big) \\ = (\nabla_e A^{a_1 \cdots a_k}{}_{b_1 \cdots b_l})\, B^{c_1 \cdots c_{k'}}{}_{d_1 \cdots d_{l'}} + A^{a_1 \cdots a_k}{}_{b_1 \cdots b_l}\, (\nabla_e B^{c_1 \cdots c_{k'}}{}_{d_1 \cdots d_{l'}}). \end{aligned} \tag{92}$$

Axiom (3): Commutativity with contraction. For all $A \in \mathcal{T}(k,l)$ (with $k \geq 1$, $l \geq 1$):

$$\nabla_X (C_{j,j'}\, A) = C_{j,j'} (\nabla_X A). \tag{93}$$

In abstract index notation, contraction is implemented by repeating an index, and the axiom states that ∇_d commutes with this operation.

Axiom (4): Reduction to directional derivative on scalars. For all $f \in \mathscr{F}(\mathcal{M})$:

$$\boxed{\nabla_X (f) = X(f)}. \tag{94}$$

In abstract index notation: $X(f) = X^a \nabla_a f$.

Remark. Axiom (4) says that on scalar functions the covariant derivative reduces to the ordinary directional derivative. No extra data is needed to differentiate scalars.

Axiom (5): Torsion-free. For all $X, Y \in \mathscr{X}(\mathcal{M})$:

$$\boxed{\nabla_X Y - \nabla_Y X - [X, Y] = 0} \quad . \tag{95}$$

Equivalently, in abstract index notation: for all $f \in \mathscr{F}(\mathcal{M})$,

$$\nabla_a \nabla_b f = \nabla_b \nabla_a f \ . \tag{96}$$

Remark. We have now assembled a "wish list" of seven properties (axioms (0)–(5)) that we would like any covariant derivative operator to satisfy. There is some redundancy among these axioms—we have not aimed for the most economical characterisation, but rather for a complete catalogue of the properties we will need. Of course, it is all very well to write down such a list; the question is whether anything *exists* that satisfies it, and if so, how much freedom remains. We address existence and non-uniqueness in §7.4 below.

7.3. Simple consequences. Two useful identities follow directly from the axioms.

Theorem 7.1 (Leibniz rule on products)**.** *For all $X \in \mathscr{X}(\mathcal{M})$, $Y \in \mathscr{X}(\mathcal{M})$, and $f \in \mathscr{F}(\mathcal{M})$:*

$$\nabla_X (f Y) = X(f)\, Y + f\, \nabla_X Y \ . \tag{97}$$

Proof. View fY as the tensor product $f \otimes Y$ (with $f \in \mathcal{T}(0,0)$, $Y \in \mathcal{T}(1,0)$). By axiom (2):

$$\nabla_X (f \otimes Y) = (\nabla_X f) \otimes Y + f \otimes (\nabla_X Y) \, .$$

By axiom (4), $\nabla_X f = X(f)$. The result follows. □

Theorem 7.2 (Lie bracket via ∇)**.** *For all $v, w \in \mathscr{X}(\mathcal{M})$:*

$$\boxed{[v, w]^b = v^a \nabla_a w^b - w^a \nabla_a v^b} \quad . \tag{98}$$

Proof. By axiom (5), $\nabla_v w - \nabla_w v = [v, w]$. Writing $\nabla_v \equiv v^a \nabla_a$ and $\nabla_w \equiv w^a \nabla_a$ gives (98). □

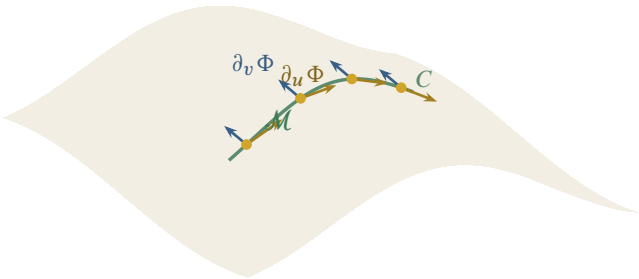
Remark. Equation (98) shows that the Lie bracket, which is defined purely in terms of the manifold structure, can be computed using *any* torsion-free connection. The result is independent of the choice of ∇.

7.4. Existence and non-uniqueness. Having written down our list of axioms, we must verify that something actually satisfies them—otherwise the theory would be vacuous. It turns out that there are *many* derivative operators satisfying axioms (0)–(5). We first exhibit the simplest one, then characterise the freedom in choosing ∇.

7.4.1. *The ordinary derivative.* Let ψ be a chart on $\mathcal{M}$ with coordinate functions $(x^1, \ldots, x^n)$. For a tensor $T \in \mathcal{T}(k, l)$ with components $T^{\mu_1 \cdots \mu_k}{}_{\nu_1 \cdots \nu_l}$ in the coordinate basis, define ∂_a by letting it act via the partial derivative $\partial/\partial x^\sigma$ on these components.

Remark. The operator ∂_a satisfies axioms (0)–(5), but it is **chart-dependent**: it is defined only with respect to the chart ψ. A different chart generally gives a different operator ∂_a. A true covariant derivative is chart-independent.

The chart-α derivative differentiates the *components* of a tensor field, but the basis itself varies from point to point. The figure below illustrates this on the recurring example surface $\mathcal{M} = \{z = h(x, y)\}$: the coordinate basis $\partial/\partial x^\mu = \partial_\mu \Phi$ is sketched at four points along a curve. Although each pair has the same chart components $(1, 0)$ and $(0, 1)$ in $\mathbb{R}^2$, the lifted vectors on $\mathcal{M}$ tilt with the surface. The Christoffel symbols introduced below in §8.3 encode this twist.



7.4.2. *Difference of two derivative operators.* Suppose ∇_a and $\tilde{\nabla}_a$ are two derivative operators both satisfying axioms (0)–(5).

Step 1: On scalars. For all $f \in \mathscr{F}(\mathcal{M})$, axiom (4) gives

$$(\tilde{\nabla}_a - \nabla_a) f = \tilde{\nabla}_a f - \nabla_a f = 0 . \tag{99}$$

Step 2: On covectors. Consider the map $(\tilde{\nabla}_a - \nabla_a) \colon \mathcal{T}(0,1) \to \mathcal{T}(0,2)$. Let $f \in \mathscr{F}(\mathcal{M})$ and $\omega \in \mathcal{T}(0,1)$. Then $f\omega_a \in \mathcal{T}(0,1)$, and by the Leibniz rule (91) and (99):

$$(\tilde{\nabla}_a - \nabla_a)(f\omega_b) = f\,(\tilde{\nabla}_a - \nabla_a)\omega_b . \tag{100}$$

Theorem 7.3. *The quantity $(\tilde{\nabla}_a - \nabla_a)\omega_b$ depends only on the value of ω_b at the point p.*

Proof. Let $\omega'_b \in \mathcal{T}(0,1)$ agree with ω_b at p, i.e. $\omega'_b|_p = \omega_b|_p$. We must show that

$$(\tilde{\nabla}_a - \nabla_a)\omega'_c\big|_p = (\tilde{\nabla}_a - \nabla_a)\omega_c\big|_p .$$

Write the difference $\omega'_b - \omega_b$ in terms of smooth functions f_α vanishing at p and smooth covector fields $\mu_b^{(\alpha)}$:

$$\omega'_b - \omega_b = \sum_\alpha f_\alpha\, \mu_b^{(\alpha)} , \qquad f_\alpha|_p = 0 .$$

(Such a decomposition exists by the smoothness of $\omega' - \omega$ and the fact that it vanishes at p.) Applying $(\tilde{\nabla}_a - \nabla_a)$ and using (100):

$$(\tilde{\nabla}_a - \nabla_a)(\omega'_b - \omega_b)\big|_p = \sum_\alpha f_\alpha|_p\, (\tilde{\nabla}_a - \nabla_a)\mu_b^{(\alpha)}\big|_p = 0 ,$$

since $f_\alpha|_p = 0$. □

Remark. This result is rather striking. A single derivative operator ∇_a applied to a covector field ω_b certainly depends on the behaviour of ω_b in a neighbourhood of p (it involves first-order information, just as an ordinary derivative does). Yet the *difference* of two derivative operators applied to ω_b depends only on $\omega_b|_p$ itself—all the "derivative-like" information cancels. In other words, the difference $\tilde{\nabla}_a - \nabla_a$ is not really a differential operator at all: it is a purely algebraic, pointwise linear map on tensors.

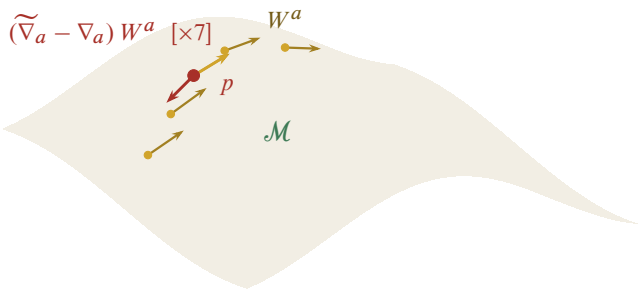
7.5. The C tensor. By Theorem 7.3, the map $(\tilde{\nabla}_a - \nabla_a)$ on covectors at p is a map

$$V_p^* \longrightarrow V_p^* \otimes V_p^* .$$

By duality ($V_p^{**} \cong V_p$), this can be interpreted as an element of $V_p \otimes V_p^* \otimes V_p^*$. We write its components as $C^c{}_{ab}$ and record:

$$\boxed{\nabla_a \omega_b = \tilde{\nabla}_a \omega_b - C^c{}_{ab}\, \omega_c} . \tag{101}$$

Remark. **Warning:** if one takes $\tilde{\nabla}_a = \partial_a$, then $C^c{}_{ab}$ does *not* transform as a tensor field of type $\mathcal{T}(1,2)$ under coordinate changes. Only the *difference* of two genuine covariant derivatives transforms tensorially.



The figure illustrates the algebraic nature of $C^c{}_{ab}$ on the recurring example manifold (Lecture 3). The vector field W has chart-α components $W^u \equiv 1$, $W^v \equiv 0$, so it is constant in chart α and $\partial_a W^b = 0$ identically. The Levi-Civita connection (§8.2 below) of the induced metric gives a nonzero derivative $\nabla_X W$ at the focal point p; the entire vector shown is the C-tensor contribution $C^\sigma{}_{\alpha\nu}\, X^\alpha W^\nu$, which depends only on the value of W at p.

7.5.1. *Symmetry of $C^c{}_{ab}$.* Let $f \in \mathscr{F}(\mathcal{M})$ and consider the covector $\omega_b = \nabla_b f = \tilde{\nabla}_b f$ (by axiom (4) both operators agree on scalars). Then

$$\nabla_a \nabla_b f = \tilde{\nabla}_a \tilde{\nabla}_b f - C^c{}_{ab}\, \nabla_c f\,. \tag{102}$$

The left-hand side is symmetric in a, b (torsion-free, axiom (5)), and the first term on the right-hand side is likewise symmetric. Therefore

$$\boxed{C^c{}_{ab} = C^c{}_{ba}}\,. \tag{103}$$

7.5.2. *Action on vectors.* We now determine how $(\tilde{\nabla}_a - \nabla_a)$ acts on vectors $t^b \in \mathcal{T}(1,0)$. Let $\omega \in \mathcal{T}(0,1)$ and $t \in \mathcal{T}(1,0)$. The contraction $\omega_b\, t^b$ is a scalar, so by axiom (4):

$$(\tilde{\nabla}_a - \nabla_a)(\omega_b\, t^b) = 0\,. \tag{104}$$

Expanding by the Leibniz rule:

$$t^b\, (\tilde{\nabla}_a - \nabla_a)\omega_b + \omega_b\, (\tilde{\nabla}_a - \nabla_a) t^b = 0\,.$$

Rearranging (101) gives $(\tilde{\nabla}_a - \nabla_a)\omega_b = +C^c{}_{ab}\, \omega_c$. Substituting:

$$t^b\, C^c{}_{ab}\, \omega_c + \omega_b\, (\tilde{\nabla}_a - \nabla_a) t^b = 0\,.$$

Relabelling the dummy index $c \leftrightarrow b$ in the first term and requiring this to hold for all ω_b, we obtain $(\tilde{\nabla}_a - \nabla_a) t^b = -C^b{}_{ac}\, t^c$, i.e.

$$\boxed{\nabla_a t^b = \tilde{\nabla}_a t^b + C^b{}_{ac}\, t^c}\,. \tag{105}$$

Difference of connections on vectors

Two torsion-free derivative operators ∇_a and $\tilde{\nabla}_a$ are related on vector fields by

$$\nabla_a t^b = \tilde{\nabla}_a t^b + C^b{}_{ac}\, t^c\,,$$

where $C^c{}_{ab} = C^c{}_{ba}$ is symmetric in its lower indices.

7.6. Extension to general tensors. The Leibniz rule and axiom (4) determine the action of $(\tilde{\nabla}_a - \nabla_a)$ on *all* tensor fields. For a general $T \in \mathcal{T}(k,l)$:

(106)
$$\nabla_a T^{b_1\cdots b_k}{}_{c_1\cdots c_l} = \tilde{\nabla}_a T^{b_1\cdots b_k}{}_{c_1\cdots c_l} + \sum_{j=1}^{k} C^{b_j}{}_{ad}\, T^{b_1\cdots d\cdots b_k}{}_{c_1\cdots c_l} - \sum_{j=1}^{l} C^{d}{}_{ac_j}\, T^{b_1\cdots b_k}{}_{c_1\cdots d\cdots c_l}$$

In the first sum, d replaces b_j; in the second sum, d replaces c_j.

Remark. The rule is: one $+C$ term for each upper index, one $-C$ term for each lower index. The action of $\tilde{\nabla}_a - \nabla_a$ on an arbitrary tensor of type (k,l) is thus completely determined by the single object $C^c{}_{ab} \in V_p \otimes V_p^* \otimes V_p^*$.

Theorem 7.4 (Converse)**.** *Conversely, given* any *derivative operator* $\tilde{\nabla}_a$ *satisfying axioms (0)–(5) and* any *smooth tensor field* $C^c{}_{ab}$ *symmetric in* a, b*, the operator* ∇_a *defined by* (106) *is also a derivative operator satisfying axioms (0)–(5).*

The space of connections

The set of all torsion-free connections on $\mathcal{M}$ is an *affine space*: fixing any one connection $\tilde{\nabla}_a$, every other connection is obtained by adding a symmetric tensor $C^c{}_{ab}$. There is no preferred "origin"—no canonical choice of connection—without additional structure such as a metric.

7.7. Christoffel symbols. The most important special case arises when we choose $\tilde{\nabla}_a = \partial_a$, the ordinary derivative in some coordinate system. The corresponding $C^c{}_{ab}$ is then written $\Gamma^c{}_{ab}$ and called the **Christoffel symbols** (of the second kind).

Definition 7.2 (Christoffel symbols)**.** Let ∂_a denote the ordinary (coordinate) derivative in a chart ψ. The **Christoffel symbols** $\Gamma^c{}_{ab}$ are defined by the relation

(107)
$$\nabla_a t^b = \partial_a t^b + \Gamma^b{}_{ac}\, t^c\,.$$

In coordinate components (replacing abstract indices $a, b, c, \ldots$ by coordinate indices $\mu, \nu, \lambda, \ldots$), this reads

(108)
$$\nabla_\mu t^\nu = \frac{\partial t^\nu}{\partial x^\mu} + \Gamma^\nu{}_{\mu\lambda}\, t^\lambda\,.$$

Remark. Although $\Gamma^c{}_{ab}$ has three indices, it does **not** transform as a tensor under coordinate changes. Under a change of coordinates $x^\mu \to x'^{\mu'}$, the Christoffel symbols pick up an inhomogeneous term involving second derivatives of the coordinate transformation. It is only the *difference* of two connections that transforms tensorially.

For a covector field ω_b, the corresponding formula is

(109)
$$\nabla_\mu \omega_\nu = \frac{\partial \omega_\nu}{\partial x^\mu} - \Gamma^\lambda{}_{\mu\nu}\, \omega_\lambda\,.$$

For a general tensor $T \in \mathcal{T}(k,l)$:

$$\nabla_\sigma T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\nu_l} = \frac{\partial T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\nu_l}}{\partial x^\sigma} + \sum_{j=1}^{k} \Gamma^{\mu_j}{}_{\sigma\lambda}\, T^{\mu_1\cdots\lambda\cdots\mu_k}{}_{\nu_1\cdots\nu_l} - \sum_{j=1}^{l} \Gamma^{\lambda}{}_{\sigma\nu_j}\, T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\lambda\cdots\nu_l}\,. \tag{110}$$

Exercise 7.1. Using (110), write out the explicit expression for $\nabla_\sigma g_{\mu\nu}$ in terms of partial derivatives and Christoffel symbols.

Exercise 7.2. Let x^μ and $x'^{\mu'}$ be two coordinate systems. Derive the transformation law for the Christoffel symbols $\Gamma^\nu{}_{\mu\lambda}$ under the coordinate change and verify that it contains an inhomogeneous (non-tensorial) term.

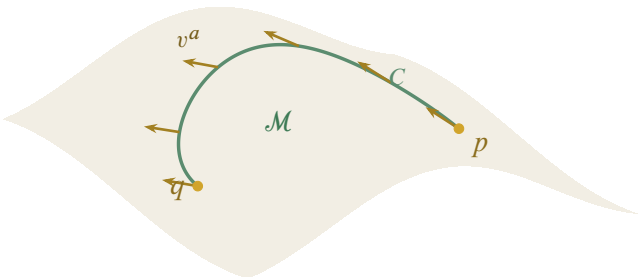
We have seen that the set of torsion-free derivative operators on a manifold forms an affine space, parameterised by a symmetric tensor field $C^c{}_{ab}$. Without additional structure, there is no preferred choice of connection. In the next lecture we will see that the data of a *metric* g_{ab} on $\mathcal{M}$ is enough to uniquely single out one particular derivative operator—the **Levi-Civita connection**—by imposing the additional requirement $\nabla_a g_{bc} = 0$ (metric compatibility). That connection will play a central role in the formulation of general relativity.

8. Parallel transport continued

In the previous lecture we introduced derivative operators (covariant derivatives) on a manifold $\mathcal{M}$. We now use them to define *parallel transport*—the notion of moving a vector (or tensor) along a curve "without rotating it."

The central goal is to *engineer* a device: given a derivative operator ∇_a, a curve C from p to q, and a vector $v^a \in V_p$, we want to produce a corresponding vector in V_q. That is, we want to build a linear map $U_C : V_p \to V_q$ — a **parallel transporter** — from the data of the connection alone. When a metric is present, requiring that this transporter preserves inner products singles out a unique derivative operator: the *Levi-Civita connection*. We derive its explicit form in coordinates (the Christoffel symbols) and conclude with the geodesic equation.

8.1. Parallel transport. Let $\mathcal{M}$ be a manifold equipped with a derivative operator ∇_a. Let C be a smooth curve in $\mathcal{M}$ with tangent vector t^a.



The curve C connects a point p to a point q. We wish to define a linear map

$$U_C : V_p \to V_q$$

that "transports" vectors from p to q along C.

Definition 8.1 (Parallel transport of a vector)**.** Let C be a smooth curve in $\mathcal{M}$ with tangent vector t^a. A vector field v^a defined along C is said to be **parallelly transported** along C if

$$\boxed{t^a \nabla_a v^b = 0} \tag{111}$$

at every point of C.

More generally, if $T^{a_1\cdots a_k}{}_{c_1\cdots c_l}$ is a tensor field of type (k,l) defined along C, we say it is parallelly transported if

$$t^a \nabla_a T^{a_1\cdots a_k}{}_{c_1\cdots c_l} = 0\,. \tag{112}$$

8.1.1. *Parallel transport in coordinates.* Let ψ be a coordinate chart determining the ordinary derivative operator ∂_a. Recall that the covariant derivative of a vector is related to the ordinary derivative by

$$\nabla_a v^b = \partial_a v^b + \Gamma^b{}_{ac}\, v^c\,.$$

In the chart ψ, the tangent vector to the curve has components $t^\mu = \mathrm{d}x^\mu/\mathrm{d}t$. The parallel transport condition $t^a \nabla_a v^b = 0$ becomes

$$\frac{\mathrm{d}x^\mu}{\mathrm{d}t}\frac{\partial v^\nu}{\partial x^\mu} + \frac{\mathrm{d}x^\mu}{\mathrm{d}t}\,\Gamma^\nu{}_{\mu\lambda}\, v^\lambda = 0\,.$$

The first factor in each term combines to give $\mathrm{d}v^\nu/\mathrm{d}t$ along the curve. Hence:

$$\boxed{\frac{\mathrm{d}v^\nu}{\mathrm{d}t} + \frac{\mathrm{d}x^\mu}{\mathrm{d}t}\,\Gamma^\nu{}_{\mu\lambda}\, v^\lambda = 0} \tag{113}$$

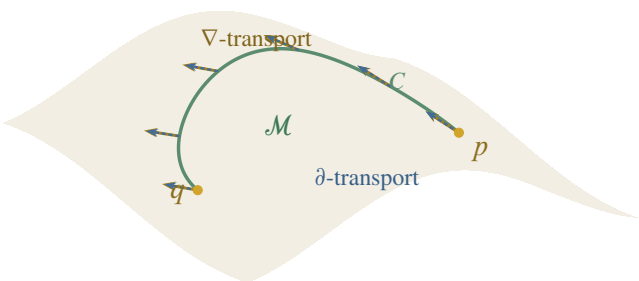
Remark. The parallel transport condition (113) only involves the values of v^a on the curve C. We do *not* need v^a to be defined as a vector field on all of $\mathcal{M}$—it suffices for v^a to be given along C.

Remark. Equation (113) is a system of linear first-order ordinary differential equations for the components $v^\nu(t)$. By the standard existence and uniqueness theorem for ODEs, given an initial vector $v^a|_p$, there exists a unique solution along C. This defines the **parallel transporter**

$$U_C\colon V_p \to V_q\,, \qquad v^a(p) \mapsto \tilde{v}^a(q)\,,$$

which is a linear isomorphism (linearity follows from the linearity of the ODE).

The transporter U_C depends on which connection ∇_a we choose. The figure below contrasts two transports of the same initial vector along the same curve: the chart-α partial derivative ∂_a (which trivially keeps chart components constant) and the Levi-Civita connection of the surface metric. The two prescriptions agree at p and diverge along C — "parallel" is not absolute, it is a choice of ∇_a.



8.2. The Levi-Civita connection. So far, there is no preferred derivative operator on a bare manifold—we have an infinite-dimensional family to choose from. If the manifold $\mathcal{M}$ is additionally equipped with a metric g_{ab}, however, we can make a *canonical* choice of derivative operator ∇_a. The guiding physical requirement is that parallel transport should preserve lengths and angles: what kind of "parallel" transport would it be if vectors changed their length or the angle between them varied as we slid them along a curve?

8.2.1. *Metric compatibility.* Consider two vectors $v^a, w^a \in V_p$ that are parallelly transported along a curve C with tangent t^a:

$$t^a \nabla_a v^b = 0\,, \qquad t^a \nabla_a w^b = 0\,.$$

We require that the inner product $(v, w)_p = g_{ab}\, v^a w^b$ is preserved during transport. That is,

$$t^a \nabla_a \big(g_{bc}\, v^b w^c\big) = 0\,.$$

Applying the Leibniz rule:

$$0 = (t^a \nabla_a g_{bc})\, v^b w^c + g_{bc}\, (t^a \nabla_a v^b)\, w^c + g_{bc}\, v^b\, (t^a \nabla_a w^c)\,. \tag{114}$$

The last two terms vanish by the parallel transport condition. Since this must hold for *all* vectors v^b, w^c, all curves, and all tangent vectors t^a, we conclude:

$$\boxed{\nabla_a g_{bc} = 0} \tag{115}$$

Definition 8.2 (Metric compatibility)**.** A derivative operator ∇_a is said to be **metric-compatible** (or **compatible with the metric** g_{ab}) if $\nabla_a g_{bc} = 0$.

Why metric compatibility?

The metric encodes the notions of length and angle. If $\nabla_a g_{bc} = 0$, then parallel transport preserves the inner product: two vectors that are orthogonal at p remain orthogonal after transport to q, and their lengths do not change. This is the natural analogue of the familiar fact that, in Euclidean space, translating a vector does not alter its length or direction.

8.2.2. *Existence and uniqueness of the Levi-Civita connection.*

Theorem 8.1 (Levi-Civita connection)**.** *Let g_{ab} be a metric on $\mathcal{M}$. Then there exists a* ***unique*** *torsion-free derivative operator ∇_a satisfying $\nabla_a g_{bc} = 0$.*

Proof. Let $\tilde{\nabla}_a$ be *any* derivative operator (for instance, the ordinary derivative ∂_a associated with some coordinate chart). Any other derivative operator ∇_a differs from $\tilde{\nabla}_a$ by a tensor $C^d{}_{ab}$:

$$\nabla_a \omega_b = \tilde{\nabla}_a \omega_b - C^d{}_{ab}\, \omega_d\,.$$

Since ∇_a is torsion-free, $C^d{}_{ab} = C^d{}_{ba}$.

Step 1: Impose metric compatibility. The condition $\nabla_a g_{bc} = 0$ reads

$$0 = \nabla_a g_{bc} = \tilde{\nabla}_a g_{bc} - C^d{}_{ab}\, g_{dc} - C^d{}_{ac}\, g_{bd}\,. \tag{116}$$

Writing $C_{cab} = g_{dc}\, C^d{}_{ab}$, this becomes

$$\tilde{\nabla}_a g_{bc} = C_{cab} + C_{bac}\,. \tag{117}$$

Step 2: Cyclic permutations. Write (117) with three different index orderings:

$$\tilde{\nabla}_a g_{bc} = C_{cab} + C_{bac}\,, \tag{118}$$

$$\tilde{\nabla}_b g_{ac} = C_{cba} + C_{abc}\,, \tag{119}$$

$$\tilde{\nabla}_c g_{ab} = C_{bca} + C_{acb}\,. \tag{120}$$

Form the combination (118) + (119) − (120):

$$\tilde{\nabla}_a g_{bc} + \tilde{\nabla}_b g_{ac} - \tilde{\nabla}_c g_{ab} = C_{cab} + C_{bac} + C_{cba} + C_{abc} - C_{bca} - C_{acb}\,.$$

Using the torsion-free symmetry $C_{cab} = C_{cba}$, the right-hand side simplifies to $2C_{cab}$. Therefore

$$C_{cab} = \tfrac{1}{2}\big(\tilde{\nabla}_a g_{bc} + \tilde{\nabla}_b g_{ac} - \tilde{\nabla}_c g_{ab}\big)\,. \tag{121}$$

Raising the first index:

$$C^c{}_{ab} = \tfrac{1}{2}\, g^{cd}\big(\tilde{\nabla}_a g_{bd} + \tilde{\nabla}_b g_{ad} - \tilde{\nabla}_d g_{ab}\big)\,. \tag{122}$$

Since $C^c{}_{ab}$ is uniquely determined by $\tilde{\nabla}_a$ and g_{ab}, the derivative operator ∇_a is unique. □

Remark. The result is remarkable: the single condition $\nabla_a g_{bc} = 0$ is neither too strong (it does not eliminate all derivative operators) nor too weak (it does not leave any freedom). It is "just right"—the metric uniquely determines the connection. From now on, whenever a metric g_{ab} is given, we choose ∇_a to be the unique Levi-Civita connection determined by g_{ab}.

8.3. Christoffel symbols. In the special case where $\tilde{\nabla}_a = \partial_a$ is the ordinary derivative associated with a coordinate chart ψ, the tensor $C^c{}_{ab}$ becomes the Christoffel symbol $\Gamma^c{}_{ab}$. Since $\partial_a g_{bd}$ simply differentiates the metric components, we obtain (in abstract index notation):

$$\Gamma^c{}_{ab} = \tfrac{1}{2}\, g^{cd}\big(\partial_a g_{bd} + \partial_b g_{ad} - \partial_d g_{ab}\big)\,. \tag{123}$$

In the coordinate chart ψ with coordinates $(x^1, \dots, x^n)$, this reads:

$$\boxed{\Gamma^\rho{}_{\mu\nu} = \frac{1}{2}\sum_\sigma g^{\rho\sigma}\left(\frac{\partial g_{\sigma\mu}}{\partial x^\nu} + \frac{\partial g_{\sigma\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\sigma}\right)} \tag{124}$$

Christoffel symbols from the metric

The Christoffel symbols (124) are determined entirely by the metric and its first derivatives. They are symmetric in the lower two indices: $\Gamma^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\nu\mu}$. Note that they are *not* the components of a tensor (see §8.6 below).

8.4. Example: $\mathbb{R}^2$ in Cartesian and polar coordinates.

8.4.1. *Cartesian coordinates.*

Example 8.1 (Flat metric in Cartesian coordinates)**.** Let $\mathcal{M} = \mathbb{R}^2$ with Cartesian coordinates (x, y). A vector is expanded as

$$v = \sum_{\mu=x,y} v^\mu \left.\frac{\partial}{\partial x^\mu}\right|_p\,.$$

The Euclidean metric is

$$g_{\mu\nu}\,\mathrm{d}x^\mu \otimes \mathrm{d}x^\nu = \mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y\,,$$

so $g_{\mu\nu} = \delta_{\mu\nu}$ and all derivatives of $g_{\mu\nu}$ vanish. Hence

$$\Gamma^\lambda{}_{\mu\nu} = 0\,.$$

Choose $\tilde{\nabla}_a = \partial_a$ with respect to Cartesian coordinates. Then $\nabla_a = \partial_a$, and the parallel transport equation $t^a \nabla_a v^b = 0$ becomes

$$t^\nu \frac{\partial v^\mu}{\partial x^\nu} = 0 \quad\Longrightarrow\quad v^\mu \text{ is constant along } C\,.$$

This is the familiar result: in flat space with Cartesian coordinates, parallel transport simply keeps the components of the vector constant.

8.4.2. *Polar coordinates.*

Example 8.2 (Flat metric in polar coordinates)**.** Now introduce polar coordinates (r, φ) on $\mathbb{R}^2$ via $(x, y) = (r\cos\varphi,\ r\sin\varphi)$. Using $\mathrm{d}x = \cos\varphi\,\mathrm{d}r - r\sin\varphi\,\mathrm{d}\varphi$ and $\mathrm{d}y = \sin\varphi\,\mathrm{d}r + r\cos\varphi\,\mathrm{d}\varphi$, the metric becomes

$$g = \mathrm{d}r \otimes \mathrm{d}r + r^2\,\mathrm{d}\varphi \otimes \mathrm{d}\varphi\,. \tag{125}$$

In matrix form:

$$g_{\mu'\nu'} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \qquad g^{\mu'\nu'} = \begin{pmatrix} 1 & 0 \\ 0 & 1/r^2 \end{pmatrix}.$$

The factor r^2 in front of $\mathrm{d}\varphi \otimes \mathrm{d}\varphi$ has a simple geometric meaning: a small angular displacement $\mathrm{d}\varphi$ produces a larger arc-length displacement the further one is from the origin, so the metric must weight the angular direction by r.

Christoffel symbols. From (124), the only nonvanishing derivatives of $g_{\mu'\nu'}$ involve $\partial g_{\varphi\varphi}/\partial r = 2r$. A direct computation gives:

$$\Gamma^r{}_{\varphi\varphi} = -r\,, \qquad \Gamma^\varphi{}_{r\varphi} = \Gamma^\varphi{}_{\varphi r} = \frac{1}{r}\,, \tag{126}$$

with all other components vanishing.

Exercise 8.1. Verify the Christoffel symbols (126) by direct computation from (124).

8.4.3. *Parallel transport along a radial line.*

Take the curve C to be a displacement along the radial direction: the tangent vector is $t^a = (\partial/\partial r)^a$, so $t^r = 1$ and $t^\varphi = 0$.

The parallel transport equation (113) yields two component equations:

$$\frac{\mathrm{d}v^r}{\mathrm{d}r} + \Gamma^r{}_{rr}\,v^r + \Gamma^r{}_{r\varphi}\,v^\varphi = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}v^r}{\mathrm{d}r} = 0\,, \tag{127}$$

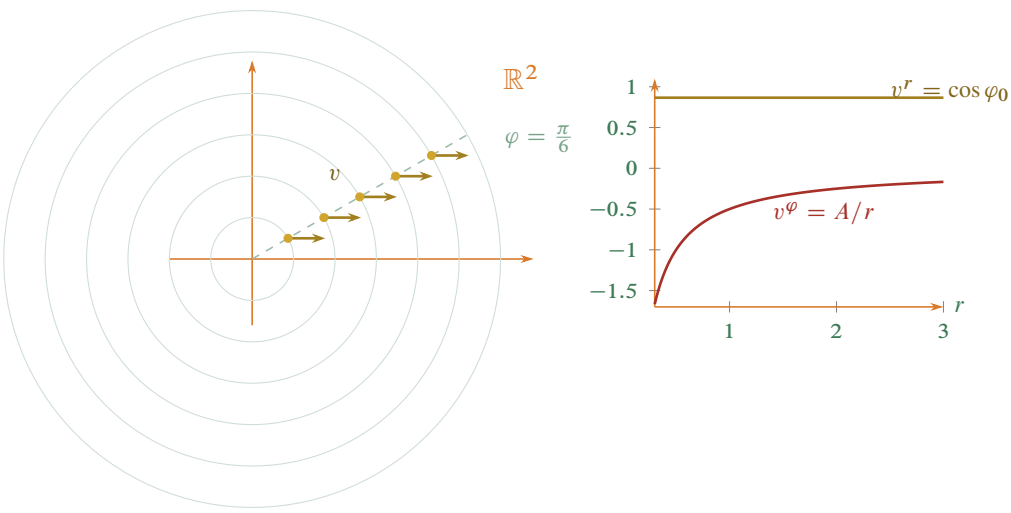
$$\frac{\mathrm{d}v^\varphi}{\mathrm{d}r} + \Gamma^\varphi{}_{rr}\,v^r + \Gamma^\varphi{}_{r\varphi}\,v^\varphi = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}v^\varphi}{\mathrm{d}r} = -\frac{v^\varphi}{r}\,. \tag{128}$$

From (127), v^r is constant. Equation (128) is separable:

$$\frac{1}{v^\varphi}\,\mathrm{d}v^\varphi = -\frac{1}{r}\,\mathrm{d}r \quad\Longrightarrow\quad \log v^\varphi = -\log r + \text{const} \quad\Longrightarrow\quad v^\varphi = \frac{A}{r}\,,$$

where A is a constant determined by the initial conditions.

Remark. The components v^φ change along the curve, but this is entirely because the coordinate basis vectors $\partial/\partial r$ and $\partial/\partial\varphi$ themselves change from point to point. The vector v^a is "constant" in the Cartesian sense—the parallel transport equation correctly accounts for the variation of the basis.



The figure shows the radial transport explicitly for $\varphi_0 = \pi/6$. Each arrow on the left is the same Cartesian vector $(1, 0)$, drawn at successive radii along the ray. On the right, $v^r = \cos\varphi_0$ is constant in r, while v^φ has the hyperbolic profile A/r predicted by (128).

8.5. Covariant derivative of one-forms and connection one-forms. Suppose ∇_a is a derivative operator. Let ω be a covector (a tensor of type $(0, 1)$) and let ψ be a coordinate chart. In the coordinate basis, the covariant derivative of the dual basis element $\mathrm{d}x^\nu$ in the direction $\partial/\partial x^\mu$ is

$$\nabla_{\partial/\partial x^\mu}(\mathrm{d}x^\nu) = -\Gamma^\nu{}_{\mu\lambda}\,\mathrm{d}x^\lambda\,. \tag{129}$$

For each choice of μ and ν, this defines a one-form called a **connection one-form**.

8.6. Transformation of connection coefficients. Let (U, ψ) and (V, ψ') be two overlapping charts with coordinates $x^\mu = \psi(p)$ and $y^\alpha = \psi'(p)$. Write $\Gamma^\gamma{}_{\alpha\beta}$ for the Christoffel symbols in the x-chart and $\tilde{\Gamma}^\gamma{}_{\alpha\beta}$ for those in the y-chart.

Starting from the definition $\Gamma^\gamma{}_{\alpha\beta}\,\partial/\partial x^\gamma = \nabla_{\partial/\partial x^\alpha}(\partial/\partial x^\beta)$, one can express the basis vectors $\partial/\partial y^\alpha = (\partial x^\mu/\partial y^\alpha)\,\partial/\partial x^\mu$ and compute:

$$\tilde{\Gamma}^\gamma{}_{\alpha\beta} = \frac{\partial x^\lambda}{\partial y^\alpha}\frac{\partial x^n}{\partial y^\beta}\frac{\partial y^\gamma}{\partial x^\nu}\Gamma^\nu{}_{\lambda n} + \frac{\partial^2 x^\nu}{\partial y^\alpha\,\partial y^\beta}\frac{\partial y^\gamma}{\partial x^\nu} \tag{130}$$

Remark. The first term in (130) is the standard tensorial transformation law. The second term, involving second derivatives of the coordinate transformation, shows that $\Gamma^\nu{}_{\mu\lambda}$ does *not* transform as a tensor. It is precisely this inhomogeneous term that allows $\nabla_a v^b$ to be a tensor even though v^b and $\Gamma^b{}_{ac}$ individually are not.

Exercise 8.2. Show that $\nabla_X Y$ is independent of the choice of coordinates: if $X = X^\mu\,\partial/\partial x^\mu$, $Y = Y^\nu\,\partial/\partial x^\nu$ in one chart and $X = \tilde{X}^\alpha\,\partial/\partial y^\alpha$, $Y = \tilde{Y}^\beta\,\partial/\partial y^\beta$ in another, then $X^\mu\nabla_{\partial/\partial x^\mu}(Y^\nu\,\partial/\partial x^\nu) = \tilde{X}^\alpha\nabla_{\partial/\partial y^\alpha}(\tilde{Y}^\beta\,\partial/\partial y^\beta)$.

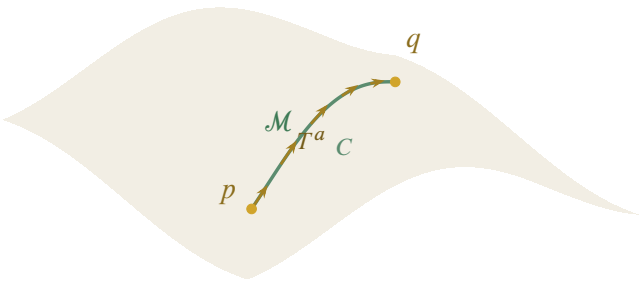
8.7. Geodesics. We now arrive at one of the central concepts in general relativity.

Definition 8.3 (Geodesic)**.** Let $(\mathcal{M}, g_{ab})$ be a manifold with metric. A **geodesic** is a curve whose tangent vector T^a is parallelly transported along itself:

$$T^a\nabla_a T^b = 0 \tag{131}$$

Geodesics as straightest curves

A geodesic is the "straightest possible" curve one can trace through $\mathcal{M}$. On a curved manifold there is in general no global notion of a "straight line," but the geodesic equation captures the idea locally: the tangent vector does not turn relative to the connection. In flat space, geodesics reduce to straight lines.



8.7.1. *Geodesic equation in coordinates.* Choose a coordinate chart ψ. A curve is described by functions $x^\mu(t)$, and the tangent vector has components $T^\mu = \mathrm{d}x^\mu/\mathrm{d}t$. Applying the parallel transport equation (113) with $v^\mu = T^\mu$:

$$\frac{\mathrm{d}T^\mu}{\mathrm{d}t} + \Gamma^\mu{}_{\sigma\nu}\frac{\mathrm{d}x^\sigma}{\mathrm{d}t}T^\nu = 0\,.$$

Since $T^\mu = \mathrm{d}x^\mu/\mathrm{d}t$, we write $\mathrm{d}T^\mu/\mathrm{d}t = \mathrm{d}^2x^\mu/\mathrm{d}t^2$ to obtain:

$$\boxed{\frac{\mathrm{d}^2x^\mu}{\mathrm{d}t^2} + \sum_{\sigma,\nu}\Gamma^\mu{}_{\sigma\nu}\frac{\mathrm{d}x^\sigma}{\mathrm{d}t}\frac{\mathrm{d}x^\nu}{\mathrm{d}t} = 0} \tag{132}$$

Remark. Equation (132) is a system of n coupled second-order ordinary differential equations for the n functions $x^\mu(t)$. Given initial position $x^\mu(0)$ and initial velocity $\mathrm{d}x^\mu/\mathrm{d}t|_{t=0}$, the ODE existence and uniqueness theorem guarantees a unique geodesic (at least locally).

Exercise 8.3. Show that the geodesics of Euclidean $\mathbb{R}^n$ (with the standard flat metric in Cartesian coordinates) are straight lines.

Exercise 8.4. Compute the Christoffel symbols for the round metric $g = \mathrm{d}\theta\otimes\mathrm{d}\theta + \sin^2\theta\,\mathrm{d}\varphi\otimes\mathrm{d}\varphi$ on S^2, and write down the geodesic equations. Verify that great circles satisfy these equations.

9. Abstract index notation review; introduction to curvature

In this lecture we consolidate the abstract index notation (AIN) introduced earlier and develop its computational machinery (contraction, outer products, raising and lowering indices, symmetrisation). AIN is a very helpful and compact tool for discussing tensors: it will save us from repeatedly writing out expressions in coordinate-basis components, while still permitting explicit calculations when needed. We then turn to the central geometric question of general relativity: *what is curvature?* Our ultimate goal is to formulate a precise, intrinsic notion of curvature that can be equated to the density of stress-energy—the heart of Einstein's field equations.

9.1. Review of abstract index notation. Recall from Lecture 6 that a tensor $T \in \mathcal{T}(k,l)$ can be viewed in two equivalent ways:

(i) as an element of the tensor product vector space $\underbrace{V_p\otimes\cdots\otimes V_p}_{k}\otimes\underbrace{V_p^*\otimes\cdots\otimes V_p^*}_{l}$, or

(ii) as a multilinear map $\underbrace{V_p^*\times\cdots\times V_p^*}_{k}\times\underbrace{V_p\times\cdots\times V_p}_{l}\longrightarrow\mathbb{R}$.

Notation. In the abstract index notation, we write $T^{a_1\cdots a_k}{}_{b_1\cdots b_l}$ to denote the tensor T of type (k,l). The lower-case Latin indices $a,b,c,\ldots$ label **arguments and their type** (contravariant vs. covariant), *not* components with respect to a particular basis.

Remark. To obtain actual *components*, one must choose a coordinate chart and expand in the coordinate basis. Components are denoted with Greek indices: $T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\nu_l}$. Thus the abstract expression $T^{ab}{}_c$ represents the tensor itself, whereas $T^{\mu\nu}{}_\rho$ denotes a specific array of numbers in a given basis.

9.1.1. *Tensor operations in abstract index notation.* The two basic operations—contraction and outer product—have clean expressions in AIN.

Contraction. A repeated abstract index (one upper, one lower) denotes contraction. If $T \in \mathcal{T}(k,l)$ with $k \geq 1, l \geq 1$, then

$$C_{j,j'}\, T \;\longleftrightarrow\; T^{a_1 \cdots \underset{j}{c} \cdots a_{k-1}}{}_{b_1 \cdots \underset{j'}{c} \cdots b_{l-1}} \;\in\; \mathcal{T}(k-1, l-1)\,. \tag{133}$$

The repeated index c is summed over (in the abstract sense); the result has type $(k-1,\, l-1)$.

Outer product. Let $T \in \mathcal{T}(k,l)$ and $S \in \mathcal{T}(k',l')$. The outer product is simply the juxtaposition of their abstract indices:

$$T^{a_1 \cdots a_k}{}_{b_1 \cdots b_l}\; S^{a'_1 \cdots a'_{k'}}{}_{b'_1 \cdots b'_{l'}} \;\in\; \mathcal{T}(k+k', l+l')\,. \tag{134}$$

Example 9.1. Let v^a be a vector and ω_a a covector. The contraction $v^a\, \omega_a \in \mathcal{T}(0,0) \cong \mathbb{R}$ is the scalar $\omega(v)$. Meanwhile, the outer product $v^a\, \omega_b \in \mathcal{T}(1,1)$ is a type-$(1,1)$ tensor (a linear map $V_p \to V_p$).

9.2. The metric in abstract index notation.

The metric tensor $g \in \mathcal{T}(0,2)$ is written in AIN as

$$g_{ab}\,. \tag{135}$$

Since g is nondegenerate, its inverse exists. We denote the inverse metric by

$$(g^{-1})^{ab} \;\equiv\; g^{ab} \;\in\; \mathcal{T}(2,0)\,. \tag{136}$$

Lemma 9.1. *The metric and its inverse satisfy*

$$\boxed{g^{ab}\, g_{bc} = \delta^a{}_c}\,, \tag{137}$$

where $\delta^a{}_c$ is the identity tensor (Kronecker delta) in $\mathcal{T}(1,1)$.

Proof. Consider the outer product $g^{ab}\, g_{cd} \in \mathcal{T}(2,2)$. Contracting on the pair (b,c) yields a tensor of type $(1,1)$. In any coordinate basis, this contraction gives

$$\sum_{\nu} g^{\mu\nu}\, g_{\nu\rho} = \delta^\mu{}_\rho\,,$$

since $g^{\mu\nu}$ is the matrix inverse of $g_{\mu\nu}$. In abstract index notation, this reads $g^{ab}\, g_{bc} = \delta^a{}_c$. □

9.2.1. *Raising and lowering indices.* The metric provides a canonical isomorphism $V_p \to V_p^*$. In AIN, this operation is simply contraction with g_{ab} or g^{ab}.

Definition 9.1 (Lowering an index)**.** Given a vector v^a, define the covector

$$\boxed{v_a \;\equiv\; g_{ac}\, v^c}\,. \tag{138}$$

Definition 9.2 (Raising an index)**.** Given a covector ω_a, define the vector

$$\boxed{\omega^a \;\equiv\; g^{ac}\, \omega_c}\,. \tag{139}$$

Remark. These operations are consistent: lowering and then raising returns the original tensor, since $g^{ab}\, g_{bc}\, v^c = \delta^a{}_c\, v^c = v^a$.

More generally, one can raise or lower any individual index of a higher-rank tensor. For instance, given $T^{abc}{}_{def} \in \mathcal{T}(3,3)$, lowering the first upper index gives

$$g_{aa'}\, T^{a'bc}{}_{def} \;\equiv\; T^{bc}{}_{a\,def} \;\in\; \mathcal{T}(2,4)\,. \tag{140}$$

This is consistent: the result is obtained by repeatedly applying g_{ab} and g^{ab} in any order.

9.3. Symmetric and antisymmetric tensors. Let $T,\ T' \in \mathcal{T}(k,l)$. Their sum is again a tensor of type (k,l), written in AIN as

$$T^{a_1\cdots a_k}{}_{b_1\cdots b_l} + T'^{a_1\cdots a_k}{}_{b_1\cdots b_l} \ \in\ \mathcal{T}(k,l)\,.$$

Definition 9.3 (Symmetrisation)**.** For a type-$(0,2)$ tensor T_{ab}, the **symmetric part** is

$$T_{(ab)} \ \equiv\ \tfrac{1}{2}\big(T_{ab} + T_{ba}\big)\,. \tag{141}$$

Definition 9.4 (Antisymmetrisation)**.** For a type-$(0,2)$ tensor T_{ab}, the **antisymmetric part** is

$$T_{[ab]} \ \equiv\ \tfrac{1}{2}\big(T_{ab} - T_{ba}\big)\,. \tag{142}$$

Any type-$(0,2)$ tensor decomposes uniquely as $T_{ab} = T_{(ab)} + T_{[ab]}$.

9.3.1. *General symmetrisation and antisymmetrisation.* For a tensor with l covariant indices, the general formulas involve the symmetric group S_l.

Definition 9.5 (General symmetrisation)**.**

$$T_{(a_1\cdots a_l)} \ \equiv\ \frac{1}{l!}\sum_{\pi\in S_l} T_{a_{\pi(1)}\cdots a_{\pi(l)}}\,. \tag{143}$$

Definition 9.6 (General antisymmetrisation)**.**

$$T_{[a_1\cdots a_l]} \ \equiv\ \frac{1}{l!}\sum_{\pi\in S_l} \varepsilon(\pi)\, T_{a_{\pi(1)}\cdots a_{\pi(l)}}\,, \tag{144}$$

where $\varepsilon(\pi) = +1$ for even permutations and $\varepsilon(\pi) = -1$ for odd permutations.

Example 9.2 (Mixed symmetrisation)**.** Round and square brackets can be applied independently to different groups of indices. For instance,

$$T^{(ab)c}{}_{[de]} = \tfrac{1}{4}\big(T^{abc}{}_{de} + T^{bac}{}_{de} - T^{abc}{}_{ed} - T^{bac}{}_{ed}\big)\,. \tag{145}$$

Definition 9.7 (Differential forms)**.** A tensor $T_{a_1\cdots a_l}$ of type $(0,l)$ that is **totally antisymmetric**,

$$T_{a_1\cdots a_l} = T_{[a_1\cdots a_l]}\,,$$

is called a **differential l-form**.

Remark. On an n-dimensional manifold, the space of l-forms is $\binom{n}{l}$-dimensional. In particular, l-forms vanish identically for $l > n$.

9.4. Introduction to curvature. We now turn to one of the most important concepts in general relativity: *curvature*. The key question is:

How can we detect that a manifold is curved without embedding it in a higher-dimensional space?

If a manifold were embedded in a higher-dimensional flat space, curvature would be easy to see—one can simply "look" at how the manifold bends inside the ambient space. But in general relativity, spacetime is **not** embedded in any ambient space: we are "stuck inside" the manifold. We therefore need a purely intrinsic criterion—one that makes no reference to any embedding—to detect and quantify curvature.

The key observation, which will serve as the foundation for the Riemann curvature tensor, is that *in flat manifolds parallel transport is path-independent, whereas in curved manifolds it is not*.

9.4.1. *Parallel transport and path dependence.*

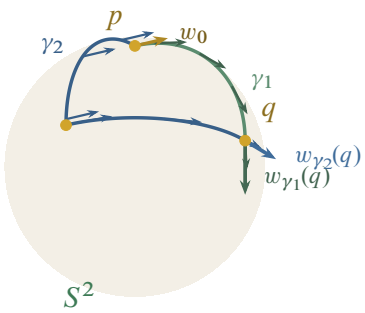
Curvature from parallel transport

On a *flat* manifold (such as $\mathbb{R}^n$ with the Euclidean metric), parallel transport of a vector from a point p to a point q is **independent of the path** chosen. On a *curved* manifold (such as the two-sphere S^2), parallel transport **depends on the path**: transporting a vector along two different paths from p to q generally yields two different vectors at q.

Curvature measures precisely this path dependence.

Consider two illustrative cases:

- **Flat space $\mathbb{R}^n$:** Given two paths γ_1, γ_2 from p to q, parallel-transporting any vector $v \in V_p$ along γ_1 or γ_2 produces the *same* vector at q. The result is path-independent.
- **The two-sphere S^2:** Transport a vector v at the north pole along two different great-circle paths to a point on the equator. The resulting vectors at that point will in general *differ*. The discrepancy is a direct manifestation of the curvature of S^2.



The figure above is the canonical illustration: the initial vector $w_0 = (1, 0, 0)$ at the north pole p is parallel-transported to $q = (1, 0, 0)$ along two paths. Along the meridian γ_1 (which is itself a geodesic and to which w_0 is initially tangent) the vector ends pointing south, $w_{\gamma_1}(q) = (0, 0, -1)$. Along γ_2 (a meridian to $(0, 1, 0)$ followed by an arc of the equator) the vector ends pointing along the equator, $w_{\gamma_2}(q) = (0, -1, 0)$. The two end vectors are perpendicular — the angle between them equals the area $\pi/2$ of the spherical triangle on the unit sphere, exactly as predicted by the Gauss–Bonnet theorem.

9.4.2. *The parallel transporter.* To make the notion of parallel transport precise, we introduce additional geometric data beyond the manifold and metric.

Definition 9.8 (Parallel transporter (heuristic))**.** Given a curve γ from p to q on a manifold $\mathcal{M}$, a **parallel transporter** along γ is a linear map

$$U_\gamma \colon V_p \longrightarrow V_q \tag{146}$$

that "moves" tangent vectors from p to q along γ.

Remark. The tangent spaces V_p and V_q at different points are *distinct* vector spaces with no canonical identification between them. The parallel transporter U_γ provides exactly this missing identification—but it depends on the choice of path γ. When the path dependence is nontrivial, the manifold is curved.

9.4.3. *The problem of differentiation on manifolds.* In Euclidean space $\mathbb{R}^n$, we can differentiate a vector field $v^\mu(x)$ simply by taking partial derivatives $\partial v^\mu / \partial x^\nu$, because the additive structure of $\mathbb{R}^n$ allows us to *compare vectors at different points* by shifting them.

On a general manifold $\mathcal{M}$, no such global comparison is available. To define a derivative of a vector field, we need to compare the vector $v|_q$ at a nearby point $q = p + \Delta x$ with a vector at q obtained by parallel-transporting $v|_p$ along an infinitesimal path from p to q.

9.4.4. *Connection coefficients and the covariant derivative.* Consider a vector field $v = v^\mu \, \partial/\partial x^\mu$ and an infinitesimal displacement Δx^ν from a point p. The parallel-transported vector $\tilde{v}$ at the nearby point q has components

$$\tilde{v}^\mu = v^\mu - v^\lambda \, \Gamma^\mu{}_{\nu\lambda}(x) \, \Delta x^\nu \,, \tag{147}$$

where the $\Gamma^\mu{}_{\nu\lambda}$ are the **connection coefficients** (or **Christoffel symbols**).

Remark. The connection coefficients encode the infinitesimal parallel transport rule: they tell us how much the components of a vector "rotate" when transported an infinitesimal distance Δx^ν in the ν-direction.

The key observation is that, to every infinitesimal parallel transporter U_γ, we can associate a derivative-type operator on vector fields. Define:

$$\nabla_\nu \left(v^\mu \left. \frac{\partial}{\partial x^\mu} \right|_p \right) = \lim_{\Delta x^\nu \to 0} \frac{v^\mu(x + \Delta x) - \tilde{v}^\mu(x + \Delta x)}{\Delta x^\nu} \left. \frac{\partial}{\partial x^\mu} \right|_q \,. \tag{148}$$

Substituting (147) into this expression, we obtain

The covariant derivative of a vector field

The **covariant derivative** of a vector field v^μ in the direction $\partial/\partial x^\nu$ is

$$\nabla_\nu v^\mu = \frac{\partial v^\mu}{\partial x^\nu} + v^\lambda \, \Gamma^\mu{}_{\nu\lambda} \,. \tag{149}$$

In abstract index notation, this is written

$$\nabla_b \, v^a = \partial_b \, v^a + v^c \, \Gamma^a{}_{bc} \,. \tag{150}$$

Remark. The first term $\partial_b \, v^a$ is the ordinary partial derivative of the components—this is *not* tensorial by itself. The second term $v^c \, \Gamma^a{}_{bc}$ compensates for the fact that the basis vectors themselves change from point to point. Together, $\nabla_b \, v^a$ transforms as a tensor of type $(1, 1)$.

Remark. The connection coefficients $\Gamma^\mu{}_{\nu\lambda}$ do **not** transform as the components of a tensor. Under a coordinate change, they acquire an inhomogeneous term involving second derivatives of the coordinate transformation. It is the *combination* (149) that transforms tensorially.

From parallel transport to curvature

We have seen that a parallel transporter determines a covariant derivative ∇_a, and vice versa. In the next lectures, we will show that curvature is encoded in the **failure of covariant derivatives to commute**: the Riemann curvature tensor $R^a{}_{bcd}$ is defined via

$$(\nabla_c \, \nabla_d - \nabla_d \, \nabla_c) \, v^a = R^a{}_{bcd} \, v^b \,.$$

When $R^a{}_{bcd} = 0$, parallel transport is path-independent and the manifold is flat. Nonvanishing $R^a{}_{bcd}$ signals genuine curvature.

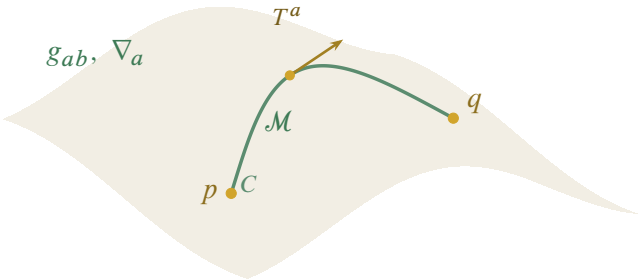
10. Geodesics continued; curvature

In the previous lectures we defined geodesics as curves whose tangent vector is parallelly transported along itself. This definition, while mathematically precise, may seem somewhat abstract: it characterises geodesics in terms of the derivative operator. In this lecture we provide a *second*, potentially more intuitive characterisation of geodesics—namely, that they are the curves which **extremise the arc length** (or proper time) functional between two fixed points.

This variational point of view connects the abstract machinery of parallel transport to the familiar principle of least action.

After establishing this variational characterisation, we turn to the central object of Riemannian geometry: the **Riemann curvature tensor**. Our discussion of parallel transport has given us the tools to define curvature intrinsically—without reference to any embedding space—and this intrinsic notion of curvature is precisely the ingredient that Einstein's field equations will equate to the distribution of matter and energy.

10.1. Arc length and proper time. Let $(\mathcal{M}, g_{ab})$ be a manifold with metric, and let ∇_a be the Levi-Civita connection. Consider a smooth curve C from a point p to a point q in $\mathcal{M}$, parametrised by t, with tangent vector T^a.



Definition 10.1 (Arc length)**.** Assume for now that g_{ab} has Riemannian signature $(+, +, \ldots, +)$. The **length** (or **arc length**) of C with respect to g_{ab} is

$$\ell = \int_a^b \sqrt{g_{ab}\, T^a T^b}\, \mathrm{d}t \quad , \tag{151}$$

where T^a is the tangent vector to C and t is the curve parameter, with $C(a) = p$ and $C(b) = q$.

Intuitively, one should think of this formula as follows: divide the curve C into many small segments, approximate each segment by a straight line whose direction is the tangent vector T^a at that point, compute the length of each tangent vector using the metric (this is the "infinitesimal length" $\sqrt{g_{ab}\, T^a T^b}\, \mathrm{d}t$), then add up all the infinitesimal lengths and take the limit as the subdivision becomes infinitely fine.

For a Lorentzian metric with signature $(-, +, +, \ldots, +)$, the quantity $g_{ab}\, T^a T^b$ need not be positive. We distinguish three cases:

Definition 10.2 (Causal character of curves)**.** Let C be a smooth curve with tangent T^a on a Lorentzian manifold $(\mathcal{M}, g_{ab})$.

- C is **timelike** if $g_{ab}\, T^a T^b < 0$ everywhere along C.
- C is **null** (or **lightlike**) if $g_{ab}\, T^a T^b = 0$ everywhere along C.
- C is **spacelike** if $g_{ab}\, T^a T^b > 0$ everywhere along C.

Definition 10.3 (Proper time)**.** For a **timelike** curve C on a Lorentzian manifold, the **proper time** along C is

$$\tau = \int_a^b \sqrt{-g_{ab}\, T^a T^b}\, \mathrm{d}t \quad . \tag{152}$$

It is worth emphasising that not every curve on a Lorentzian manifold falls neatly into one of these three classes. One can certainly write down curves that are null for a while, then become spacelike, and later turn timelike. Such curves are mathematically legitimate, but they are *unphysical*: no particle, no solution to any physical equation of motion, can follow a worldline that

changes causal character. Moreover, as we now show, geodesics are never of this mixed type. If a curve changes from, say, timelike to spacelike, then we declare its "length" to be simply *undefined*—there is no consistent way to assign a single real number to such a curve.

A key observation is that the causal character of a geodesic cannot change.

Theorem 10.1 (Constancy of the norm along a geodesic)**.** *Let C be a geodesic with tangent T^a. Then the quantity $g_{ab}\, T^a T^b$ is constant along C. In particular, a geodesic cannot change from timelike to null, or from null to spacelike, etc.*

Proof. Compute the derivative of the norm along C:

$$T^c \nabla_c \big(g_{ab}\, T^a T^b\big) = (T^c \nabla_c g_{ab})\, T^a T^b + g_{ab}\, (T^c \nabla_c T^a)\, T^b + g_{ab}\, T^a\, (T^c \nabla_c T^b)\,. \tag{153}$$

The first term vanishes because the Levi-Civita connection is metric-compatible: $\nabla_c g_{ab} = 0$. The second and third terms vanish by the geodesic equation $T^c \nabla_c T^a = 0$. Hence

$$T^c \nabla_c \big(g_{ab}\, T^a T^b\big) = 0\,,$$

i.e. the inner product $(T, T) = g_{ab}\, T^a T^b$ is constant along the geodesic. □

Causal character is preserved

Since for a geodesic T^a is parallelly transported along itself, and parallel transport (with respect to a metric-compatible connection) preserves inner products, the "squared speed" $g_{ab}\, T^a T^b$ cannot change sign. A geodesic that starts timelike remains timelike forever.

10.2. Reparametrisation invariance. One might worry that the arc length or proper time depend on the particular parameter chosen to specify where we are along the curve. If they did, the "length" of a curve would reflect our arbitrary choice of parametrisation rather than a geometric property of the curve itself—not a desirable feature. Fortunately, the arc length and proper time are **reparametrisation invariant**: they do not depend on the choice of curve parameter.

Theorem 10.2 (Reparametrisation invariance of arc length)**.** *Let $s = s(t)$ be a monotone reparametrisation of C, and let S^a denote the tangent vector with respect to s. Then ℓ computed using parameter s equals ℓ computed using parameter t.*

Proof. The new tangent vector is related to the old one by

$$S^a = \frac{dt}{ds}\, T^a\,.$$

Therefore

$$\ell' = \int \sqrt{g_{ab}\, S^a S^b}\, \mathrm{d}s = \int \sqrt{g_{ab}\, T^a T^b}\, \left|\frac{dt}{ds}\right|\, \mathrm{d}s = \int \sqrt{g_{ab}\, T^a T^b}\, \mathrm{d}t = \ell\,.$$

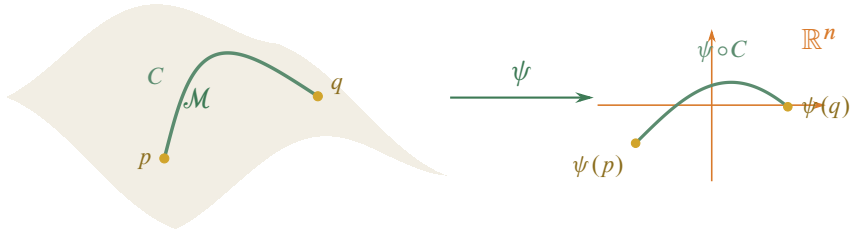
The same argument applies to the proper time τ. □

10.3. Geodesics as extremal curves. We now prove the fundamental result connecting geodesics to the calculus of variations. The key idea is to think of the arc length ℓ as a **functional**: a machine that accepts a curve as input and returns a real number. The domain of this functional is the (infinite-dimensional) space of all sufficiently smooth curves from p to q, and we ask: *for which curves is ℓ stationary under small variations?* The answer turns out to be precisely the geodesics.

Theorem 10.3 (Geodesics extremise arc length)**.** *A curve C joining two points p and q extremises the arc length functional ℓ if and only if C is a geodesic.*

Setup. Let p and q lie in a common chart $\psi\colon \mathcal{M} \to \mathbb{R}^n$, and assume C is spacelike. In coordinates, the arc length is

$$\ell = \int_a^b \sqrt{\sum_{\mu,\nu} g_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}t} \frac{\mathrm{d}x^\nu}{\mathrm{d}t}}\, \mathrm{d}t\,, \qquad C(a) = p,\ C(b) = q\,. \tag{154}$$



10.3.1. *Variational derivation.* Consider an infinitesimal variation of the curve:

$$x^\mu(t) \longmapsto x^\mu(t) + \delta x^\mu(t)\,, \qquad \delta x^\mu(a) = \delta x^\mu(b) = 0\,.$$

The endpoints are held fixed. We assume a parametrisation with **unit speed**: $g_{ab}\, T^a T^b = 1$.

Proof of Theorem 10.3. The extremality condition $\delta\ell = 0$ for all variations δx^α yields, after integration by parts,

$$0 = -\sum_\alpha g_{\alpha\beta} \frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}t^2} - \sum_{\alpha,\lambda} \frac{\partial g_{\alpha\beta}}{\partial x^\lambda} \frac{\mathrm{d}x^\alpha}{\mathrm{d}t} \frac{\mathrm{d}x^\lambda}{\mathrm{d}t} + \frac{1}{2} \sum_{\alpha,\lambda} \frac{\partial g_{\alpha\lambda}}{\partial x^\beta} \frac{\mathrm{d}x^\alpha}{\mathrm{d}t} \frac{\mathrm{d}x^\lambda}{\mathrm{d}t}\,. \tag{155}$$

Multiplying through by $g^{\mu\beta}$ and using the formula for the Christoffel symbols, this is equivalent to

$$\boxed{\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}t^2} + \sum_{\alpha,\lambda} \Gamma^\mu{}_{\alpha\lambda} \frac{\mathrm{d}x^\alpha}{\mathrm{d}t} \frac{\mathrm{d}x^\lambda}{\mathrm{d}t} = 0} \tag{156}$$

This is precisely the **geodesic equation**. Hence a curve extremises arc length if and only if it is a geodesic. □

Remark (Local equation, global information). There is something rather striking about this result. The geodesic equation (156) is a *local* differential equation: given the initial point p and an initial tangent vector, one can solve it step by step, determining the curve in a small neighbourhood of each point. Yet this purely local equation somehow "knows" which curve extremises the length between two potentially distant points p and q. Of course, the initial tangent direction encodes some global information about which direction to exit from p so as to arrive at q, but it is still remarkable that a second-order ODE in the curve parameter captures a global variational principle.

Geodesics from the Lagrangian

The geodesic equation can also be obtained by applying the Euler–Lagrange equations to the **Lagrangian**

$$L = \sum_{\mu,\nu} g_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}t} \frac{\mathrm{d}x^\nu}{\mathrm{d}t}\,, \tag{157}$$

via $\partial L/\partial x^\alpha - \mathrm{d}/\mathrm{d}t\,(\partial L/\partial \dot{x}^\alpha) = 0$. This provides a very efficient method for computing the Christoffel symbols: one simply reads them off from the Euler–Lagrange equations.

Exercise 10.1. Use the Lagrangian method to derive the geodesic equations on the sphere S^2 with metric $g = \mathrm{d}\theta \otimes \mathrm{d}\theta + \sin^2\theta \, \mathrm{d}\varphi \otimes \mathrm{d}\varphi$, and read off the Christoffel symbols.

10.4. Extrema versus minima.

Remark (Extremising $\neq$ minimising). It is essential to distinguish *extremising* from *minimising*. Even for an ordinary function of one variable, the condition $\mathrm{d}f = 0$ is satisfied at maxima, minima, and saddle points alike; moreover, a function may have many local minima, only one of which is the global minimum. The same subtleties arise in the calculus of variations. The geodesic equation tells us that $\delta\ell = 0$, but this alone does not guarantee that we have found a minimum (or a maximum)—only a stationary point of the length functional.

For a **Riemannian** metric (positive-definite signature), a geodesic does locally *minimise* arc length: among all nearby curves, the geodesic is the shortest. The situation is considerably more subtle for **Lorentzian** signature. Consider two timelike-separated events p and q. One might guess that the timelike geodesic from p to q has the smallest proper time, but in fact the opposite is closer to the truth: timelike curves with **arbitrarily small** proper time always exist. To see this, imagine a "zigzag" worldline that repeatedly accelerates to nearly the speed of light, turns around, accelerates back, and so on. Along each nearly-null segment the proper time is almost zero, so the total proper time can be made as small as desired.

Consequently, if a curve of *greatest* proper time between p and q exists, it must be a timelike geodesic—because any variation can only decrease the proper time. However, a timelike geodesic need not maximise proper time *globally*; it is only a local extremum, much as a function can have many local maxima.

The twin paradox revisited

In special relativity, the twin who travels along a non-geodesic (accelerated) worldline accumulates less proper time than the twin who follows the geodesic (inertial) worldline. This is because the inertial worldline *maximises* proper time among all timelike curves joining the same two events.

10.5. Curvature: the Riemann tensor.

We now turn to the second major topic of this lecture: **curvature**. With the machinery of manifolds, tensors, derivative operators, and parallel transport in hand, we have finally assembled enough mathematical foundations to discuss the curvature of a manifold in a fully intrinsic fashion—without reference to any ambient space in which the manifold might be embedded. This intrinsic notion of curvature is a key ingredient in Einstein's general theory of relativity; by the end of the next lecture we will have all the geometric tools needed to formulate the field equations.

The curvature of a manifold $\mathcal{M}$ equipped with a connection ∇_a measures the **path-dependence of parallel transport**. On a flat manifold, parallel-transporting a vector around a closed loop returns the original vector. On a curved manifold, the transported vector generally *differs* from the original. The Riemann curvature tensor provides a precise, intrinsic measure of this failure.

Curvature as non-commutativity

The Riemann curvature tensor measures the failure of successive covariant derivatives to commute. In flat space, $\nabla_a \nabla_b = \nabla_b \nabla_a$ when acting on any tensor; curvature is the obstruction to this commutativity. The way to connect this mathematical definition to the physical picture of path-dependent parallel transport is to recall that an infinitesimal parallel transport operation is in one-to-one correspondence with a covariant derivative. The commutator $\nabla_a \nabla_b - \nabla_b \nabla_a$ therefore captures the difference between two infinitesimal

parallel transports performed in opposite orders—precisely the path-dependence around an infinitesimal loop.

10.5.1. *The commutator of covariant derivatives on dual vectors.* We begin by studying the action of the commutator $\nabla_a\nabla_b - \nabla_b\nabla_a$ on an arbitrary dual vector field ω_c.

Theorem 10.4. *For any $f \in \mathscr{F}(\mathcal{M})$ and any dual vector field ω_c,*

$$(\nabla_a\nabla_b - \nabla_b\nabla_a)(f\omega_c) = f\,(\nabla_a\nabla_b - \nabla_b\nabla_a)\omega_c\,. \tag{158}$$

Consequently, $(\nabla_a\nabla_b - \nabla_b\nabla_a)\omega_c$ depends only on the value of ω_c at the point p.

Proof. Apply the Leibniz rule:

$$\begin{aligned}\nabla_a\nabla_b(f\omega_c) &= \nabla_a\big((\nabla_b f)\,\omega_c + f\,\nabla_b\omega_c\big)\\ &= (\nabla_a\nabla_b f)\,\omega_c + (\nabla_b f)\,\nabla_a\omega_c + (\nabla_a f)\,\nabla_b\omega_c + f\,\nabla_a\nabla_b\omega_c\,.\end{aligned} \tag{159}$$

The corresponding expression with a and b swapped is

$$\nabla_b\nabla_a(f\omega_c) = (\nabla_b\nabla_a f)\,\omega_c + (\nabla_a f)\,\nabla_b\omega_c + (\nabla_b f)\,\nabla_a\omega_c + f\,\nabla_b\nabla_a\omega_c\,. \tag{160}$$

Subtracting (160) from (159) and using the torsion-free condition $\nabla_a\nabla_b f = \nabla_b\nabla_a f$ (axiom (5)):

$$(\nabla_a\nabla_b - \nabla_b\nabla_a)(f\omega_c) = f\,(\nabla_a\nabla_b - \nabla_b\nabla_a)\omega_c\,.$$

Since the operator $(\nabla_a\nabla_b - \nabla_b\nabla_a)$ is $\mathscr{F}(\mathcal{M})$-linear on dual vector fields, by the same argument as in Theorem 7.3 (localisation), it depends only on the pointwise value of ω_c. □

10.5.2. *Definition of the Riemann tensor.* Since the map $\omega_c \mapsto (\nabla_a\nabla_b - \nabla_b\nabla_a)\omega_c$ is a linear map $\mathcal{T}(0,1) \to \mathcal{T}(0,3)$ that depends only on the pointwise value of ω_c, there exists a tensor $R_{abc}{}^d \in \mathcal{T}(1,3)$ such that:

Definition 10.4 (Riemann curvature tensor)**.** The **Riemann curvature tensor** $R_{abc}{}^d$ is defined by

$$\boxed{(\nabla_a\nabla_b - \nabla_b\nabla_a)\omega_c = R_{abc}{}^d\,\omega_d} \tag{161}$$

for all dual vector fields ω_c.

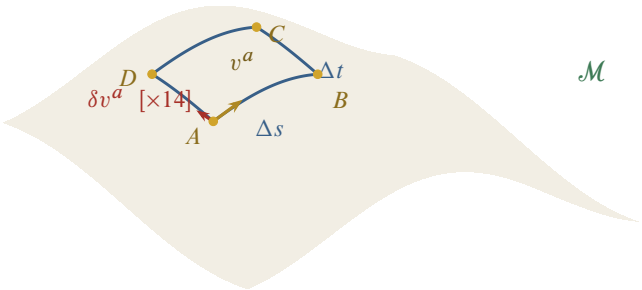
Remark. The Riemann tensor is a tensor of type $(1,3)$. It is determined entirely by the derivative operator ∇_a (and hence, when ∇_a is the Levi-Civita connection, entirely by the metric g_{ab}). If $R_{abc}{}^d = 0$, the connection is **flat**: covariant derivatives commute on all tensor fields, and parallel transport is path-independent.

Exercise 10.2. Show that for a vector field v^c:

$$(\nabla_a\nabla_b - \nabla_b\nabla_a)v^c = -R_{abd}{}^c\,v^d\,.$$

Hint: Apply the commutator to the scalar $\omega_c\,v^c$ and use the Leibniz rule.

10.6. Curvature and parallel transport around a loop. We now give a geometric interpretation of the Riemann tensor as measuring the change in a vector under parallel transport around an infinitesimal closed loop.



Consider a two-dimensional surface $S \subset \mathcal{M}$ parametrised by coordinates (s, t). Let $T^a = (\partial/\partial t)^a$ and $S^a = (\partial/\partial s)^a$ be the coordinate tangent vectors. We parallel-transport a vector v^a around the infinitesimal loop:

$$(0,0) \xrightarrow{\text{(i)}} (0, \Delta t) \xrightarrow{\text{(ii)}} (\Delta s, \Delta t) \xrightarrow{\text{(iii)}} (\Delta s, 0) \xrightarrow{\text{(iv)}} (0,0)\,.$$

We test the change in v^a by contracting with an arbitrary dual vector ω_a. Since v^a is parallelly transported, only ω_a varies along each leg.

Legs (i) and (iii). Along leg (i) (transport in the t-direction at $s = 0$), the change in $v^a\omega_a$ is

$$\delta_1 = \Delta t \cdot T^b \nabla_b (v^a \omega_a)\Big|_{(\Delta t/2,\,0)} = \Delta t \cdot v^a\, T^b \nabla_b \omega_a\Big|_{(\Delta t/2,\,0)}\,,$$

where we used $T^b \nabla_b v^a = 0$ (parallel transport). Along leg (iii) (transport in the $(-t)$-direction at $s = \Delta s$):

$$\delta_3 = -\Delta t \cdot v^a\, T^b \nabla_b \omega_a\Big|_{(\Delta t/2,\,\Delta s)}\,.$$

Combining:

(162)
$$\delta_1 + \delta_3 = \Delta t \Big(v^a\, T^b \nabla_b \omega_a\Big|_{(\Delta t/2,\,0)} - v^a\, T^b \nabla_b \omega_a\Big|_{(\Delta t/2,\,\Delta s)} \Big) = -\Delta s\, \Delta t\; S^c \nabla_c \big(v^a\, T^b \nabla_b \omega_a\big)\,.$$

Legs (ii) and (iv). By an analogous calculation (transport in the s- and $(-s)$-directions):

$$\delta_2 + \delta_4 = \Delta s\, \Delta t\; T^c \nabla_c \big(v^a\, S^b \nabla_b \omega_a\big)\,. \tag{163}$$

Total change. The net change in $v^a\omega_a$ around the loop is

$$\begin{aligned}
\delta(v^a\omega_a) &= (\delta_1 + \delta_3) + (\delta_2 + \delta_4) \\
&= \Delta s\, \Delta t\; v^a \big(T^c\, S^b\, \nabla_c \nabla_b \omega_a - S^c\, T^b\, \nabla_c \nabla_b \omega_a\big) \\
&= \Delta s\, \Delta t\; v^a\, T^c\, S^b\, (\nabla_c \nabla_b - \nabla_b \nabla_c)\omega_a\,. \qquad (164)
\end{aligned}$$

By the definition of the Riemann tensor (161):

$$\delta(v^a\omega_a) = \Delta s\, \Delta t\; v^a\, T^c\, S^b\, R_{cba}{}^{d}\, \omega_d\,. \tag{165}$$

Since this holds for *all* dual vectors ω_a, we conclude:

Remark (Path independence to first order). An important subtlety in the derivation above is that parallel transport around the loop is path-independent *to first order*: the contributions $\delta_1 + \delta_3$ vanish as $\Delta s \to 0$, and $\delta_2 + \delta_4$ vanish as $\Delta t \to 0$. It is only at *second* order—in the product $\Delta s\, \Delta t$—that the path-dependence appears. This is why the Riemann tensor, which encodes curvature, is a *second-order* quantity: curvature is invisible at the scale of a single infinitesimal displacement and manifests only when two independent directions are probed.

Parallel transport around an infinitesimal loop

The change in a vector v^a after parallel transport around an infinitesimal loop spanned by T^a and S^a is

$$\delta v^a = \Delta s\, \Delta t\; v^d\, T^c\, S^b\, R_{cbd}{}^a \tag{166}$$

Remark. Equation (166) is the precise statement that curvature equals the path-dependence of parallel transport. On a flat manifold, $R_{abc}{}^d = 0$ and $\delta v^a = 0$: parallel transport around any closed loop returns the original vector.

Exercise 10.3. Consider the unit sphere S^2 with the round metric. Parallel-transport a vector around the triangle formed by the equator from $\varphi = 0$ to $\varphi = \pi/2$, the meridian from the equator to the north pole, and the meridian from the north pole back to the starting point. Verify that the vector rotates by $\pi/2$, consistent with the curvature of S^2.

Exercise 10.4. Show that in a coordinate chart, the components of the Riemann tensor are given by

$$R^\rho{}_{\sigma\mu\nu} = \frac{\partial \Gamma^\rho{}_{\nu\sigma}}{\partial x^\mu} - \frac{\partial \Gamma^\rho{}_{\mu\sigma}}{\partial x^\nu} + \Gamma^\rho{}_{\mu\lambda}\,\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\,\Gamma^\lambda{}_{\mu\sigma}\,.$$

Hint: Apply the definition (161) to ω_c expressed in the coordinate basis and use the formula for $\nabla_\mu \omega_\nu$ in terms of Christoffel symbols.

In the next lecture we continue our study of curvature: we establish the symmetry properties and the Bianchi identity of the Riemann tensor, introduce derived quantities (the Ricci tensor, scalar curvature, and Einstein tensor), and relate curvature to the *geodesic deviation equation*—the precise statement that nearby geodesics diverge or converge in the presence of curvature.

11. Curvature continued

In the previous lecture we introduced the Riemann curvature tensor $R_{abc}{}^d$ via the commutator of covariant derivatives acting on a covector field, and gave it a geometric interpretation as the path-dependence of parallel transport around an infinitesimal loop. In this lecture we extend the action of the Riemann tensor to vectors and general tensors, establish its fundamental symmetry and Bianchi identities, and define three derived quantities—the Ricci tensor, scalar curvature, and Einstein tensor—that will be central to Einstein's field equations. We then derive the **geodesic deviation equation**, which relates the Riemann tensor to the relative acceleration of nearby geodesics: this is the physical manifestation of curvature that Einstein's theory identifies with gravity. Finally, we give explicit coordinate formulas for computing curvature from the metric.

11.1. Action of the Riemann tensor on vectors and tensors.

Recall that for a covector field ω_c:

$$(\nabla_a \nabla_b - \nabla_b \nabla_a)\omega_c = R_{abc}{}^d\, \omega_d\,. \tag{167}$$

We now determine the action of the commutator $[\nabla_a, \nabla_b] \equiv \nabla_a \nabla_b - \nabla_b \nabla_a$ on vectors and general tensor fields.

11.1.1. *Action on a vector field.* Let t^c be an arbitrary vector field and ω_c an arbitrary covector field. The contraction $t^c \omega_c$ is a scalar, so by the torsion-free property:

$$0 = (\nabla_a \nabla_b - \nabla_b \nabla_a)(t^c \omega_c)\,. \tag{168}$$

Expanding by the Leibniz rule:

$$\begin{aligned} 0 &= \nabla_a(\omega_c\,\nabla_b t^c + t^c\,\nabla_b\omega_c) - \nabla_b(\omega_c\,\nabla_a t^c + t^c\,\nabla_a\omega_c) \\ &= \omega_c\,(\nabla_a\nabla_b - \nabla_b\nabla_a)t^c + t^c\,(\nabla_a\nabla_b - \nabla_b\nabla_a)\omega_c \\ &= \omega_c\,(\nabla_a\nabla_b - \nabla_b\nabla_a)t^c + t^c\,\omega_d\,R_{abc}{}^d\,. \end{aligned} \tag{169}$$

Since ω_c is arbitrary, we conclude:

$$\boxed{(\nabla_a\nabla_b - \nabla_b\nabla_a)t^c = -R_{abd}{}^c\,t^d} \tag{170}$$

Note the crucial minus sign: for vectors the Riemann tensor appears with the opposite sign compared to covectors.

11.1.2. *Action on a general tensor field.*

Exercise 11.1. By repeated application of the Leibniz rule, show that for a tensor field $T^{c_1\cdots c_k}{}_{d_1\cdots d_l}$ of type (k,l):

(171)

$$\boxed{(\nabla_a\nabla_b - \nabla_b\nabla_a)T^{c_1\cdots c_k}{}_{d_1\cdots d_l} = -\sum_{j=1}^{k} R_{abe}{}^{c_j}\,T^{c_1\cdots e\cdots c_k}{}_{d_1\cdots d_l} + \sum_{j=1}^{l} R_{abd_j}{}^{e}\,T^{c_1\cdots c_k}{}_{d_1\cdots e\cdots d_l}}$$

In the first sum, the dummy index e replaces c_j in the j-th contravariant slot; in the second sum, e replaces d_j in the j-th covariant slot.

Remark. The pattern is: one $-R$ term for each upper index (just as for a single vector), one $+R$ term for each lower index (just as for a single covector). This is the universal formula for the "failure of covariant derivatives to commute" on tensors of arbitrary type.

11.2. Symmetries of the Riemann tensor.

We now establish four fundamental properties of $R_{abc}{}^d$. Properties (1)–(2) hold for *any* torsion-free derivative operator; properties (3)–(4) require in addition that ∇_a be the metric-compatible (Levi-Civita) connection.

11.2.1. *Property 1: Antisymmetry in the first two indices.*

Theorem 11.1.

$$R_{abc}{}^d = -R_{bac}{}^d\,. \tag{172}$$

Proof. This follows directly from the definition $R_{abc}{}^d\,\omega_d = (\nabla_a\nabla_b - \nabla_b\nabla_a)\omega_c$. The commutator $[\nabla_a, \nabla_b]$ is manifestly antisymmetric in a and b. □

11.2.2. *Property 2: First Bianchi identity (algebraic Bianchi identity).*

Theorem 11.2 (First Bianchi identity)**.** *For any torsion-free derivative operator:*

$$R_{[abc]}{}^d = 0\,. \tag{173}$$

Proof. Let ω_a be an arbitrary covector field and ∇_a a torsion-free derivative operator. We claim that

$$\nabla_{[a}\nabla_b\omega_{c]} = 0\,. \tag{174}$$

To verify (174), let $\tilde{\nabla}_a = \partial_a$ be the ordinary derivative in some coordinate chart. Then $\nabla_b\omega_c = \partial_b\omega_c - C^d{}_{bc}\,\omega_d$, and hence

$$\nabla_a\nabla_b\omega_c = \partial_a\partial_b\omega_c + C^d{}_{ab}\,\partial_{|d|}\omega_c + C^e{}_{ac}\,C^d{}_{|e|b}\,\omega_d + \cdots\,.$$

Antisymmetrising over a, b, c and using the torsion-free condition $C^d{}_{ab} = C^d{}_{ba}$, all terms vanish, confirming (174).

Now write

$$0 = 2\,\nabla_{[a}\nabla_b\omega_{c]} = \nabla_{[a}\nabla_b\omega_{c]} - \nabla_{[b}\nabla_a\omega_{c]} = R_{[abc]}{}^d\,\omega_d\,. \tag{175}$$

Since ω_d is arbitrary, $R_{[abc]}{}^d = 0$. □

11.2.3. *Property 3: Antisymmetry in the last two indices.*

Theorem 11.3. *If ∇_a is the Levi-Civita connection ($\nabla_a g_{bc} = 0$), then*

$$R_{abcd} = -R_{abdc}\,, \tag{176}$$

where $R_{abcd} \equiv g_{de}\,R_{abc}{}^e$.

Proof. Since $\nabla_a g_{cd} = 0$, applying the commutator formula to the metric tensor gives

$$0 = (\nabla_a\nabla_b - \nabla_b\nabla_a)g_{cd} = R_{abc}{}^e\,g_{ed} + R_{abd}{}^e\,g_{ce} = R_{abcd} + R_{abdc}\,. \tag{177}$$

Hence $R_{abcd} = -R_{abdc}$. □

Remark. Together, properties (1)–(3) imply that R_{abcd} is antisymmetric in the pair (a, b) and in the pair (c, d), and symmetric under exchange of the two pairs: $R_{abcd} = R_{cdab}$.

11.2.4. *Property 4: Differential Bianchi identity.*

Theorem 11.4 (Bianchi identity)**.** *For any torsion-free derivative operator:*

$$\nabla_{[a}\,R_{bc]d}{}^e = 0\,. \tag{178}$$

Proof sketch. Let ω_d be an arbitrary covector field. Consider the two expressions:

(i) $(\nabla_a\nabla_b - \nabla_b\nabla_a)\nabla_c\omega_d = R_{abc}{}^e\,\nabla_e\omega_d + R_{abd}{}^f\,\nabla_c\omega_f$.

(ii) $\nabla_a(\nabla_b\nabla_c - \nabla_c\nabla_b)\omega_d = \nabla_a(R_{bcd}{}^e\,\omega_e) = \omega_e\,\nabla_a R_{bcd}{}^e + R_{bcd}{}^e\,\nabla_a\omega_e$.

Antisymmetrise both expressions over a, b, c. In (i), the term $R_{[abc]}{}^e\,\nabla_e\omega_d$ vanishes by the first Bianchi identity, and the remaining terms cancel with corresponding terms from (ii). This yields

$$\omega_e\,\nabla_{[a}\,R_{bc]d}{}^e = 0\,.$$

Since ω_e is arbitrary: $\nabla_{[a}\,R_{bc]d}{}^e = 0$. □

11.3. Ricci tensor, scalar curvature, and the Einstein tensor.

11.3.1. *Geometric interpretation of the Riemann tensor.* Given two vectors $v^a, w^a \in V_p$, the quantity $R(v, w, \cdot, \cdot)$ defines a linear transformation $M_a{}^b$ of V_p. The two vectors v^a and w^a span a two-dimensional subspace of V_p; the Riemann tensor "probes" the curvature of $\mathcal{M}$ in the corresponding two-dimensional plane.

Riemann as sectional curvature probe

Think of the Riemann tensor as a "box" or machine with four slots. You feed it two vectors v^a and w^a; these define a two-dimensional plane in the tangent space at p—the plane of the infinitesimal loop around which you will parallel transport. What remains is a linear transformation $M_a{}^b = R(v, w, \cdot, \cdot)$ that tells you precisely how a vector gets rotated when parallel-transported around that loop (to second order in the loop's area). The Riemann tensor thus encodes *all* second-order curvature information of the manifold: every possible loop direction, every possible vector to transport.

In many applications one does not need all this information. The Ricci tensor and scalar curvature, defined below, are successive contractions that capture progressively coarser summaries of the curvature—averaged over some or all loop directions.

11.3.2. *The Ricci tensor.*

Definition 11.1 (Ricci tensor)**.** The **Ricci tensor** is the contraction of the Riemann tensor over its second and fourth indices:

$$R_{ac} = R_{abc}{}^{b} \,. \tag{179}$$

Remark. By the symmetries of R_{abcd} (properties 1–3), the Ricci tensor is symmetric: $R_{ab} = R_{ba}$.

11.3.3. *Scalar curvature.*

Definition 11.2 (Scalar curvature)**.** The **scalar curvature** (or **Ricci scalar**) is the trace of the Ricci tensor:

$$\boxed{R = R_a{}^a = g^{ab} R_{ab}} \tag{180}$$

11.3.4. *The contracted Bianchi identity and the Einstein tensor.* We now contract the differential Bianchi identity (178) to obtain a fundamental divergence condition.

Step 1: First contraction. Write out the Bianchi identity explicitly:

$$\nabla_a R_{bcd}{}^a + \nabla_b R_{cd} - \nabla_c R_{bd} = 0 \,,$$

where we used $R_{bcd}{}^a$ contracted on the first and last indices to relate to the Ricci tensor.

Step 2: Second contraction. Contract with g^{bd}:

$$\nabla_a R_c{}^a + \nabla_b R_c{}^b - \nabla_c R = 0 \,,$$

which gives

$$2\,\nabla_a R_c{}^a - \nabla_c R = 0 \,. \tag{181}$$

This can be rewritten as $\nabla^a G_{ab} = 0$, where:

Definition 11.3 (Einstein tensor)**.** The **Einstein tensor** is defined by

$$\boxed{G_{ab} = R_{ab} - \tfrac{1}{2}\, R\, g_{ab}} \tag{182}$$

Divergence-free Einstein tensor

The Einstein tensor is divergence-free:

$$\nabla^a G_{ab} = 0 \,. \tag{183}$$

This identity holds purely as a consequence of the Bianchi identity—it is a geometric identity, not a field equation. It will play a central role when we write down Einstein's field equations.

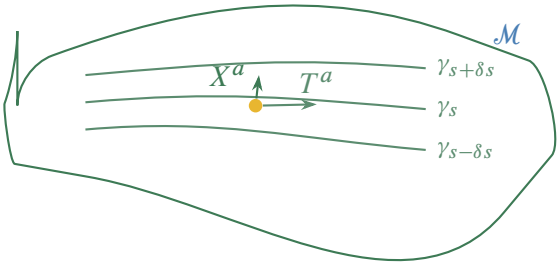
11.4. Geodesic deviation.

We now come to what is arguably the most physically important interpretation of the Riemann curvature tensor: it controls how nearby geodesics accelerate toward or away from each other. Imagine two geodesics that start out infinitesimally close and pointing in the same direction. On a flat manifold they remain parallel forever; on a curved manifold they may diverge or converge. The rate of this relative acceleration is captured by the **geodesic deviation equation**, and it is this equation that will play a central role in motivating Einstein's field equations.

11.4.1. *Setup: a one-parameter family of geodesics.* Consider a one-parameter family of geodesics $\gamma_s(t)$, where $s \in (-\varepsilon, \bar{\varepsilon})$ labels the different geodesics and t is an affine parameter along each one. We require:

(1) γ_s is a geodesic for every s, and
(2) the map $(s,t) \mapsto \gamma_s(t)$ is smooth, one-to-one, with smooth inverse.

The image of this family is a two-dimensional submanifold $\Sigma \subset \mathcal{M}$, and (s,t) serve as a coordinate system on Σ.



Define two vector fields on Σ:

- $T^a = (\partial/\partial t)^a$: the tangent vector to each geodesic. Since γ_s is a geodesic: $T^a \nabla_a T^b = 0$.
- $X^a = (\partial/\partial s)^a$: the **deviation vector**, pointing from $\gamma_s(t)$ to the infinitesimally nearby geodesic $\gamma_{s+\delta s}(t)$.

Since T^a and X^a are coordinate vector fields on Σ, they commute:

$$[T, X]^a = 0 \qquad \Longleftrightarrow \qquad T^b \nabla_b X^a = X^b \nabla_b T^a \,. \tag{184}$$

Exercise 11.2. Using (184) and the geodesic equation $T^a \nabla_a T^b = 0$, show that $X^a T_a$ is constant along each geodesic. (By a reparametrisation of s, we may set $X^a T_a = 0$, so that X^a is purely transverse.)

11.4.2. *Relative velocity and acceleration.* Define the **relative velocity** of nearby geodesics:

$$v^a = T^b \nabla_b X^a \,. \tag{185}$$

This measures the rate of change of the deviation vector X^a along the geodesic.

The **relative acceleration** is

$$A^a = T^c \nabla_c v^a = T^c \nabla_c (T^b \nabla_b X^a) \,. \tag{186}$$

We now simplify (186) using the commutativity condition (184):

$$\begin{aligned} A^a &= T^c \nabla_c (X^b \nabla_b T^a) \\ &= (T^c \nabla_c X^b)(\nabla_b T^a) + X^b\, T^c \nabla_c \nabla_b T^a \,. \end{aligned} \tag{187}$$

Using the commutativity relation again on the first term, $(T^c \nabla_c X^b) = (X^c \nabla_c T^b)$:

$$A^a = (X^c \nabla_c T^b)(\nabla_b T^a) + X^b\, T^c \nabla_c \nabla_b T^a \,. \tag{188}$$

The first term equals $X^c \nabla_c (T^b \nabla_b T^a) - X^b$(extra terms). But $T^b \nabla_b T^a = 0$ (geodesic equation), so

$$(X^c \nabla_c T^b)(\nabla_b T^a) = X^c \nabla_c (T^b \nabla_b T^a) - X^b (T^c \nabla_b \nabla_c T^a - \text{cancelled}) \,.$$

The simplification, using the definition of the Riemann tensor $T^c \nabla_c \nabla_b T^a - T^c \nabla_b \nabla_c T^a = -R_{cbd}{}^a\, T^c T^d$, and the geodesic equation, yields:

$$A^a = -R_{cbd}{}^a\, X^b\, T^c\, T^d \tag{189}$$

This is the **geodesic deviation equation** (also called the **Jacobi equation**).

Tidal forces

In general relativity, freely falling particles follow geodesics. The geodesic deviation equation says that the relative acceleration between two nearby freely falling particles is proportional to the Riemann tensor. This is the precise formulation of *tidal forces*: the Riemann tensor is what an observer actually *measures* as the differential gravitational acceleration across a small region of spacetime.

Curvature ⇔ geodesic deviation

A manifold is flat ($R_{abc}{}^d = 0$ everywhere) if and only if $A^a = 0$ for *all* one-parameter families of geodesics. Conversely, if there exist relatively accelerating geodesics, then $R_{abc}{}^d \neq 0$: the manifold is curved.

Remark. A word of caution: the condition requires *smooth families* of geodesics, not just two arbitrary geodesics that happen to diverge. Even in flat space one can pick two geodesics pointing in wildly different directions that move apart; this says nothing about curvature. What matters is that geodesics which start infinitesimally close *and* initially parallel still accelerate away from each other—that is the signature of genuine curvature.

11.5. Computing curvature in coordinates. Given a manifold $\mathcal{M}$ with metric g_{ab}, the Riemann tensor $R_{abc}{}^d$ is uniquely determined by the Levi-Civita connection ∇_a. We now derive the explicit coordinate expression.

11.5.1. *Derivation.* Choose a coordinate chart (ψ, x^μ). For a covector field ω_c, recall that the covariant derivative is

$$\nabla_b \omega_c = \partial_b \omega_c - \Gamma^d{}_{bc}\, \omega_d\,.$$

Applying a second covariant derivative:

$$\nabla_a \nabla_b \omega_c = \partial_a(\partial_b \omega_c - \Gamma^d{}_{bc}\, \omega_d) - \Gamma^e{}_{ab}(\partial_e \omega_c - \Gamma^d{}_{ec}\, \omega_d) - \Gamma^e{}_{ac}(\partial_b \omega_e - \Gamma^d{}_{be}\, \omega_d)\,. \tag{190}$$

The Riemann tensor is obtained by antisymmetrising:

$$R_{abc}{}^d\, \omega_d = 2\, \nabla_{[a} \nabla_{b]} \omega_c = (-2\, \partial_{[a} \Gamma^d{}_{b]c} + 2\, \Gamma^e{}_{c[a}\, \Gamma^d{}_{b]e})\, \omega_d\,.$$

Since ω_d is arbitrary, we read off the coordinate components:

$$R^\sigma{}_{\mu\nu\rho} = \frac{\partial \Gamma^\sigma{}_{\nu\rho}}{\partial x^\mu} - \frac{\partial \Gamma^\sigma{}_{\mu\rho}}{\partial x^\nu} + \sum_\alpha \left(\Gamma^\alpha{}_{\nu\rho}\, \Gamma^\sigma{}_{\mu\alpha} - \Gamma^\alpha{}_{\mu\rho}\, \Gamma^\sigma{}_{\nu\alpha}\right) \tag{191}$$

Remark. The Riemann tensor involves the Christoffel symbols and their first derivatives, hence the metric and its first *and* second derivatives. This is consistent with the fact that the Christoffel symbols involve first derivatives of the metric, and the Riemann tensor involves first derivatives of the Christoffel symbols.

Remark (Practical complexity). In a four-dimensional spacetime, $R^{\sigma}{}_{\mu\nu\rho}$ has $4^4 = 256$ components. Each involves Christoffel symbols (themselves computed from the metric), plus a summation over an internal index α. A brute-force computation thus requires on the order of a thousand individual evaluations—and if a human has even a one-percent error rate per evaluation, one should expect roughly ten mistakes in the final answer. The symmetry properties (antisymmetry in the first and last pairs of indices, the Bianchi identity) reduce the number of independent components to 20 in four dimensions, but the intermediate algebra remains formidable. In practice, one should exploit every available symmetry and, whenever possible, verify results with a computer algebra system.

11.5.2. *Ricci tensor in coordinates.* By contraction, the Ricci tensor in coordinates is

$$R_{\mu\rho} = \sum_{\nu} R^{\nu}{}_{\mu\nu\rho} = \sum_{\nu}\left(\frac{\partial \Gamma^{\nu}{}_{\nu\rho}}{\partial x^{\mu}} - \frac{\partial \Gamma^{\nu}{}_{\mu\rho}}{\partial x^{\nu}} + \sum_{\alpha}\left(\Gamma^{\alpha}{}_{\nu\rho}\,\Gamma^{\nu}{}_{\mu\alpha} - \Gamma^{\alpha}{}_{\mu\rho}\,\Gamma^{\nu}{}_{\nu\alpha}\right)\right). \tag{192}$$

11.6. Useful formulas. We collect several results that simplify practical computations involving the Christoffel symbols.

11.6.1. *The metric determinant.* Let $g = \det(g_{\mu\nu})$ denote the determinant of the metric in a coordinate chart. A standard identity from matrix calculus gives:

$$\sum_{\nu,\alpha} g^{\nu\alpha}\,\frac{\partial g_{\nu\alpha}}{\partial x^{\mu}} = \frac{1}{g}\,\frac{\partial g}{\partial x^{\mu}} = \frac{\partial \log|g|}{\partial x^{\mu}}. \tag{193}$$

11.6.2. *Trace of the Christoffel symbols.*

Theorem 11.5. *The trace of the Christoffel symbols over the first and second indices is*

$$\Gamma^{\alpha}{}_{\alpha\mu} = \sum_{\nu}\Gamma^{\nu}{}_{\nu\mu} = \frac{1}{2}\sum_{\nu,\alpha} g^{\nu\alpha}\,\frac{\partial g_{\nu\alpha}}{\partial x^{\mu}} = \frac{\partial \log\sqrt{|g|}}{\partial x^{\mu}}. \tag{194}$$

Proof. From the definition of the Christoffel symbols (194):

$$\Gamma^{\alpha}{}_{\alpha\mu} = \frac{1}{2}\,g^{\alpha\sigma}\left(\frac{\partial g_{\sigma\alpha}}{\partial x^{\mu}} + \frac{\partial g_{\sigma\mu}}{\partial x^{\alpha}} - \frac{\partial g_{\alpha\mu}}{\partial x^{\sigma}}\right).$$

The last two terms cancel when summed over α (by relabelling $\alpha \leftrightarrow \sigma$ and using the symmetry $g^{\alpha\sigma} = g^{\sigma\alpha}$), leaving

$$\Gamma^{\alpha}{}_{\alpha\mu} = \frac{1}{2}\,g^{\alpha\sigma}\,\frac{\partial g_{\sigma\alpha}}{\partial x^{\mu}} = \frac{1}{2g}\,\frac{\partial g}{\partial x^{\mu}} = \frac{\partial \log\sqrt{|g|}}{\partial x^{\mu}}.$$ □

11.6.3. *Divergence formula.*

Theorem 11.6 (Covariant divergence in coordinates)**.** *For a vector field T^a, the covariant divergence is*

$$\boxed{\nabla_a T^a = \sum_{\mu}\frac{1}{\sqrt{|g|}}\,\frac{\partial\left(\sqrt{|g|}\,T^{\mu}\right)}{\partial x^{\mu}}} \tag{195}$$

Proof. By definition,

$$\nabla_a T^a = \partial_a T^a + \Gamma^{a}{}_{ab}\,T^b = \sum_{\mu}\frac{\partial T^{\mu}}{\partial x^{\mu}} + \sum_{\mu}\Gamma^{\mu}{}_{\mu\nu}\,T^{\nu}.$$

Substituting the Christoffel trace (194):

$$\nabla_a T^a = \sum_\mu \frac{\partial T^\mu}{\partial x^\mu} + \sum_\nu \frac{1}{\sqrt{|g|}} \frac{\partial \sqrt{|g|}}{\partial x^\nu} T^\nu = \sum_\mu \frac{1}{\sqrt{|g|}} \frac{\partial(\sqrt{|g|}\, T^\mu)}{\partial x^\mu},$$

where the last step uses the product rule. □

Remark. The divergence formula (195) is extremely useful in practice. It shows that the covariant divergence can be computed without explicitly evaluating Christoffel symbols: one needs only the metric determinant and partial derivatives.

Exercise 11.3. Using the divergence formula (195), compute $\nabla_a T^a$ for the metric of $\mathbb{R}^3$ in spherical coordinates (r, θ, φ) and verify that it reproduces the standard expression for the divergence in spherical coordinates.

With the Riemann tensor, its symmetries, the derived curvature quantities (Ricci tensor, scalar curvature, Einstein tensor), and the geodesic deviation equation in hand, we have completed the mathematical foundations of differential geometry needed for general relativity. After roughly ten lectures of building up this machinery, we are finally ready to turn—in the next lecture—to Einstein's general theory of relativity itself: the Lie derivative, the Newtonian limit, and the formulation of the field equations.

12. Lie derivatives and the Newtonian limit

Having developed the full apparatus of differential geometry—manifolds, tangent spaces, tensor fields, covariant derivatives, and curvature—we are finally ready to formulate Einstein's general theory of relativity. In this lecture we state the fundamental hypotheses of GR and discuss the principles that guide the construction of equations of motion in curved spacetime. We then introduce the **Lie derivative**, which makes precise the sense in which diffeomorphisms are the gauge symmetry of general relativity, and leads naturally to **Killing vector fields** and the connection between spacetime symmetries and conservation laws. Finally, we state **Einstein's field equations** relating spacetime curvature to the distribution of matter, linearise them around flat spacetime, and show that in the appropriate limit they reproduce Newtonian gravity: both Poisson's equation for the gravitational potential and Newton's equation of motion for test particles.

12.1. The fundamental hypotheses of general relativity.

The equivalence principle, developed in Lecture 2, tells us that no experiment can locally distinguish a gravitational field from uniform acceleration. A far-reaching consequence is that it is impossible, in general, to set up a family of inertial observers to measure the gravitational "force"—any observer falls along with the test body, and there is no natural background motion to compare with. This motivates us to abandon the concept of gravitational force altogether and instead encode gravity in the geometry of spacetime.

Fundamental hypothesis of general relativity

(1) Spacetime is a four-dimensional manifold $\mathcal{M}$ equipped with a Lorentz metric g_{ab} of signature $(-, +, +, +)$.
(2) The manifold $\mathcal{M}$ need not be topologically $\mathbb{R}^4$, and the metric g_{ab} need not be flat.
(3) The world lines of freely falling bodies are geodesics of g_{ab}.

Remark. While the gravitational force cannot be measured directly in a general spacetime, the *relative acceleration* of nearby geodesics—quantified by the geodesic deviation equation (189)—is always well defined. We can therefore speak meaningfully of *tidal forces* in general relativity. This observation will prove critical when we derive Einstein's field equations below.

When can we measure "gravitational force"?

In situations with time-translation symmetry—for instance, the gravitational field of a single isolated body like the Earth—one can set up a preferred family of static observers and define a gravitational force relative to them. But the moment two or more bodies are in relative motion, the time-translation symmetry is broken and no preferred background exists. General relativity is designed to handle this fully general situation.

12.2. Equations of motion in curved spacetime. Given the fundamental hypothesis, how do we determine the laws of physics in curved spacetime? Two guiding principles, while not mathematically rigorous, are powerful enough to essentially determine the correct equations of motion in most situations of physical interest.

12.2.1. *General covariance and minimal substitution.*

Definition 12.1 (Principle of general covariance)**.** Only the metric g_{ab} and quantities derivable from it may appear as spacetime quantities in the laws of physics.

In particular, no reference to a preferred coordinate system or background structure is permitted.

Definition 12.2 (Correspondence principle)**.** The equations of physics in general relativity must reduce to their special-relativistic form when $g_{ab} = \eta_{ab}$.

Together, these two principles suggest the following **minimal substitution rules** for passing from special to general relativity:

$$\eta_{ab} \longrightarrow g_{ab}\,, \qquad \partial_a \longrightarrow \nabla_a\,, \tag{196}$$

where ∇_a is the Levi-Civita connection of g_{ab}.

Remark. The minimal substitution rules are a starting point, not a theorem. Additional curvature-coupling terms (e.g. a term $\alpha\, R\,\phi$ in a scalar field equation) may be permitted by both principles, since such terms vanish in the flat limit. One must supply independent arguments—experiment or internal consistency—to fix such ambiguities.

12.2.2. *Free particles, energy, and the stress-energy tensor.* Physical quantities continue to be represented by tensor fields, just as in special relativity. The key updates concern the equations of motion.

Free particle motion. A free particle follows a time-like geodesic:

$$u^a \nabla_a u^b = 0\,, \tag{197}$$

where u^a is the unit four-velocity tangent to the world line, normalised so that $g_{ab}\, u^a u^b = -1$. When a force acts, the equation generalises to $f^a = m\, u^b \nabla_b u^a$, where m is the rest mass. The four-momentum is $p^a = m\, u^a$.

Energy. An observer with four-velocity v^a, present at an event on the particle's world line, measures the particle's energy to be

$$E = -p_a\, v^a = -g_{ab}\, p^a v^b\,. \tag{198}$$

This definition requires the observer and the particle to be at the *same* spacetime event: on a curved manifold, parallel transport is path-dependent, so there is no canonical way to compare

vectors at distant points. A given observer cannot, in general, define the energy of a distant particle.

Continuous matter. A continuous distribution of matter is described by the stress-energy tensor T_{ab}. For a **perfect fluid** with energy density ρ, pressure p, and four-velocity field u^a:

$$T_{ab} = (\rho + p)\, u_a u_b + p\, g_{ab}\,. \tag{199}$$

Applying the minimal substitution rule to the special-relativistic conservation law $\partial^a T_{ab} = 0$ yields

$$\boxed{\nabla^a T_{ab} = 0} \tag{200}$$

Why $\nabla^a T_{ab} = 0$ is not global conservation

In special relativity, $\partial^a T_{ab} = 0$ implies global conservation of energy and momentum via Gauss's theorem: one sets up a family of inertial observers with parallel four-velocities, defines a current $J^a = -T^{ab}\, v_b$, and integrates its divergence over a spacetime region. In general relativity, the analogous construction requires v^a to satisfy Killing's equation $\nabla_{(a} v_{b)} = 0$—but such vector fields need not exist on a general curved spacetime. We will return to this point in §12.4.3.

12.3. The Lie derivative.

12.3.1. *Diffeomorphism invariance in general relativity.* For a diffeomorphism $\phi\colon \mathcal{M} \to \mathcal{M}$, the spacetimes $(\mathcal{M}, g_{ab})$ and $(\mathcal{M}, \phi^* g_{ab})$ describe the same physics: all operationally defined quantities are built from tensors in a coordinate-free manner, so the Jacobian factors introduced by ϕ cancel in every contraction and tensor transformation law.

Gauge freedom of general relativity

Diffeomorphisms comprise the gauge freedom of general relativity. Two metrics related by a diffeomorphism represent the same physical spacetime.

To study *infinitesimal* diffeomorphisms, we need a notion of derivative that measures how a tensor field changes along a flow. This is the **Lie derivative**.

12.3.2. *Definition and elementary properties.* Let v^a be a smooth vector field on $\mathcal{M}$ and ϕ_t the one-parameter group of diffeomorphisms it generates (its *flow*, introduced in Lecture 5). The pullback ϕ_t^* acts on tensor fields by dragging them back along the flow.

Definition 12.3 (Lie derivative)**.** The **Lie derivative** of a tensor field $T^{a_1\cdots a_k}{}_{b_1\cdots b_l}$ along v^a is

$$\mathcal{L}_v T^{a_1\cdots a_k}{}_{b_1\cdots b_l} = \lim_{t\to 0} \frac{1}{t}\Big(\phi_t^*\, T^{a_1\cdots a_k}{}_{b_1\cdots b_l}\big|_{\phi_t(p)} - T^{a_1\cdots a_k}{}_{b_1\cdots b_l}\big|_p\Big)\,. \tag{201}$$

The Lie derivative has three fundamental properties:

(1) **Tensorial**: $\mathcal{L}_v T$ is a tensor field of the same type as T.
(2) **Leibniz rule**: for any tensor fields S and T, $\mathcal{L}_v(S \otimes T) = (\mathcal{L}_v S) \otimes T + S \otimes (\mathcal{L}_v T)$.
(3) **Linearity**: $\mathcal{L}_v(\alpha S + \beta T) = \alpha\, \mathcal{L}_v S + \beta\, \mathcal{L}_v T$ for $\alpha, \beta \in \mathbb{R}$.

On scalar fields. For $f \in C^\infty(\mathcal{M})$:

$$\mathcal{L}_v f = v(f) = v^a \partial_a f\,. \tag{202}$$

On vector fields. For a vector field w^a, the Lie derivative is the Lie bracket:

$$\mathcal{L}_v w^a = [v, w]^a = v^b \nabla_b w^a - w^b \nabla_b v^a \,. \tag{203}$$

Exercise 12.1. Prove (203) from the definition (201).

On covector fields. For a covector field μ_a, apply the Leibniz rule to the scalar $\mu_a w^a$ for an arbitrary vector field w^a:

$$\mathcal{L}_v(\mu_a w^a) = w^a \mathcal{L}_v \mu_a + \mu_a \, [v, w]^a \,.$$

Since $\mathcal{L}_v(\mu_a w^a) = v^b \nabla_b(\mu_a w^a)$ by (202), expanding the right-hand side and comparing terms yields (since w^a is arbitrary):

$$\mathcal{L}_v \mu_a = v^b \nabla_b \mu_a + \mu_b \, \nabla_a v^b \,. \tag{204}$$

12.3.3. *The Lie derivative of a general tensor field.* Extending the pattern of (203) and (204) to a tensor of type (k, l):

Theorem 12.1 (Lie derivative of a general tensor)**.** *For any torsion-free derivative operator ∇_a:*
(205)

$$\mathcal{L}_v T^{a_1 \cdots a_k}{}_{b_1 \cdots b_l} = v^c \nabla_c T^{a_1 \cdots a_k}{}_{b_1 \cdots b_l} - \sum_{j=1}^{k} T^{a_1 \cdots c \cdots a_k}{}_{b_1 \cdots b_l} \, \nabla_c v^{a_j} + \sum_{j=1}^{l} T^{a_1 \cdots a_k}{}_{b_1 \cdots c \cdots b_l} \, \nabla_{b_j} v^c$$

In the first sum, the index c replaces a_j in the j-th contravariant slot; in the second sum, c replaces b_j in the j-th covariant slot.

Remark. Although individual terms in (205) depend on the choice of ∇_a, the Christoffel-symbol contributions cancel in the sum. The Lie derivative is defined purely in terms of the flow of v^a and requires no connection.

Exercise 12.2. Verify (205) by applying the Leibniz rule repeatedly, using the results for scalars (202), vectors (203), and covectors (204).

12.4. Killing vector fields and conservation laws.

12.4.1. *The Lie derivative of the metric.* Since $\nabla_a g_{bc} = 0$ for the Levi-Civita connection, applying (205) to the metric gives

$$\mathcal{L}_v g_{ab} = \nabla_a v_b + \nabla_b v_a \tag{206}$$

where $v_a = g_{ab} \, v^b$.

Lie derivative as metric strain

The quantity $\mathcal{L}_v g_{ab}$ measures how the metric changes as one flows along v^a. If $\mathcal{L}_v g_{ab} = 0$, then distances and angles are preserved by the flow: it is an isometry. Formula (206) shows that this condition depends only on the symmetrised covariant derivative of v^a—the curved-space analogue of the strain tensor in continuum mechanics.

12.4.2. *Killing's equation and isometries.*

Definition 12.4 (Killing vector field)**.** A vector field v^a is a **Killing vector field** if it generates an isometry: $\mathcal{L}_v g_{ab} = 0$. By (206), this is equivalent to **Killing's equation**:

$$\nabla_{(a} v_{b)} = 0 \tag{207}$$

Each solution of Killing's equation corresponds to a continuous symmetry of the spacetime geometry. In Minkowski space, there are ten independent Killing vector fields: four translations (generating conservation of energy-momentum) and six Lorentz transformations (generating conservation of angular momentum).

12.4.3. *Conservation laws in general relativity.* In special relativity, the condition $\partial^a T_{ab} = 0$ leads to global conservation laws via Gauss's theorem. The procedure requires a family of inertial observers with parallel four-velocities ($\partial_a v^b = 0$), which allows one to define a conserved current $J^a = -T^{ab} v_b$ satisfying $\partial_a J^a = 0$.

In general relativity, the analogous condition $\nabla^a T_{ab} = 0$ does *not* by itself imply a global conservation law. To construct a divergence-free current, we need a vector field v^a satisfying Killing's equation (207).

No global energy conservation in general

On a general curved spacetime, Killing vector fields need not exist. In particular, de Sitter spacetime admits no global time-like Killing vector field. Consequently, there is no global notion of conserved energy, and the equation $\nabla^a T_{ab} = 0$ does *not* imply global energy conservation.

However, when a Killing vector ξ^a does exist, we obtain a genuinely conserved current:

$$J^a = T^{ab} \xi_b \,, \qquad \nabla_a J^a = 0 \,. \tag{208}$$

Exercise 12.3. Prove that if ξ^a is a Killing vector field and $\nabla^a T_{ab} = 0$, then $\nabla_a J^a = 0$ where $J^a = T^{ab} \xi_b$.

Remark. Physically, the absence of global energy conservation reflects the fact that the gravitational field does work on matter via tidal forces, transferring energy between the curvature of spacetime and the matter content. Only when a time-translation symmetry exists can one define a total energy that is conserved. Equation (200) may still be regarded as expressing *local* conservation of energy-momentum in a sufficiently small, approximately flat region.

12.5. Einstein's field equations. We have equations of motion for matter fields ($\nabla^a T_{ab} = 0$), but we do not yet have an equation determining the metric itself. Since general relativity treats spacetime as dynamical—its geometry responds to the presence of matter—we need an equation of motion for g_{ab}. This is where Mach's principle enters: the geometry of spacetime is determined by the distribution of matter and energy.

12.5.1. *From tidal forces to spacetime curvature.* The key clue comes from comparing the Newtonian description of tidal forces with the geodesic deviation equation. In Newtonian gravity, two test particles on nearby orbits in a gravitational potential ϕ experience a relative (tidal) acceleration

$$a^i = -\frac{\partial^2 \phi}{\partial x^i \, \partial x^j} \, \Delta x^j \,, \tag{209}$$

where Δx^j is the separation vector. In general relativity, the geodesic deviation equation (189) gives

$$A^a = -R_{cbd}{}^a \, X^b \, T^c \, T^d \, .$$

Comparing these suggests the correspondence

$$R_{cbd}{}^a \, T^c T^d \longleftrightarrow \frac{\partial^2 \phi}{\partial x^i \, \partial x^j} \, . \tag{210}$$

In Newtonian theory, ϕ is determined by the mass density ρ via Poisson's equation $\nabla^2 \phi = 4\pi\rho$. Since the energy density observed by an observer with four-velocity v^a is $\rho = T_{ab} \, v^a v^b$, the correspondence (210) points toward an equation relating curvature to T_{ab}. Contracting over the velocity directions yields a correspondence between the Ricci tensor and the stress-energy tensor.

12.5.2. *The Einstein equation.* A first guess, following the analogy above, would be $R_{ab} = 8\pi \, T_{ab}$. However, this is *inconsistent* with the equation of motion $\nabla^a T_{ab} = 0$. The contracted Bianchi identity (181) gives

$$\nabla^a R_{ab} = \tfrac{1}{2} \, \nabla_b R \, ,$$

which vanishes only when the Ricci scalar R is constant—a condition not satisfied by most interesting spacetimes.

The resolution is to use the Einstein tensor $G_{ab} = R_{ab} - \frac{1}{2} \, R \, g_{ab}$, which *is* divergence-free ($\nabla^a G_{ab} = 0$, cf. (183)) as a consequence of the Bianchi identity.

Einstein's field equations

$$G_{ab} = R_{ab} - \tfrac{1}{2} \, R \, g_{ab} = 8\pi \, T_{ab} \tag{211}$$

This equation relates the curvature of spacetime (left-hand side) to the distribution of matter and energy (right-hand side). The factor 8π is fixed by requiring agreement with Newtonian gravity in the appropriate limit.

Einstein's path to the field equations

Einstein first proposed $R_{ab} = 8\pi \, T_{ab}$ but quickly realised the inconsistency with energy-momentum conservation. The resolution—replacing the Ricci tensor with the Einstein tensor—was guided by the contracted Bianchi identity. The final form (211) was published by Einstein in November 1915. The mathematical identity $\nabla^a G_{ab} = 0$ guarantees that Einstein's equation is *automatically* consistent with the conservation of energy-momentum—the geometry and the matter talk to each other in a self-consistent way.

Remark. Einstein's equation (211) is a system of ten coupled, nonlinear partial differential equations for the ten independent components of g_{ab}. Finding exact solutions is extraordinarily difficult; we will succeed only in a few highly symmetric cases. Much of the remainder of this course is devoted to exploring the consequences of this equation.

12.6. Linearised gravity and gauge invariance. To connect Einstein's equation with Newtonian gravity, we consider the regime where the gravitational field is weak: the metric is close to the flat Minkowski metric.

12.6.1. *Metric perturbations and the linearised Einstein tensor.* Write the metric as a small perturbation of flat spacetime:

$$g_{ab} = \eta_{ab} + \gamma_{ab}\,, \qquad |\gamma_{ab}| \ll 1\,. \tag{212}$$

All indices are raised and lowered with η_{ab}. Define the **trace-reversed perturbation**:

$$\bar{\gamma}_{ab} = \gamma_{ab} - \tfrac{1}{2}\,\eta_{ab}\,\gamma\,, \qquad \gamma = \eta^{ab}\,\gamma_{ab}\,. \tag{213}$$

To first order in γ_{ab}, Einstein's equation (211) becomes the **linearised Einstein equation**:

$$G^{(1)}_{ab} = -\tfrac{1}{2}\,\partial^c\partial_c\,\bar{\gamma}_{ab} + \partial^c\partial_{(a}\,\bar{\gamma}_{b)c} - \tfrac{1}{2}\,\eta_{ab}\,\partial^c\partial^d\,\bar{\gamma}_{cd} = 8\pi\,T_{ab}\,. \tag{214}$$

12.6.2. *Gauge freedom and the Lorenz gauge.* An infinitesimal diffeomorphism generated by a vector field ξ^a transforms the metric perturbation as

$$\gamma_{ab} \longrightarrow \gamma_{ab} + \partial_a\xi_b + \partial_b\xi_a\,. \tag{215}$$

This is the linearised version of the diffeomorphism invariance discussed in §12.3.1: it is the **gauge freedom** of linearised gravity, directly analogous to the gauge freedom $A_a \to A_a + \partial_a\Lambda$ of electromagnetism.

We exploit this freedom to simplify the field equations. Choose ξ^a satisfying the wave equation

$$\partial^b\partial_b\,\xi_a = -\partial^b\bar{\gamma}_{ab}\,. \tag{216}$$

After this gauge transformation, the **Lorenz gauge condition**

$$\partial^b\bar{\gamma}_{ab} = 0 \tag{217}$$

is satisfied, and the linearised Einstein equation (214) reduces to

$$\boxed{\partial^c\partial_c\,\bar{\gamma}_{ab} = -16\pi\,T_{ab}} \tag{218}$$

Remark. In vacuum ($T_{ab} = 0$), equation (218) is the wave equation $\Box\,\bar{\gamma}_{ab} = 0$ for a massless spin-2 field propagating on flat spacetime. Its solutions are **gravitational waves**—ripples in the fabric of spacetime—which we will study in a later lecture.

12.7. The Newtonian limit.

We now show that Einstein's equation reduces to Newtonian gravity in the appropriate limit, thereby fixing the constant 8π in (211) and providing a powerful consistency check.

12.7.1. *Assumptions.* The Newtonian limit is the regime where:

(a) **Gravity is weak**: the metric is well approximated by (212) with $|\gamma_{ab}| \ll 1$.
(b) **Motion is slow**: all velocities are much less than c.
(c) **Stresses are small**: material stresses (pressure) are much less than the mass-energy density ρ.

These assumptions allow us to choose a global inertial coordinate system in which the stress-energy tensor takes the form

$$T_{ab} \approx \rho\,t_a\,t_b\,, \tag{219}$$

where $t^a = (\partial/\partial x^0)^a$ is the unit time-like vector. Since sources are slowly varying, time derivatives of the perturbation $\bar{\gamma}_{ab}$ are negligible compared to spatial derivatives.

12.7.2. *Recovery of Poisson's equation.* Under these assumptions, the wave equation (218) reduces to the Laplace/Poisson equations:

$$\nabla^2 \bar{\gamma}_{\mu\nu} = -16\pi\, T_{\mu\nu}\,.$$

For $(\mu, \nu) \neq (0, 0)$, the right-hand side vanishes, giving $\nabla^2 \bar{\gamma}_{\mu\nu} = 0$. With the boundary condition $\bar{\gamma}_{\mu\nu} \to 0$ at spatial infinity, the unique solution is $\bar{\gamma}_{\mu\nu} = 0$ for $(\mu, \nu) \neq (0, 0)$.

The $(0, 0)$ component gives

$$\nabla^2 \bar{\gamma}_{00} = -16\pi\rho\,. \tag{220}$$

Defining the **Newtonian gravitational potential**

$$\phi = -\tfrac{1}{4}\,\bar{\gamma}_{00}\,, \tag{221}$$

equation (220) becomes

$$\boxed{\nabla^2 \phi = 4\pi\rho} \tag{222}$$

which is **Poisson's equation**—the fundamental field equation of Newtonian gravity.

The full metric perturbation in the Newtonian limit is

$$\gamma_{ab} = \bar{\gamma}_{ab} - \tfrac{1}{2}\,\eta_{ab}\,\bar{\gamma} = -(4\,t_a t_b + 2\,\eta_{ab})\,\phi\,, \tag{223}$$

giving the line element

$$ds^2 = -(1 + 2\phi)\,dt^2 + (1 - 2\phi)(dx^2 + dy^2 + dz^2)\,. \tag{224}$$

12.7.3. *Recovery of Newton's equation of motion.* It remains to show that freely falling particles obey Newton's law. The geodesic equation in coordinates reads

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu{}_{\rho\sigma}\,\frac{dx^\rho}{d\tau}\,\frac{dx^\sigma}{d\tau} = 0\,.$$

For a slowly moving test body, $dx^\mu/d\tau \approx (1, 0, 0, 0)$ and $\tau \approx t$, so

$$\frac{d^2 x^\mu}{dt^2} \approx -\Gamma^\mu{}_{00}\,. \tag{225}$$

From the metric (223), the relevant Christoffel symbols are (neglecting time derivatives):

$$\Gamma^i{}_{00} = -\tfrac{1}{2}\,\eta^{ij}\,\frac{\partial \gamma_{00}}{\partial x^j} = \frac{\partial \phi}{\partial x^i}\,, \qquad i = 1, 2, 3\,. \tag{226}$$

Substituting into (225):

$$\boxed{\frac{d^2 x^i}{dt^2} = -\frac{\partial \phi}{\partial x^i}} \tag{227}$$

In vector notation, $\boldsymbol{a} = -\nabla\phi$: a test particle accelerates in the direction of decreasing gravitational potential. This is precisely **Newton's law of gravitation**.

General relativity contains Newtonian gravity

In the limit of weak gravitational fields, slow motion, and small stresses, Einstein's field equations (211) reduce to Poisson's equation (222) for the gravitational potential, and the geodesic equation reduces to Newton's law of motion (227). Newtonian gravity is contained within general relativity as a limiting case.

The meaning of 8π

The factor of 8π in Einstein's equation (211) is not arbitrary: it is uniquely fixed by the requirement that the Newtonian limit reproduces Poisson's equation with the correct coefficient $4\pi\rho$. This is the quantitative content of Mach's principle in general relativity.

With Einstein's field equations in hand and the Newtonian limit verified, we have established the complete framework of general relativity. In the lectures that follow, we will explore the consequences of Einstein's equation in greater depth: its linearised solutions (gravitational radiation), exact solutions in the presence of spherical symmetry (the Schwarzschild solution), and cosmological models.

13. Properties of Einstein's equations and maps between manifolds

With Einstein's field equations (211) in hand, we now investigate their structure and consequences in greater detail. We begin by deriving useful algebraic identities—the trace of the field equations and the specialisation to pressureless matter—and then discuss three structural properties that distinguish general relativity from other field theories: the extreme nonlinearity of the equations, the fact that the stress-energy tensor cannot be regarded as an independent source, and the remarkable self-consistency by which the field equations contain the geodesic hypothesis. We then provide the detailed derivation of the linearised quantities used in the previous lecture, and develop the mathematical formalism of **maps between manifolds**—pushforward and pullback—which provides the rigorous foundation for the diffeomorphism invariance that underlies the gauge freedom of general relativity.

13.1. Algebraic consequences of the field equations.

A natural first step when presented with a new equation of motion is to explore what further equations can be derived from it by elementary operations. For a rank-two tensor equation, the most immediate operation is contraction with the metric.

13.1.1. *The trace and the alternate form.*

Contracting Einstein's equation (211) with g^{ab}:

$$g^{ab}G_{ab} = g^{ab}\left(R_{ab} - \tfrac{1}{2}\,R\,g_{ab}\right) = R - 2\,R = -R = 8\pi\,g^{ab}T_{ab} = 8\pi\,T\,,$$

where $T = T^a{}_a = g^{ab}\,T_{ab}$ is the trace of the stress-energy tensor. We obtain

$$\boxed{R = -8\pi\,T} \tag{228}$$

Substituting (228) back into (211) eliminates the Ricci scalar in favour of the stress-energy trace:

$$\boxed{R_{ab} = 8\pi\left(T_{ab} - \tfrac{1}{2}\,g_{ab}\,T\right)} \tag{229}$$

Remark. Equation (229) is sometimes more convenient than the original form (211), since the left-hand side involves only the Ricci tensor. It is particularly useful in settings of high symmetry where the trace T takes a simple form.

Geometry versus matter

Equation (229) appears to state that the geometry (encoded in R_{ab}) is determined by the matter content (encoded in T_{ab}). While suggestive, this interpretation must be treated with great care: the metric g_{ab} enters both sides of the equation, since it is needed to define

the stress-energy tensor itself, to raise and lower indices, and to compute the Ricci tensor. The information flows in both directions simultaneously—geometry constrains matter and matter constrains geometry—and one must solve the entire system self-consistently. We return to this point in §13.2.2 below.

13.1.2. *The dust limit.* Consider the physically important special case where an observer at rest with respect to the matter measures an energy density ρ that dominates all stresses. For a pressureless perfect fluid (**dust**) with four-velocity v^a:

$$T_{ab} = \rho\, v_a v_b\,, \tag{230}$$

so that $T = g^{ab}\, T_{ab} = \rho\, v^a v_a = -\rho$ (using the normalisation $v^a v_a = -1$). Substituting into (229) and contracting both sides with $v^a v^b$:

$$R_{ab}\, v^a v^b = 8\pi\big(T_{ab}\, v^a v^b - \tfrac{1}{2}\, T\, g_{ab}\, v^a v^b\big) = 8\pi\big(\rho - \tfrac{1}{2}\,\rho\big) = 4\pi\rho\,, \tag{231}$$

where we used $T_{ab}\, v^a v^b = \rho\,(v_a v^a)^2 = \rho$, $g_{ab}\, v^a v^b = -1$, and $T = -\rho$.

The result $R_{ab}\, v^a v^b = 4\pi\rho$ is the curved-spacetime analogue of the Newtonian relation $\nabla^2\phi = 4\pi\rho$: it expresses how the convergence of geodesics (quantified by the Ricci tensor contracted along the time direction) is driven by the local energy density.

13.2. Three structural properties of Einstein's equations.

Before we continue solving Einstein's equations, three features of the field equations deserve careful discussion. These features distinguish general relativity from other classical field theories and are essential for developing the correct physical intuition.

13.2.1. *Extreme nonlinearity.* Einstein's field equation (211) is a system of ten coupled, second-order, *highly nonlinear* partial differential equations for the ten independent components of the metric $g_{\mu\nu}$ (in any coordinate chart). The nonlinearity arises because the Christoffel symbols involve first derivatives of $g_{\mu\nu}$, the Riemann tensor involves products of Christoffel symbols (hence products of first derivatives), and the Ricci tensor involves derivatives of these products (hence second derivatives multiplied by the metric inverse, which is itself a nonlinear function of $g_{\mu\nu}$).

Comparison with other theories

The Schrödinger equation is linear; its difficulty lies in the enormous dimension of the Hilbert space. Maxwell's equations are also linear: given the sources, the electromagnetic field is uniquely determined by superposition. Einstein's equations enjoy neither of these simplifications. They are nonlinear in a way that makes even existence and uniqueness of solutions a delicate mathematical question. It must be said that any nonlinear equation can in principle be recast as an infinite system of linear equations—so one might argue that the difficulties of general relativity and quantum mechanics are not entirely incomparable—but in practice the nonlinearity of Einstein's equations makes them among the most challenging in all of physics.

13.2.2. *The stress-energy tensor is not a simple source.* In electrodynamics, the four-current J^a acts as a source for Maxwell's equations: one specifies J^a independently and then solves for the electromagnetic field. It is tempting to draw an analogy with general relativity and treat T_{ab} as a "source" that independently determines the metric. This analogy is *misleading*.

The stress-energy tensor of most physical systems—perfect fluids, electromagnetic fields, scalar fields—is itself defined in terms of the metric g_{ab}. For the perfect fluid (199), for instance,

T_{ab} depends on g_{ab} through the lowered four-velocity $u_a = g_{ab}\, u^b$ and through the metric appearing explicitly in $p\, g_{ab}$.

Self-consistent system

One cannot interpret T_{ab} before solving Einstein's equation. The metric and the matter fields must be determined simultaneously as a self-consistent solution of the coupled system. This stands in sharp contrast to electrodynamics, where the source can be specified independently of the field.

13.2.3. *Self-consistency: the geodesic hypothesis.* Recall that the fundamental hypotheses of general relativity (§12.1) include the statement that freely falling test bodies move on geodesics. Einstein's field equations imply the conservation law (200), $\nabla^a T_{ab} = 0$, as an automatic consequence of the Bianchi identity. Remarkably, this conservation law in turn implies the geodesic hypothesis for pressureless matter.

Consider a dust of non-interacting test particles with $T_{ab} = \rho\, u_a u_b$. Expanding the divergence:

$$\begin{aligned} 0 = \nabla^a T_{ab} &= \nabla^a(\rho\, u_a u_b) \\ &= u_b\, \nabla^a(\rho\, u_a) + \rho\, u_a\, \nabla^a u_b\,. \end{aligned} \tag{232}$$

Contract (300) with u^b. Since $u^b u_b = -1$, differentiating gives $u^b\, \nabla_c u_b = 0$ for any direction c, and hence $u^b\, \rho\, u_a\, \nabla^a u_b = 0$. The contraction therefore yields

$$-\nabla^a(\rho\, u_a) = 0\,.$$

This is the **continuity equation**: the mass current $\rho\, u^a$ is conserved. Substituting back into (300):

$$\rho\, u^a\, \nabla_a u_b = 0\,. \tag{233}$$

Wherever $\rho \neq 0$, this gives $u^a \nabla_a u^b = 0$: *the dust particles follow geodesics*.

Einstein's equations contain the geodesic hypothesis

For a pressureless dust, the conservation law $\nabla^a T_{ab} = 0$—which is an automatic consequence of Einstein's field equations via the Bianchi identity—implies that the dust grains move on geodesics of the spacetime metric. The geodesic hypothesis is not an independent assumption: it is contained within the field equations themselves.

Self-consistency as evidence for correctness

This kind of self-consistency is extraordinarily rare in physics. We *began* by postulating that freely falling test bodies move on geodesics, and used this together with physical reasoning to arrive at Einstein's field equations. The field equations then turn around and *imply* the geodesic hypothesis, without it having been explicitly built in. This "bootstrapping" property is one of the strongest pieces of internal evidence that general relativity is on the right track: usually when one writes down equations of motion for a complex system, checking them for self-consistency with auxiliary assumptions reveals contradictions. That Einstein's equations pass this test is, as one might say, breathtakingly impressive.

13.3. Detailed derivation of the linearised field equations.

Solving Einstein's field equations is extraordinarily difficult, and the task will occupy us for the remainder of this course. We will

approach solutions through three limiting regimes: *weak gravitational fields* (linearised gravity and gravitational radiation), *high symmetry* (the Schwarzschild solution), and *cosmological uniformity* (Friedmann–Lemaître–Robertson–Walker models).

In the previous lecture (§12.6) we stated the linearised Einstein equation and used it to recover the Newtonian limit. We now provide the step-by-step derivation, filling in the computational details.

13.3.1. *Setup and index conventions.* Recall the weak-field ansatz (212): $g_{ab} = \eta_{ab} + \gamma_{ab}$ with $|\gamma_{ab}| \ll 1$.

Remark (The meaning of "small"). In a coordinate-free formulation, the statement "γ_{ab} is small" has no intrinsic meaning: one must specify a standard of comparison. For linearised gravity, we define smallness relative to a *global inertial coordinate system*—the coordinates in which the background metric takes the form $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$—and require that all components $\gamma_{\mu\nu}$ be much less than 1 in this system. The very existence of such a global coordinate system is a non-trivial assumption: it restricts both the topology of $\mathcal{M}$ and the strength of the gravitational field.

Throughout this section, all indices are raised and lowered with the flat metric η_{ab}, with one important exception: g^{ab} always denotes the *true inverse* of the full metric. To linear order, the inverse metric is

$$g^{ab} = \eta^{ab} - \gamma^{ab} + O(\gamma^2)\,. \tag{234}$$

Exercise 13.1. Verify (234) by showing that $g^{ab}\, g_{bc} = \delta^a{}_c$ to linear order:

$$(\eta^{ab} - \gamma^{ab})(\eta_{bc} + \gamma_{bc}) = \delta^a{}_c - \gamma^a{}_c + \gamma^a{}_c - \gamma^{ab}\gamma_{bc} = \delta^a{}_c + O(\gamma^2)\,.$$

13.3.2. *Christoffel symbols to linear order.* The Christoffel symbols of the Levi-Civita connection are

$$\Gamma^c{}_{ab} = \tfrac{1}{2}\, g^{cd}\big(\partial_a g_{bd} + \partial_b g_{ad} - \partial_d g_{ab}\big)\,.$$

Substituting $g_{ab} = \eta_{ab} + \gamma_{ab}$ and $g^{cd} = \eta^{cd} - \gamma^{cd}$, and noting that $\partial_a \eta_{bc} = 0$:

$$\Gamma^c{}_{ab} = \tfrac{1}{2}\, \eta^{cd}\big(\partial_a \gamma_{bd} + \partial_b \gamma_{ad} - \partial_d \gamma_{ab}\big) + O(\gamma^2)\,. \tag{235}$$

The γ^{cd} contribution to g^{cd} would multiply terms already of order γ, producing $O(\gamma^2)$ corrections that we discard.

13.3.3. *The Ricci and Einstein tensors to linear order.* The Riemann tensor in coordinates (191) involves terms linear in derivatives of Γ_{ab} and terms quadratic in Γ_{ab}. Since the Christoffel symbols are already $O(\gamma)$, the quadratic terms are $O(\gamma^2)$ and may be dropped. The Ricci tensor to linear order is therefore

$$R^{(1)}_{ab} = \partial_c\, \Gamma^c{}_{ab} - \partial_a\, \Gamma^c{}_{cb}\,. \tag{236}$$

Substituting (235) and simplifying (where $\gamma = \gamma^c{}_c = \eta^{cd}\, \gamma_{cd}$ denotes the trace):

$$R^{(1)}_{ab} = \partial^c \partial_{(a}\, \gamma_{b)c} - \tfrac{1}{2}\, \partial^c \partial_c\, \gamma_{ab} - \tfrac{1}{2}\, \partial_a \partial_b\, \gamma\,. \tag{237}$$

Exercise 13.2. Derive (237) from (235) and (236).

To linear order, $G^{(1)}_{ab} = R^{(1)}_{ab} - \frac{1}{2}\, \eta_{ab}\, R^{(1)}$ (the metric multiplying the Ricci scalar may be replaced by η_{ab}, since $\gamma_{ab}\, R^{(1)}$ is $O(\gamma^2)$). Computing $R^{(1)} = \eta^{ab}\, R^{(1)}_{ab}$ and combining terms:

$$G^{(1)}_{ab} = \partial^c \partial_{(a}\, \gamma_{b)c} - \tfrac{1}{2}\, \partial^c \partial_c\, \gamma_{ab} - \tfrac{1}{2}\, \partial_a \partial_b\, \gamma - \tfrac{1}{2}\, \eta_{ab}\big(\partial^c \partial^d\, \gamma_{cd} - \partial^c \partial_c\, \gamma\big)\,. \tag{238}$$

This expression simplifies considerably when written in terms of the trace-reversed perturbation (213), $\bar{\gamma}_{ab} = \gamma_{ab} - \frac{1}{2}\, \eta_{ab}\, \gamma$:

$$G^{(1)}_{ab} = -\tfrac{1}{2}\, \partial^c \partial_c\, \bar{\gamma}_{ab} + \partial^c \partial_{(a}\, \bar{\gamma}_{b)c} - \tfrac{1}{2}\, \eta_{ab}\, \partial^c \partial^d\, \bar{\gamma}_{cd}\,. \tag{239}$$

Setting $G^{(1)}_{ab} = 8\pi\, T_{ab}$ recovers the linearised Einstein equation (214) stated in the previous lecture.

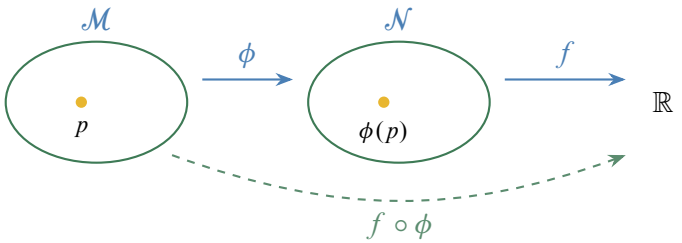
Exercise 13.3. Verify that substituting $\gamma_{ab} = \bar{\gamma}_{ab} + \frac{1}{2}\,\eta_{ab}\,\bar{\gamma}$ (where $\bar{\gamma} = \eta^{ab}\,\bar{\gamma}_{ab} = -\gamma$) into (238) yields (239).

13.4. Maps between manifolds. The gauge freedom of linearised gravity (§12.6.2) arose from infinitesimal diffeomorphisms. To place this on firm mathematical footing, we now develop the theory of maps between manifolds and their action on geometric objects. This formalism also underpins the Lie derivative introduced in §12.3, where we specialised to the case of a diffeomorphism from $\mathcal{M}$ to itself generated by a vector field.

13.4.1. *Pullback of functions.* Let $\mathcal{M}$ and $\mathcal{N}$ be smooth manifolds and $\phi\colon \mathcal{M} \to \mathcal{N}$ a smooth map. Given any smooth function $f\colon \mathcal{N} \to \mathbb{R}$, the composition

$$\phi^* f = f \circ \phi\colon \mathcal{M} \to \mathbb{R} \tag{240}$$

defines a smooth function on $\mathcal{M}$, called the **pullback** of f by ϕ.



The pullback operates in the direction *opposite* to ϕ: while ϕ maps $\mathcal{M} \to \mathcal{N}$, the pullback ϕ^* maps functions on $\mathcal{N}$ to functions on $\mathcal{M}$.

13.4.2. *Pushforward of tangent vectors.* For any tangent vector $v \in T_p\mathcal{M}$, we define the **pushforward** $\phi_* v \in T_{\phi(p)}\mathcal{N}$ by its action on smooth functions f on $\mathcal{N}$:

$$(\phi_* v)(f) = v(f \circ \phi)\,. \tag{241}$$

Exercise 13.4. Verify that $\phi_* v$ so defined is a tangent vector at $\phi(p)$: check that it is $\mathbb{R}$-linear and satisfies the Leibniz rule with respect to pointwise multiplication of functions on $\mathcal{N}$.

The pushforward defines a linear map

$$\phi_*\colon T_p\mathcal{M} \longrightarrow T_{\phi(p)}\mathcal{N}\,. \tag{242}$$

Relation to the Jacobian. Let $\{x^\mu\}$ be coordinates on $\mathcal{M}$ and $\{y^\nu\}$ coordinates on $\mathcal{N}$. The matrix of ϕ_* in the corresponding coordinate bases is

$$[\phi_*]^\nu{}_\mu = \frac{\partial y^\nu}{\partial x^\mu}\,, \tag{243}$$

which is the **Jacobian matrix** of the map ϕ.

Exercise 13.5. Derive (243) from (241) by evaluating $\big(\phi_*\, \partial/\partial x^\mu\big)(y^\nu)$.

13.4.3. *Pullback of covectors.* Given a covector $\mu \in T^*_{\phi(p)}\mathcal{N}$, the **pullback covector** $\phi^*\mu \in T^*_p\mathcal{M}$ is defined by demanding consistency with the natural pairing:

$$(\phi^*\mu)(v) = \mu(\phi_* v) \qquad \text{for all } v \in T_p\mathcal{M}\,. \tag{244}$$

This defines a linear map $\phi^*\colon T^*_{\phi(p)}\mathcal{N} \to T^*_p\mathcal{M}$, going in the direction *opposite* to ϕ_*.

13.4.4. *Extension to pure tensors.* The pullback extends naturally to covariant tensors of type $(0, l)$. For $T_{a_1 \cdots a_l}$ a tensor on $\mathcal{N}$:

$$(\phi^* T)_{a_1 \cdots a_l}\, v_1^{a_1} \cdots v_l^{a_l} = T_{a_1 \cdots a_l}\, (\phi_* v_1)^{a_1} \cdots (\phi_* v_l)^{a_l} \tag{245}$$

for all $v_1, \ldots, v_l \in T_p\mathcal{M}$. Similarly, the pushforward extends to contravariant tensors of type $(k, 0)$. For $T^{a_1 \cdots a_k}$ a tensor on $\mathcal{M}$:

$$(\phi_* T)^{a_1 \cdots a_k}\, (\mu_1)_{a_1} \cdots (\mu_k)_{a_k} = T^{a_1 \cdots a_k}\, (\phi^* \mu_1)_{a_1} \cdots (\phi^* \mu_k)_{a_k} \tag{246}$$

for all $\mu_1, \ldots, \mu_k \in T^*_{\phi(p)}\mathcal{N}$.

Remark. Note the directions of these maps: ϕ_* pushes tangent vectors and contravariant tensors *forward* ($\mathcal{M} \to \mathcal{N}$), while ϕ^* pulls covectors and covariant tensors *back* ($\mathcal{N} \to \mathcal{M}$). For a general smooth map, there is no natural action on *mixed* tensors of type (k, l) with $k, l \geq 1$, because the pushforward and pullback point in opposite directions.

13.4.5. *Mixed tensors and diffeomorphisms.* If $\phi\colon \mathcal{M} \to \mathcal{M}$ is a **diffeomorphism** (a smooth bijection with smooth inverse), we can overcome the arrow-direction obstruction. Since ϕ^{-1} exists and is smooth, we have an additional pushforward

$$(\phi^{-1})_*\colon T_{\phi(p)}\mathcal{M} \longrightarrow T_p\mathcal{M}$$

that takes tangent vectors at $\phi(p)$ *back* to p. This allows us to define the pullback of a mixed tensor $T^{a_1 \cdots a_k}{}_{b_1 \cdots b_l}$ by:

$$\begin{aligned} &(\phi^* T)^{a_1 \cdots a_k}{}_{b_1 \cdots b_l}\, (\mu_1)_{a_1} \cdots (\mu_k)_{a_k}\, v_1^{b_1} \cdots v_l^{b_l} \\ &\qquad = T^{a_1 \cdots a_k}{}_{b_1 \cdots b_l}\, \big((\phi^{-1})^* \mu_1\big)_{a_1} \cdots \big((\phi^{-1})^* \mu_k\big)_{a_k}\, (\phi_* v_1)^{b_1} \cdots (\phi_* v_l)^{b_l}\,, \end{aligned} \tag{247}$$

for all $v_1, \ldots, v_l \in T_p\mathcal{M}$ and $\mu_1, \ldots, \mu_k \in T^*_p\mathcal{M}$.

Exercise 13.6. Show that $\phi_* = (\phi^{-1})^*$ when both are defined on tangent vectors. *Hint*: for $v \in T_p\mathcal{M}$ and any smooth function f, evaluate $\big((\phi^{-1})^* v\big)(f)$ using the pullback definition.

13.4.6. *Isometries.* Among all diffeomorphisms, those that preserve the metric occupy a special role.

Definition 13.1 (Isometry)**.** A diffeomorphism $\phi\colon \mathcal{M} \to \mathcal{M}$ is an **isometry** if the pullback of the metric equals the metric:

$$\phi^* g_{ab} = g_{ab}\,. \tag{248}$$

An isometry preserves all distances and angles: the spacetimes $(\mathcal{M}, g_{ab})$ and $(\mathcal{M}, \phi^* g_{ab})$ are geometrically identical. In Minkowski space, the isometries form the Poincaré group (translations and Lorentz transformations).

Symmetries versus gauge freedom

The isometries of a spacetime are its *symmetries*: they generate conserved quantities via the Killing vector fields of §12.4. However, the *gauge freedom* of general relativity is much larger: *all* diffeomorphisms (not just isometries) map solutions of Einstein's equations to physically equivalent solutions. The spacetimes $(\mathcal{M}, g_{ab})$ and $(\mathcal{M}, \phi^* g_{ab})$ describe the same physics for *any* diffeomorphism ϕ, regardless of whether ϕ is an isometry.

Isometries versus diffeomorphisms

Think of an isometry as a rigid motion of the manifold—it moves points around but preserves the "shape" of spacetime. A general diffeomorphism may distort the metric, but

because general relativity is formulated in terms of coordinate-free geometric quantities, the distorted and undistorted spacetimes make identical physical predictions. The distinction is analogous to that between a rotation (which preserves a vector's length) and a general coordinate transformation (which changes the components but not the vector itself).

With the formalism of maps between manifolds in hand, we have placed the gauge freedom of general relativity—and in particular the gauge transformations of linearised gravity (215)—on a rigorous mathematical footing. In the next lecture, we will apply the linearised theory developed in Lectures 12 and 13 to study **gravitational radiation**: wave-like solutions of the vacuum linearised Einstein equations that propagate at the speed of light.

14. Gravitational radiation

In the previous two lectures we linearised Einstein's field equations around flat spacetime and showed that in the Lorenz gauge (217) they reduce to the wave equation (218), $\Box\,\bar{\gamma}_{ab} = -16\pi\, T_{ab}$. We then studied the *static* Newtonian limit, where time derivatives are negligible, and recovered Newton's law of gravitation.

In general relativity, however, the gravitational field is a *dynamical* entity: unlike in Newtonian theory, the geometry itself can change with time. Just as the electromagnetic field in Maxwell's theory supports wave-like solutions, the dynamical nature of the gravitational field opens the possibility of **gravitational radiation**—wave-like solutions of the linearised vacuum Einstein equations that propagate at the speed of light. We can already study gravitational radiation at the linearised level of approximation; the full nonlinear theory adds corrections but the essential physics is captured here.

Gravitational waves as ripples in spacetime

In electrodynamics, accelerating charges produce electromagnetic waves. In general relativity, accelerating masses produce gravitational waves: ripples in the spacetime metric that propagate outward at speed c. Just as electromagnetic waves carry energy and momentum, gravitational waves carry energy away from their source, causing (for instance) binary star systems to spiral inward. The key difference is that gravitational waves are perturbations of the *metric itself*, not of a field propagating *on* a fixed background.

14.1. Residual gauge freedom.

Recall from §12.6.2 that in the Lorenz gauge we have:

(249) Gauge condition: $$\partial^a \bar{\gamma}_{ab} = 0\,,$$

(250) Field equation: $$\partial^c \partial_c\, \bar{\gamma}_{ab} = -16\pi\, T_{ab}\,.$$

The Lorenz condition does *not* completely fix the gauge. We are still free to perform gauge transformations

(251) $$\gamma_{ab} \longrightarrow \gamma_{ab} + \partial_a \xi_b + \partial_b \xi_a$$

provided the vector field ξ^a satisfies

(252) $$\partial^b \partial_b\, \xi^a = 0\,.$$

This is precisely the condition that ensures the new perturbation still satisfies the Lorenz gauge (249).

Remark. The situation is directly analogous to electrodynamics in the Lorenz gauge $\partial_\mu A^\mu = 0$: one may still perform gauge transformations $A_\mu \to A_\mu + \partial_\mu \Lambda$ provided $\Box\,\Lambda = 0$. The residual gauge freedom allows us to impose additional simplifying conditions on the perturbation.

14.2. The radiation gauge (transverse-traceless gauge). We exploit the residual freedom (252) to impose the **radiation gauge** conditions:

$$\gamma = 0\,, \qquad \gamma_{0j} = 0\,, \qquad j = 1,2,3\,, \tag{253}$$

where $\gamma = \gamma^a{}_a = \eta^{ab}\gamma_{ab}$ is the trace. The condition $\gamma = 0$ implies $\bar{\gamma}_{ab} = \gamma_{ab}$ (the trace-reversal is trivial when the trace vanishes), and the Lorenz condition $\partial^a\bar{\gamma}_{ab} = 0$ then further constrains the spatial components.

The radiation gauge is achieved by solving an initial-value problem for ξ^μ on a fixed initial surface $t = t_0$. One solves:

$$2\Big(-\frac{\partial\xi_0}{\partial t} + \nabla\cdot\boldsymbol{\xi}\Big) = -\gamma\,, \qquad \text{(fixes trace)} \tag{254}$$

$$\frac{\partial\xi_\mu}{\partial t} + \frac{\partial\xi_0}{\partial x^\mu} = -\gamma_{0\mu}\,, \qquad \mu = 1,2,3\,, \tag{255}$$

$$\nabla^2\xi_\mu + \frac{\partial}{\partial x^\mu}\Big(\frac{\partial\xi_0}{\partial t}\Big) = -\frac{\partial\gamma_{0\mu}}{\partial t}\,, \qquad \mu = 1,2,3\,. \tag{256}$$

Given initial data ξ_μ and $\partial_t\xi_\mu$ on $t = t_0$, one solves the wave equation $\Box\,\xi^a = 0$ to propagate the gauge transformation throughout spacetime. One then verifies (exercise) that the radiation gauge conditions hold not just on the initial surface, but at all times.

Complete gauge fixing in vacuum

If $T_{ab} = 0$ **everywhere** (not just in a region), then one can further achieve

$$\gamma_{00} = 0\,. \tag{257}$$

This follows because $\gamma = 0$ implies $\bar{\gamma}_{0b} = \gamma_{0b}$, and the Lorenz condition (249) then gives $\partial_t\gamma_{00} = 0$. The field equation (250) with $T_{00} = 0$ reduces to $\nabla^2\gamma_{00} = 0$, whose unique bounded solution is $\gamma_{00} = \text{const}$. A constant shift of γ_{00} is always permissible, so we set $\gamma_{00} = 0$. In this **transverse-traceless (TT) gauge**, the only non-vanishing components of γ_{ab} are the purely spatial ones γ_{ij}, which are traceless and transverse ($\partial^i\gamma_{ij} = 0$).

14.3. Plane wave solutions. We now seek wave-like solutions of the vacuum equation $\Box\,\gamma_{ab} = 0$ (recall $\bar{\gamma} = \gamma$ in the radiation gauge). The way we find waves is the way we *always* find waves: we make a plane-wave ansatz and substitute it into the field equations to see what conditions emerge. Specifically:

$$\gamma_{ab} = H_{ab}\,e^{i\,k_\mu x^\mu}\,, \tag{258}$$

where H_{ab} is a constant symmetric tensor (the **polarization tensor**) and k_μ is the wave four-vector.

Substituting into the wave equation $\Box\,\gamma_{ab} = 0$:

$$\boxed{k^\mu k_\mu = 0} \tag{259}$$

—the wave vector is **null**. Gravitational waves propagate at the speed of light.

The gauge conditions (249) and (253) impose the following constraints on the polarization tensor:

$$\text{(a)}\quad k^\mu\,H_{\mu\nu} = 0\,, \qquad \text{(Lorenz gauge)} \tag{260}$$

$$\text{(b)}\quad H_{0\nu} = 0\,, \qquad \text{(radiation gauge)} \tag{261}$$

$$\text{(c)}\quad H^\mu{}_\mu = 0\,. \qquad \text{(tracelessness)} \tag{262}$$

14.3.1. *Counting the degrees of freedom.* A symmetric $(0, 2)$ tensor in four dimensions has 10 independent components. How many survive the gauge and transversality conditions?

- Condition (a): 4 equations (one for each value of ν).
- Condition (b): 4 equations ($\nu = 0, 1, 2, 3$).
- Condition (c): 1 equation.
- Total: 9 equations, but (a) and (b) together imply $H_{0\nu}\, k^\nu = 0$, so only **8** are independent.

Two polarizations

A plane gravitational wave has

$$10 - 8 = 2 \tag{263}$$

linearly independent polarization states. These are the $+$ ("plus") and $\times$ ("cross") polarizations. Two is bigger than zero! It is a striking prediction of Einstein's theory that the gravitational field has dynamical, wave-like solutions—a feature entirely absent from Newtonian gravity.

Exercise 14.1. Show that conditions (a) and (b) are not independent by verifying that both imply $H_{0\nu}\, k^\nu = 0$.

14.4. No gravitational waves in $(2+1)$ dimensions. The degree-of-freedom count is dimension-dependent, and it is particularly interesting—and highly relevant for current investigations into quantum gravity—to ask whether gravitational waves exist in lower dimensions. In $(2+1)$-dimensional spacetime:

- A symmetric $(0, 2)$ tensor has 6 independent components.
- The gauge conditions impose 6 independent constraints.
- Result: $6 - 6 = 0$ degrees of freedom—**no gravitational waves** in $(2+1)$ dimensions.

We could, in fact, have guessed this result without any calculation. In $(2+1)$ dimensions, the Riemann tensor is completely determined by the Ricci tensor and scalar:

$$R_{abcd} = g_{ac}\, R_{bd} + g_{bd}\, R_{ac} - g_{bc}\, R_{ad} - g_{ad}\, R_{bc} - \tfrac{1}{2}\,(g_{ac}\, g_{bd} - g_{ad}\, g_{bc})\, R\,. \tag{264}$$

This identity holds because the **Weyl tensor** C_{abcd} (the completely trace-free part of the Riemann tensor) vanishes identically in three dimensions. To see why, recall that the Riemann tensor may be decomposed into a part determined by R_{ab} and R, plus a completely trace-free remainder—the Weyl tensor. In $(2+1)$ dimensions a component count shows that the Weyl tensor has zero independent components, so it must vanish.

Exercise 14.2. Verify the vanishing of the Weyl tensor in $(2+1)$ dimensions by counting the independent components of the Riemann tensor (using its symmetries) and comparing with the number of independent components of the Ricci tensor.

Why the Weyl tensor matters

The Riemann tensor can be decomposed into a part determined by the Ricci tensor (and hence, via Einstein's equations, by the local matter content) and a trace-free part—the Weyl tensor—that encodes the "free" gravitational field. In four dimensions, the Weyl tensor has 10 independent components, which carry the two polarization degrees of freedom of gravitational waves. In three dimensions, the Weyl tensor has 0 components: there is no room for freely propagating gravitational degrees of freedom.

The vacuum Einstein equations in (2+1) dimensions give $R_{ab} = 0$ and hence $R = 0$; equation (264) then forces $R_{abcd} = 0$—the spacetime is *flat*. Gravity in (2+1) dimensions has no local degrees of freedom whatsoever.

14.5. Experimental detection. It is impossible to discuss gravitational radiation without mentioning one of the most dramatic discoveries of modern physics.

LIGO and the first direct detection

Gravitational waves were predicted by Einstein in 1916 and indirectly confirmed by the Hulse–Taylor binary pulsar observations (Nobel Prize 1993), which showed the orbital decay rate matching the energy loss to gravitational radiation predicted by general relativity. The first *direct* detection of gravitational waves was announced on 11 February 2016 by the LIGO and Virgo collaborations. The signal, designated GW150914, was detected on 14 September 2015 and matched the waveform predicted for the inspiral and merger of two black holes of approximately 36 and 29 solar masses. The detection was made possible by a laser interferometer with arm lengths of 4 km, capable of measuring displacements of order 10^{-18} m—roughly one-thousandth the diameter of a proton. A significant portion of the data analysis was carried out in Hannover, Germany, at the Albert Einstein Institute. The 2017 Nobel Prize in Physics was awarded to Rainer Weiss, Barry Barish, and Kip Thorne for this discovery.

15. Homogeneous isotropic cosmology

We now turn from perturbative solutions to a far more ambitious programme: finding *exact* solutions of Einstein's field equations that describe the large-scale structure of the entire universe. Our goal is to be as bold as possible—to extend the domain of applicability of Einstein's equations to the cosmos as a whole, to make a model of how matter is distributed throughout the universe, and then to solve the field equations.

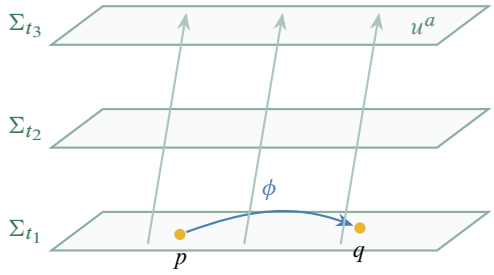
Obviously we cannot do this for an arbitrary matter distribution: Einstein's equations are far too complicated. Instead we impose massive simplifications in the form of symmetry assumptions. The key observational fact is that, on sufficiently large scales (hundreds of megaparsecs), the distribution of matter in the universe appears remarkably uniform: whether a star is here or there does not matter if one averages on galactic length scales. There is no preferred location and no preferred direction. We formalise these observations as the assumptions of **homogeneity** and **isotropy**.

Why we can model the universe

Obviously the universe is *not* homogeneous and isotropic on small scales—galaxies, stars, and planets manifestly break the symmetry. But on scales much larger than the typical separation between galaxies ($\sim$ 1 Mpc), the matter distribution averages out and looks very nearly uniform. The most compelling evidence for isotropy is the cosmic microwave background radiation, which is uniform to one part in 10^5 across the sky. We will exploit this large-scale uniformity to reduce Einstein's ten coupled PDEs to a small number of ordinary differential equations.

15.1. Homogeneity.

Definition 15.1 (Homogeneous spacetime)**.** A spacetime $(\mathcal{M}, g_{ab})$ is **homogeneous** if there exists a one-parameter foliation of $\mathcal{M}$ by spacelike hypersurfaces Σ_t such that, for each t and any two points $p, q \in \Sigma_t$, there exists an isometry of g_{ab} that maps p to q.



Physically, homogeneity says: *we do not occupy a privileged position in the universe*. At any given cosmological time t, the geometry of the spatial slice Σ_t looks the same at every point.

15.2. Isotropy. Before defining isotropy, we note an important subtlety. Imagine the universe is filled with an isotropic distribution of matter and there is at least one observer at a point p who sees the universe as isotropic. Can there be another observer at the same point who also sees isotropy? The answer is *no*: any boosted observer perceives the background matter distribution as distorted—thanks to length contraction, the stress-energy appears concentrated in the directions perpendicular to the boost, destroying isotropy. Therefore, at each point there is *at most one* observer who perceives the universe as isotropic, and our definition must refer to this preferred family of observers.

Definition 15.2 (Spatially isotropic spacetime)**.** A spacetime $(\mathcal{M}, g_{ab})$ is **spatially isotropic at each point** if there exists a congruence of timelike curves filling $\mathcal{M}$ with tangent vector field u^a such that, for all $p \in \mathcal{M}$ and any two unit spacelike vectors $s_1^a, s_2^a \in T_p\mathcal{M}$ orthogonal to u^a (i.e. $u^a s_{1\,a} = u^a s_{2\,a} = 0$), there exists an isometry of g_{ab} that leaves p and u^a fixed but maps $s_1^a \mapsto s_2^a$.

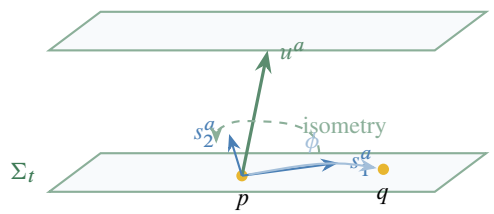
Isotropy says: *there is no preferred spatial direction*. No matter which way the preferred observer looks, the universe appears the same.

15.3. Combining homogeneity and isotropy.

Orthogonality of foliation and observers

For a spacetime that is both homogeneous and isotropic, the hypersurfaces Σ_t of the homogeneity foliation must be everywhere orthogonal to the tangent vectors u^a of the isotropic observers.

If this were not the case, the component of u^a tangent to Σ_t would define a geometrically preferred spatial direction at each point, violating isotropy.



The combination of homogeneity and isotropy is extraordinarily powerful. It implies that the spatial geometry of each slice Σ_t is a *maximally symmetric* three-dimensional Riemannian manifold—and there are only three possibilities: flat space ($\mathbb{R}^3$), the three-sphere (S^3),

or hyperbolic space (H^3). In the next lecture, we will exploit this to derive the Friedmann–Lemaître–Robertson–Walker metric and the Friedmann equations that govern the expansion of the universe.

15.4. Consequences of homogeneity and isotropy for the metric. In the previous section we defined what it means for a spacetime to be homogeneous and isotropic. Our goal now is to determine the consequences of these symmetry assumptions for the fundamental dynamical variable of general relativity: the metric g_{ab}. The hope—which will be amply justified—is that by assuming the symmetries of homogeneity and isotropy the metric is simplified to the point where Einstein's field equations become tractable. Indeed, we will reduce the problem of finding the metric of a highly symmetric model universe to that of finding *a single scalar function* $a(\tau)$.

15.4.1. *The induced metric on the spatial slices.* The full spacetime metric g_{ab} induces, via restriction, a three-dimensional Riemannian metric $h_{ab}(t)$ on each spatial hypersurface Σ_t of the homogeneity foliation. Rather than attacking the full four-dimensional problem at once, we first study the consequences of homogeneity and isotropy for this induced metric h_{ab}. Later we will embed h_{ab} back into the full metric g_{ab} and argue that very little additional freedom remains.

For the induced metric, the symmetry conditions take the following concrete forms:

- **Homogeneity**: there exists an isometry of $h_{ab}(t)$ taking any point $p \in \Sigma_t$ to any other point $q \in \Sigma_t$.
- **Isotropy**: it is impossible to construct a geometrically preferred vector in Σ_t.

We will attack the second constraint (isotropy) first and use homogeneity later to fix a remaining constant.

15.4.2. *The Riemann tensor as a linear map on two-forms.* The beauty of having developed differential geometry in arbitrary dimension in the earlier lectures is that we can immediately apply it to three-dimensional manifolds. From the induced metric h_{ab} on Σ_t we construct the three-dimensional Riemann curvature tensor $R^{(3)}{}_{abc}{}^{d}$ using the standard rules (§10.5). Raising an index with the induced metric gives

$$R^{(3)cd}_{ab} = R^{(3)d}_{ab}{}_{c'}\, h^{c'c}\,, \tag{265}$$

which we may interpret as a **linear map**

$$L\colon \mathcal{T}(0,2)\big|_p \longrightarrow \mathcal{T}(0,2)\big|_p$$

at each point $p \in \Sigma_t$.

Exercise 15.1. Show that $R^{(3)cd}_{ab}$ annihilates symmetric tensors in $\mathcal{T}(0,2)$. Conclude that L restricts to a linear map

$$L\colon W \to W\,,$$

where W is the space of **two-forms** (antisymmetric $(0,2)$ tensors) at p.

The pair-interchange symmetry of the Riemann tensor, $R_{abcd} = R_{cdab}$, implies that L is a **symmetric** linear map with respect to the inner product on W furnished by h_{ab}. On an inner product space, a symmetric linear map can be diagonalised and has real eigenvalues.

Dimension count. On a three-dimensional manifold, the space of two-forms has dimension $\binom{3}{2} = 3$, so L is a 3×3 symmetric matrix with three real eigenvalues.

15.4.3. *Isotropy forces equal eigenvalues.*

All eigenvalues of L are equal

Isotropy requires that all three eigenvalues of L are *equal*.

If any eigenvalue were distinct, its eigenvector—a two-form selected by a basis-independent prescription—would define a geometrically preferred two-form at p. But on a three-dimensional manifold, any two-form can be completed to a unique three-form (up to sign), and a three-form together with the metric determines a preferred vector via the Hodge dual. This would produce a geometrically preferred vector in Σ_t, contradicting isotropy.

Since all eigenvalues are equal, L is proportional to the identity: $L \propto \mathbb{I}$. Translating back to the Riemann tensor with raised indices:

$$R^{(3)cd}_{ab} = K\,\delta^c{}_{[a}\,\delta^d{}_{b]}\,, \tag{266}$$

for some scalar K. Lowering the indices c and d:

$$\boxed{R^{(3)}_{abcd} = K\big(h_{ca}\,h_{db} - h_{cb}\,h_{da}\big)} \tag{267}$$

15.4.4. *K is constant: the Bianchi identity argument.* Is the scalar K in (267) a function of position on Σ_t, or is it constant? Homogeneity immediately tells us it must be constant (an isometry mapping p to q preserves the Riemann tensor, hence $K(p) = K(q)$). But we can also prove this from isotropy alone, using the Bianchi identity.

Apply the three-dimensional covariant derivative D_e (associated to h_{ab}) to (267) and antisymmetrise:

$$0 = D_{[e}\,R^{(3)}_{ab]cd} = (D_{[e}\,K)\,h_{|c|a}\,h_{d|b]}$$

where we used $D_e\,h_{ab} = 0$ (metric compatibility). For $\dim(\Sigma_t) \geq 3$, the antisymmetrised product of metric factors on the right is non-degenerate, so the vanishing of the left-hand side requires

$$D_e\,K = 0\,. \tag{268}$$

Therefore K is *constant* on each Σ_t.

Spaces of constant curvature

A Riemannian manifold whose Riemann tensor takes the form (267) with K constant is called a **space of constant curvature**. The physical content of this result is transparent: recall that the Riemann tensor measures the failure of parallel transport operations to commute. On a space of constant curvature, no matter which way you perform the little parallelogram of parallel transport, you get the same failure to commute—the curvature is the same in every direction and at every point. This is exactly what one would expect from homogeneity and isotropy.

15.5. The three cases: representative spatial geometries.

We have established that the spatial slices Σ_t are spaces of constant curvature K. There are three qualitatively distinct cases, classified by the sign of K. In each case one can write down an explicit representative manifold and metric.

15.5.1. *$K > 0$: the three-sphere (closed universe).* The spatial slices are three-spheres S^3 of radius R (with $K = 1/R^2$), defined as embedded submanifolds of $\mathbb{R}^4$:

$$x^2 + y^2 + z^2 + w^2 = R^2\,.$$

(The ambient $\mathbb{R}^4$ is a mathematical convenience; the three-spheres are intrinsically defined three-dimensional manifolds.) In spherical coordinates (ψ, θ, ϕ) the induced metric reads:

$$ds^2_\Sigma = d\psi^2 + \sin^2\psi\,\big(d\theta^2 + \sin^2\theta\,d\phi^2\big)\,. \tag{269}$$

This describes a **closed** universe: the spatial sections are compact (finite volume, no boundary), so everything is connected to everything else—there is no "going off to infinity."

15.5.2. *$K = 0$: flat space (open universe).* The spatial slices are copies of flat $\mathbb{R}^3$:

$$ds_\Sigma^2 = dx^2 + dy^2 + dz^2 \,. \tag{270}$$

This is called an **open** universe (or **flat** universe): the spatial sections are non-compact.

Exercise 15.2. Rewrite the flat metric (270) in spherical coordinates (r, θ, ϕ) and verify that it takes the form $dr^2 + r^2\,(d\theta^2 + \sin^2\theta\, d\phi^2)$.

15.5.3. *$K < 0$: the hyperboloid (open universe).* The spatial slices are three-dimensional hyperboloids of radius R (with $K = -1/R^2$), which may be described as embedded submanifolds of $\mathbb{R}^4$ with Lorentzian signature:

$$t^2 - x^2 - y^2 - z^2 = R^2 \,.$$

In hyperbolic coordinates (ψ, θ, ϕ) the metric reads:

$$ds_\Sigma^2 = d\psi^2 + \sinh^2\psi\,\left(d\theta^2 + \sin^2\theta\, d\phi^2\right) . \tag{271}$$

This is also an **open** universe (non-compact, negatively curved).

15.6. The Friedmann–Lemaître–Robertson–Walker metric. We now embed the spatial slices back into the full four-dimensional spacetime. Since the isotropic observers move on worldlines orthogonal to Σ_t (§15.3), we can express the full 4D metric as:

$$g_{ab} = -u_a\, u_b + h_{ab}(t) \,, \tag{272}$$

where u^a is the unit tangent vector of the isotropic observers and $h_{ab}(t)$ is one of the three spatial metrics above.

All isotropic observers assign the same proper time τ to the surface Σ_t (by homogeneity, no observer is distinguished). In coordinates adapted to this proper time, the full metric becomes:

FLRW metric

$$\boxed{ds^2 = -d\tau^2 + a^2(\tau)\, ds_\Sigma^2} \tag{273}$$

where $a(\tau)$ is the **scale factor** and ds_Σ^2 is one of the three spatial metrics (269)/(270)/(271):

$$ds^2 = -d\tau^2 + a^2(\tau) \begin{cases} d\psi^2 + \sin^2\psi\,(d\theta^2 + \sin^2\theta\, d\phi^2) & (K > 0,\ \text{closed}) \\ dx^2 + dy^2 + dz^2 & (K = 0,\ \text{flat}) \\ d\psi^2 + \sinh^2\psi\,(d\theta^2 + \sin^2\theta\, d\phi^2) & (K < 0,\ \text{open}) \end{cases}$$

This is the **Friedmann–Lemaître–Robertson–Walker (FLRW) metric**.

An extraordinary simplification

By assuming homogeneity and isotropy we have reduced the task of finding the metric of our model universe to solving for just one single scalar function $a(\tau)$—the scale factor. This is a huge simplification: instead of ten coupled nonlinear PDEs for the ten components of $g_{\mu\nu}$, we hope (and it is indeed the case) that substituting the FLRW metric into Einstein's field equations with an appropriate stress-energy tensor on the right-hand side

will reduce everything to a single ordinary differential equation for $a(\tau)$. That equation—the **Friedmann equation**—is the subject of the next lecture.

15.7. Dynamics of a homogeneous and isotropic universe. In the previous section we described the *kinematics* of a homogeneous and isotropic universe: we derived the FLRW metric (273) and reduced the geometric degrees of freedom to a single unknown function, the scale factor $a(\tau)$. Now we turn to the *dynamics*: how does such a universe evolve from its inception to its final moments—if there are any?

To answer this question we need two ingredients: the left-hand side of Einstein's field equations (the Einstein tensor, which we can compute from the FLRW metric) and the right-hand side (the stress-energy tensor, which models the matter content of the universe). We begin with the latter.

15.7.1. *Stress-energy tensor for a homogeneous isotropic universe.* In our highly idealised model, we treat galaxies as grains of dust: point particles with negligible random velocities, moving on the worldlines of the isotropic observers. Since the fluctuations in galaxy velocities are small compared to the speed of light, pressure is neglected, and the stress-energy tensor of the matter content is simply that of pressureless dust:

$$T_{ab}^{(\text{dust})} = \rho\, u_a\, u_b\,, \tag{274}$$

where ρ is the average mass-energy density (one "smears out" the galaxies into a homogeneous soup) and u^a is the four-velocity of the isotropic observers.

Modelling the universe as dust

This model treats the entire visible matter content of the universe—hundreds of billions of galaxies, each containing hundreds of billions of stars—as a uniform pressureless gas. The approximation is justified on cosmological scales: the peculiar velocities of galaxies (a few hundred km/s) are tiny compared to the speed of light, and the gravitational interaction between galaxies is negligible compared to the overall expansion dynamics.

However, the visible matter is not the only contribution to the energy content of the universe. There is an experimentally determined background of thermal radiation pervading the cosmos at a temperature of approximately 3 K: the **cosmic microwave background** (CMB). We model this radiation as a perfect fluid with the equation of state

$$p_{\text{rad}} = \tfrac{1}{3}\,\rho_{\text{rad}}\,. \tag{275}$$

(Deriving this from statistical mechanics is a somewhat advanced exercise; the factor of $\frac{1}{3}$ reflects the fact that radiation exerts equal pressure in all three spatial directions.)

The total stress-energy tensor, incorporating both the galaxy (dust) content and the radiation, takes the form of a **perfect fluid**:

$$\boxed{T_{ab} = \rho\, u_a\, u_b + p\,(g_{ab} + u_a\, u_b)} \tag{276}$$

where ρ and p are adjusted to include all contributions (visible matter, radiation, and any other fields).

Exercise 15.3. Show that (276) is the most general stress-energy tensor consistent with homogeneity and isotropy. *Hint*: in the rest frame of the isotropic observers, T_{ab} must be invariant under spatial rotations. What form can a symmetric $(0, 2)$ tensor take if all spatial directions are equivalent?

15.7.2. *The cosmological constant.* We add one further term to Einstein's field equations: the **cosmological constant** Λ. The modified field equations read

$$G_{ab} + \Lambda\, g_{ab} = 8\pi\, T_{ab} \tag{277}$$

This term was introduced by Einstein himself (for very different reasons—he sought a static universe), and for decades it was hoped that $\Lambda = 0$.

The cosmological constant

Einstein introduced Λ in 1917 to obtain a static cosmological solution, which he believed was necessary on philosophical grounds. After Hubble's discovery of the expansion of the universe in 1929, Einstein reportedly called the cosmological constant his "greatest blunder." However, observations since the late 1990s—first from Type Ia supernovae, then from the CMB and large-scale structure surveys—have established that $\Lambda > 0$. A non-zero cosmological constant (or, more generally, "dark energy") is now an essential ingredient of the standard cosmological model. The cosmological constant acts effectively as a *negative pressure* term, driving the accelerated expansion of the universe observed today.

15.7.3. *Reduction to two independent equations.* In principle, Einstein's field equations (277) constitute ten simultaneous coupled nonlinear partial differential equations for the ten independent components of the metric—a system of monstrous complexity, far exceeding anything one typically encounters in other branches of physics. There is no Green's function, no superposition principle; nothing is going to save us in general.

However, one can either use perturbation theory (as we did for linearised gravity) or exploit symmetry to reduce the number of equations. In the present case, homogeneity and isotropy achieve a dramatic simplification.

Only two independent equations

Of the ten components of Einstein's field equations (277), only **two** are independent for a homogeneous and isotropic spacetime.

The argument proceeds as follows. First, note that the vectors $G^{ab}\, u_b$ and $T^{ab}\, u_b$ can have no spatial components (by isotropy—a non-vanishing spatial component would define a preferred direction). Similarly, when restricted to a spatial slice Σ_t, the same eigenvalue argument used for $R^{(3)cd}_{ab}$ in §15.4.3 implies that the spatial part of the Einstein tensor $G^{(3)}_{ab}$ is proportional to the identity (otherwise one could construct a preferred spatial vector). Consequently:

- The off-diagonal (space–time) components of (277) vanish identically.
- The diagonal spatial components are all *equal* (by the eigenvalue/isotropy argument).
- By homogeneity, the space–time components also vanish.

We are left with exactly two independent equations, which we write as:

$$G_{\tau\tau} + \Lambda\, g_{\tau\tau} = 8\pi\, \rho\,, \tag{278}$$

$$G_{**} + \Lambda\, g_{**} = 8\pi\, p\,, \tag{279}$$

where $G_{\tau\tau} = G_{ab}\, u^a u^b$ is the time–time component and $G_{**} = G_{ab}\, s^a s^b$ denotes *any* spatial diagonal component (contracted with a unit spacelike vector s^a orthogonal to u^a, with $s^a s_a = 1$); all spatial components give the same equation.

15.7.4. *The FLRW metric in unified form.* To carry out the explicit calculation of the Christoffel symbols, Riemann tensor, and Einstein tensor from the FLRW metric, we need a coordinate representation. Basis-free reasoning is wonderful for deducing logical consequences of geometric statements, but it is hopeless for doing any real calculation—for that, we must choose coordinates and compute.

It is convenient to write the three spatial metrics (269)/(270)/(271) in a single unified form using spherical coordinates (r, θ, ϕ). The FLRW metric becomes:

$$ds^2 = -d\tau^2 + a^2(\tau)\left[\frac{dr^2}{1-k\,r^2} + r^2\big(d\theta^2 + \sin^2\theta\, d\phi^2\big)\right] \tag{280}$$

where the **curvature parameter**

$$k = \begin{cases} +1 & \text{(three-sphere, closed universe)}\,, \\ 0 & \text{(flat space, open universe)}\,, \\ -1 & \text{(hyperboloid, open universe)}\,. \end{cases} \tag{281}$$

Exercise 15.4. Verify that (280) with $k = +1$ reduces to the S^3 metric (269) under the substitution $r = \sin\psi$, and that $k = -1$ gives the H^3 metric (271) under $r = \sinh\psi$.

The road ahead

We now have the metric (280) and the stress-energy tensor (276), and we know that Einstein's equations reduce to just two independent equations (278)–(279). What remains is an explicit but mechanical calculation: compute the Christoffel symbols from (280), build the Riemann and Ricci tensors, form the Einstein tensor, substitute into (278)–(279), and arrive at ordinary differential equations for $a(\tau)$. These are the **Friedmann equations**, and—remarkably—they can be solved. That calculation is the subject of the next lecture.

15.8. Solving Einstein's equations for FLRW.

We now carry out the explicit calculation: starting from the FLRW metric (280), we compute the Christoffel symbols, the Ricci tensor, the Ricci scalar, and the Einstein tensor, then substitute into the field equations (278)–(279) to obtain the **Friedmann equations**—ordinary differential equations governing the scale factor $a(\tau)$.

Basis-free reasoning was invaluable for deducing the *form* of the metric, but it is hopeless for doing any real calculation. To model a physical process, we must choose coordinates and compute.

15.8.1. *Christoffel symbols.* For the FLRW metric (280) in coordinates (τ, r, θ, ϕ), a straightforward (if lengthy) exercise yields the following non-vanishing Christoffel symbols. We write $\dot{a} = da/d\tau$.

Temporal–spatial:

$$\Gamma^\tau{}_{rr} = \frac{a\,\dot{a}}{1-k\,r^2}\,, \qquad \Gamma^\tau{}_{\theta\theta} = a\,\dot{a}\,r^2\,, \qquad \Gamma^\tau{}_{\phi\phi} = a\,\dot{a}\,r^2\sin^2\theta\,. \tag{282}$$

Spatial–temporal:

$$\Gamma^r{}_{\tau r} = \Gamma^\theta{}_{\tau\theta} = \Gamma^\phi{}_{\tau\phi} = \frac{\dot{a}}{a}\,. \tag{283}$$

Purely spatial:

$$\Gamma^r{}_{rr} = \frac{k\,r}{1-k\,r^2}\,, \qquad \Gamma^r{}_{\theta\theta} = -r\,(1-k\,r^2)\,, \qquad \Gamma^r{}_{\phi\phi} = -r\,(1-k\,r^2)\sin^2\theta\,, \tag{284}$$

$$\Gamma^{\theta}{}_{r\theta} = \Gamma^{\phi}{}_{r\phi} = \frac{1}{r}\,, \qquad \Gamma^{\theta}{}_{\phi\phi} = -\sin\theta\cos\theta\,, \qquad \Gamma^{\phi}{}_{\theta\phi} = \cot\theta\,. \tag{285}$$

All other components either vanish or are determined by the symmetry $\Gamma^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\nu\mu}$.

Remark. The individual Christoffel symbols are not difficult to compute; the main challenge is bookkeeping—detecting the inevitable sign errors and missed factors before they propagate. Computer algebra (Mathematica, SageMath, etc.) is an invaluable tool for verifying these results.

15.8.2. *Ricci tensor and scalar.* From the Christoffel symbols one computes the Ricci tensor. The non-vanishing components are (exercise):

$$R_{\tau\tau} = -3\,\frac{\ddot{a}}{a}\,, \tag{286}$$

$$R_{rr} = \frac{\ddot{a}\,a + 2\dot{a}^2 + 2k}{1 - k\,r^2}\,, \tag{287}$$

$$R_{\theta\theta} = (\ddot{a}\,a + 2\dot{a}^2 + 2k)\,r^2\,, \tag{288}$$

$$R_{\phi\phi} = (\ddot{a}\,a + 2\dot{a}^2 + 2k)\,r^2\sin^2\theta\,. \tag{289}$$

Note that the Ricci tensor is diagonal, and the spatial components all share the common factor $(\ddot{a}\,a + 2\dot{a}^2 + 2k)$ —confirming by direct calculation the isotropy that we built in by assumption.

The Ricci scalar is obtained by tracing with the inverse metric:

$$R = 6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2}\right). \tag{290}$$

15.8.3. *Einstein tensor.* The Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}\,g_{\mu\nu}\,R$ has the following non-vanishing components (exercise):

$$G_{\tau\tau} = 3\,\frac{\dot{a}^2}{a^2} + 3\,\frac{k}{a^2}\,, \tag{291}$$

$$G_{rr} = \frac{-2\ddot{a}\,a - \dot{a}^2 - k}{1 - k\,r^2}\,, \tag{292}$$

$$G_{\theta\theta} = r^2\big(-2\ddot{a}\,a - \dot{a}^2 - k\big)\,, \tag{293}$$

$$G_{\phi\phi} = r^2\sin^2\theta\big(-2\ddot{a}\,a - \dot{a}^2 - k\big)\,. \tag{294}$$

Exercise 15.5. Derive (291)–(294) from (286)–(289) and (290).

15.8.4. *The Friedmann equations.* Substituting the Einstein tensor and the perfect-fluid stress-energy tensor (276) into the field equations $G_{ab} + \Lambda\,g_{ab} = 8\pi\,T_{ab}$:

τ–τ component:

$$3\,\frac{\dot{a}^2}{a^2} + 3\,\frac{k}{a^2} - \Lambda = 8\pi\,\rho\,. \tag{i$'$}$$

r–r component (after dividing out common factors):

$$-2\,\frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} - \frac{k}{a^2} + \Lambda = 8\pi\,p\,. \tag{ii$'$}$$

The θ–θ and ϕ–ϕ components yield the same equation as (ii′), confirming that only two equations are independent.

By differentiating (i′) and combining with (ii′) to eliminate common terms, one arrives at the equivalent—and more commonly cited—pair:

Friedmann equations

(I) $$3\,\frac{\dot{a}^2}{a^2} = 8\pi\,\rho - \frac{3k}{a^2} + \Lambda\,,$$

(II) $$3\,\frac{\ddot{a}}{a} = -4\pi\,(\rho + 3p) + \Lambda\,.$$

Together with the **fluid equation** (which follows from $\nabla^a T_{ab} = 0$, or equivalently from differentiating (I) and substituting (II)):

(295) $$\dot{\rho} = -3\,\frac{\dot{a}}{a}\,(\rho + p)\,,$$

these govern the evolution of the scale factor $a(\tau)$ and the energy density $\rho(\tau)$ for any equation of state $p = p(\rho)$.

15.8.5. *Absorbing the cosmological constant as dark energy.* Observe that the cosmological constant Λ can be *eliminated* from (I)–(II) by redefining the energy density and pressure:

(296) $$\rho \longmapsto \rho + \frac{\Lambda}{8\pi}\,, \qquad p \longmapsto p - \frac{\Lambda}{8\pi}\,.$$

After this substitution, the Friedmann equations take the same form but with $\Lambda = 0$. In this way, the cosmological constant is interpreted as a form of energy with **negative pressure** equal in magnitude to its energy density:

(297) $$p_\Lambda = -\rho_\Lambda\,.$$

This exotic equation of state defines what is known as **dark energy**.

15.9. Qualitative consequences of the Friedmann equations. Before solving the Friedmann equations quantitatively, we can extract striking qualitative predictions by simple inspection.

15.9.1. *The universe cannot be static.* Suppose that $\rho > 0$ and $p \geq 0$ (i.e. we temporarily ignore the cosmological constant / dark energy contribution). Then equation (II) gives

$$\ddot{a} = -\frac{4\pi}{3}\,(\rho + 3p)\,a < 0\,.$$

Since $a > 0$, the scale factor has strictly negative second derivative: $a(\tau)$ is *concave*. In particular, $\dot{a} \neq 0$—the universe must always be either expanding ($\dot{a} > 0$) or contracting ($\dot{a} < 0$). A static universe ($\dot{a} = 0$, $\ddot{a} = 0$) is impossible for ordinary matter.

Why Einstein introduced Λ

This was precisely the result that Einstein found unacceptable in 1917: the prevailing belief was that the universe is eternal and static. The cosmological constant term was introduced to balance the gravitational attraction and permit a static solution. Of course, Hubble's observations later confirmed that the universe is indeed expanding. One should note, however, that the principle of including all terms permitted by the symmetries of the theory is sound—the motivation may have been questionable, but the cosmological constant itself is a perfectly legitimate (and, as it turns out, physically realised) contribution to the field equations.

15.9.2. *Hubble's law.* Let $\mathcal{R}$ denote the proper distance between two isotropic observers at cosmological time τ. Since the FLRW metric rescales all spatial distances by the common factor $a(\tau)$, we have $\mathcal{R}(\tau) = a(\tau)\, d$, where d is the *comoving distance* (constant in time). Differentiating:

$$v \equiv \dot{\mathcal{R}} = \frac{\dot{a}}{a}\,\mathcal{R} = H(\tau)\,\mathcal{R} \tag{298}$$

where

$$H(\tau) = \frac{\dot{a}}{a} \tag{299}$$

is the **Hubble parameter** (often called "Hubble's constant," although it is *not* constant in time—it is better described as Hubble's *function*).

Equation (298) is **Hubble's law**: the recession velocity of a distant galaxy is proportional to its distance, with proportionality constant H. For sufficiently large separations, v can exceed the speed of light. This entails no violation of relativity: in general relativity, only *locally* measured relative velocities (at a single event, between infinitesimally close observers) are required not to exceed c. Global recession velocities, determined by the geodesic distance between widely separated observers, are not so constrained.

The balloon analogy

One should not picture the expansion as originating from a distinguished centre. Rather, think of galaxies as dots painted on the surface of a balloon that is being inflated: all distances between dots increase uniformly, with no dot occupying a preferred position. The expansion is an expansion *of space itself*, not an expansion of matter into pre-existing space.

15.9.3. *The Big Bang singularity.* Given that the universe is expanding ($\dot{a} > 0$) and that $\ddot{a} < 0$ (for ordinary matter with $\rho + 3p > 0$), the concavity of $a(\tau)$ has a dramatic consequence. Even at *constant* expansion rate (which underestimates the past rate, since the expansion was faster in the past), the scale factor would reach zero at a finite time

$$T = \frac{a}{\dot{a}} = \frac{1}{H}\,.$$

Since the actual expansion was faster in the past ($\ddot{a} < 0$ means $\dot{a}$ was larger), the time to $a = 0$ is even shorter than $1/H$.

At $a = 0$, all spatial distances vanish, the energy density diverges (assuming any matter is present), and the spacetime metric becomes singular. This is the **Big Bang**: the initial singularity from which the universe began.

The Big Bang

In an expanding FLRW universe with $\rho + 3p > 0$, the scale factor $a(\tau)$ must have been zero at a finite time in the past. At this instant, all distances between events vanish and the energy density is infinite. This is a genuine singularity of the spacetime: it is impossible to extend the manifold in a non-singular, connected way to "before" the Big Bang. The Big Bang is *not* an explosion of pre-existing matter into pre-existing space—it is the origin of space and time themselves.

In the next lecture, we will solve the Friedmann equations explicitly for various equations of state and then turn to the final major topic of this course: the **Schwarzschild solution**—an

exact solution of Einstein's equations describing the geometry outside a spherically symmetric, non-rotating mass.

15.10. Exact FLRW solutions. We now solve the Friedmann equations analytically for various matter contents and spatial curvatures.

15.10.1. *Conservation laws from the fluid equation.* The fluid equation (295), $\dot{\rho} = -3(\dot{a}/a)(\rho + p)$, admits immediate first integrals for the two most important equations of state.

Dust ($p = 0$). The fluid equation becomes $\dot{\rho} + 3\rho\,\dot{a}/a = 0$, which is a total derivative in disguise:

$$\frac{d}{d\tau}(\rho\, a^3) = 0 \qquad \Longrightarrow \qquad \boxed{\rho\, a^3 = \text{const.}} \tag{300}$$

This is simply the conservation of rest mass: the energy density scales inversely with the volume $\propto a^3$, which is exactly what one expects for non-relativistic matter that is neither created nor destroyed.

Radiation ($p = \rho/3$). The fluid equation gives $\dot{\rho} + 4\rho\,\dot{a}/a = 0$, hence:

$$\boxed{\rho\, a^4 = \text{const.}} \tag{301}$$

Why a^4 for radiation?

The photon number density scales as a^{-3} (dilution by volume expansion), just as for dust. But each photon also loses energy as the universe expands: its wavelength is stretched by the scale factor, contributing an additional factor of a^{-1} (**cosmological redshift**). Together: $\rho_{\text{rad}} \propto a^{-3} \cdot a^{-1} = a^{-4}$. Consequently, *radiation dominated the energy budget of the early universe* (when a was small), even though dust dominates today.

15.10.2. *Late-time behaviour.* The qualitative late-time behaviour depends on the spatial curvature k. Assuming $\rho > 0$ and $p \geq 0$ throughout:

- $k = 0$ **or** $k = -1$ (flat or hyperbolic): Equation (I) with $\Lambda = 0$ gives $\dot{a}^2 = 8\pi\rho\, a^2/3 - k$. Since ρ decreases with a and the right-hand side never reaches zero (for $k = 0$ the right-hand side is positive whenever $\rho > 0$; for $k = -1$ the $-k = +1$ term keeps it positive), the expansion continues forever: **eternal expansion**. For $k = 0$, $\dot{a} \to 0$ as $\tau \to \infty$; for $k = -1$, $\dot{a} \to 1$.
- $k = +1$ (spherical): the curvature term $-k/a^2 = -1/a^2$ eventually dominates, driving $\dot{a}$ to zero. The universe reaches a maximum size and then *recollapses*, ending in a **Big Crunch**—the time-reverse of the Big Bang.

15.10.3. *Exact solutions.* Using the conservation laws, Friedmann equation (I) (with $\Lambda = 0$) reduces to a single ODE for $a(\tau)$. For **dust** ($p = 0$), define $C = 8\pi\rho\, a^3/3 = \text{const}$:

$$\dot{a}^2 = \frac{C}{a} - k\,. \tag{302}$$

For **radiation** ($p = \rho/3$), define $C' = 8\pi\rho\, a^4/3 = \text{const}$:

$$\dot{a}^2 = \frac{C'}{a^2} - k\,. \tag{303}$$

These are solved by separation of variables (the integrals are mildly irritating but standard). The results are collected in the following table.

	Dust ($p = 0$)	**Radiation** ($p = \rho/3$)
$k = +1$	$a = \frac{1}{2}C(1 - \cos\eta)$ $\tau = \frac{1}{2}C(\eta - \sin\eta)$	$a = \sqrt{C'}\big(1 - (1 - \tau/\sqrt{C'})^2\big)^{1/2}$
$k = 0$	$a = \left(\frac{9C}{4}\right)^{1/3} \tau^{2/3}$	$a = (4C')^{1/4}\, \tau^{1/2}$
$k = -1$	$a = \frac{1}{2}C(\cosh\eta - 1)$ $\tau = \frac{1}{2}C(\sinh\eta - \eta)$	$a = \sqrt{C'}\big((1 + \tau/\sqrt{C'})^2 - 1\big)^{1/2}$

Here η is a parameter (conformal time for the dust solutions). The $k = +1$ dust solution is given in parametric form—the cycloid—and describes a universe that expands from a Big Bang, reaches maximum size at $\eta = \pi$, and recollapses to a Big Crunch at $\eta = 2\pi$.

Exercise 15.6. Derive the $k = 0$ dust solution $a \propto \tau^{2/3}$ directly from (302) with $k = 0$.

The Friedmann–Lemaître–Robertson–Walker cosmologies

All solutions in the table above share the same qualitative feature: they begin with $a = 0$ (the Big Bang). For $k = 0$ and $k = -1$, they expand forever; for $k = +1$, the expansion halts and reverses. Despite the extreme simplicity of the model—a homogeneous isotropic universe filled with a single type of matter—these solutions capture the essential large-scale dynamics of our universe remarkably well. The FLRW cosmologies serve as the starting point for all of modern observational cosmology; perturbations around them describe the formation of galaxies and the temperature fluctuations of the CMB. This concludes our discussion of cosmology; we now turn to the other celebrated exact solution of Einstein's equations.

16. The Schwarzschild solution

Roughly speaking, there are two major families of well-known exact solutions to Einstein's field equations: the FLRW cosmologies (describing the large-scale evolution of the universe) and the **Schwarzschild solution** (describing the gravitational field outside a spherically symmetric, non-rotating body). We now turn to the latter.

16.1. Static and stationary spacetimes. We seek solutions that "don't change in time." There are two levels of precision for this intuition.

Definition 16.1 (Stationary spacetime)**.** A spacetime $(\mathcal{M}, g_{ab})$ is **stationary** if there exists a one-parameter group ϕ_t of isometries whose orbits are timelike curves ("time translations").

Equivalently: there exists a timelike Killing vector field ξ^a (i.e. $\nabla_{(a}\xi_{b)} = 0$ with $\xi^a\xi_a < 0$).

Definition 16.2 (Static spacetime)**.** A spacetime is **static** if it is stationary *and* there exists a spacelike hypersurface Σ that is orthogonal to the orbits of the isometry group.

By Frobenius' theorem, this is equivalent to requiring that the timelike Killing vector field additionally satisfies

$$\xi_{[a}\, \nabla_b\, \xi_{c]} = 0\,. \tag{304}$$

Remark. A rotating body (e.g. a Kerr black hole) generates a *stationary but not static* spacetime: it has a timelike Killing vector (time-translation symmetry), but the rotation introduces off-diagonal $dt\, d\phi$ terms in the metric that prevent the existence of a hypersurface orthogonal to the time direction everywhere.

In adapted coordinates, a static metric takes the form

$$ds^2 = -V^2(x^1, x^2, x^3)\, dt^2 + h_{\mu\nu}(x^1, x^2, x^3)\, dx^\mu\, dx^\nu\,, \tag{305}$$

where $V^2 = -\xi^a \xi_a$ is the squared norm of the Killing vector. Crucially, there are *no* cross-terms $dt\, dx^\mu$ (this is the consequence of staticity, not merely stationarity), and the components depend only on the spatial coordinates, not on t.

16.2. Spherical symmetry.

Definition 16.3 (Spherically symmetric spacetime)**.** A spacetime is **spherically symmetric** if its isometry group contains a subgroup isomorphic to $SO(3)$ whose orbits are two-spheres S^2.

The full spacetime metric g_{ab} induces a metric $\hat{h}_{ab}$ on each orbit sphere. By $SO(3)$ invariance, the induced metric must be a multiple of the standard round metric on S^2—there is simply no other $SO(3)$-invariant metric on S^2. The proportionality constant is characterised by the area A of the orbit sphere; we define the **areal radius**

$$r = \sqrt{A/4\pi}\,, \tag{306}$$

so that the induced metric on each orbit sphere is $r^2(d\theta^2 + \sin^2\theta\, d\phi^2)$.

For a spacetime that is both **static** and **spherically symmetric**, the timelike Killing field ξ^a is unique (up to normalisation). One can show that ξ^a must be orthogonal to the orbit spheres (project ξ^a onto S^2; the projection must be $SO(3)$-invariant, but the only invariant vector field on S^2 is the zero field). The orbit spheres therefore lie within the spatial hypersurfaces Σ_t.

16.3. The Schwarzschild ansatz. Combining staticity and spherical symmetry, we choose coordinates as follows: pick a fiducial orbit sphere in Σ, introduce spherical coordinates (θ, ϕ) on it, and propagate them to neighbouring spheres along geodesics orthogonal to the fiducial sphere (parameterised by the areal radius r). Using t for the Killing parameter, the full coordinate system is (t, r, θ, ϕ).

In these coordinates, the most general static spherically symmetric metric takes the form:

$$\boxed{ds^2 = -f(r)\, dt^2 + h(r)\, dr^2 + r^2\big(d\theta^2 + \sin^2\theta\, d\phi^2\big)} \tag{307}$$

where $f(r)$ and $h(r)$ are two unknown functions of the radial coordinate alone. All the symmetry content—staticity and spherical symmetry—has been used to arrive at this ansatz.

From ten unknowns to two

The general metric in four dimensions has ten independent components. Staticity eliminated the four space–time cross-terms and the dependence on t, leaving seven spatial functions. Spherical symmetry then fixed the angular part to be the standard round metric on S^2 (multiplied by r^2), reducing the unknowns to just *two functions of a single variable*: $f(r)$ and $h(r)$. In the next lecture, we will substitute this ansatz into the vacuum Einstein equations and solve for f and h, obtaining the celebrated **Schwarzschild metric**.

16.4. Solving the vacuum Einstein equations. We now carry out the purely mechanical—but highly rewarding—procedure of substituting the ansatz (307) into the vacuum Einstein equations and solving for the unknown functions $f(r)$ and $h(r)$. The remarkable thing about the Schwarzschild solution is how easy the equations are to solve, despite the formidable appearance of the intermediate expressions. One should think of the Schwarzschild solution as the *hydrogen atom of general relativity*: both pertain to spherically symmetric systems, and both yield an extraordinary amount of physics from an extraordinarily small amount of work.

16.4.1. *Metric components and their inverses.* The ansatz (307) gives the diagonal metric components

$$g_{tt} = -f(r)\,, \quad g_{rr} = h(r)\,, \quad g_{\theta\theta} = r^2\,, \quad g_{\phi\phi} = r^2 \sin^2\theta\,,$$

with inverse

$$g^{tt} = -f^{-1}\,, \quad g^{rr} = h^{-1}\,, \quad g^{\theta\theta} = r^{-2}\,, \quad g^{\phi\phi} = r^{-2}\sin^{-2}\theta\,.$$

16.4.2. *Christoffel symbols.* The non-vanishing Christoffel symbols are (exercise—the main challenge is bookkeeping, not conceptual difficulty):

$$\Gamma^r{}_{rr} = \frac{h'}{2h}\,, \qquad \Gamma^r{}_{\theta\theta} = -\frac{r}{h}\,, \qquad \Gamma^r{}_{\phi\phi} = -\frac{r\sin^2\theta}{h}\,, \tag{308}$$

$$\Gamma^r{}_{tt} = \frac{f'}{2h}\,, \qquad \Gamma^t{}_{tr} = \frac{f'}{2f}\,, \tag{309}$$

$$\Gamma^\theta{}_{r\theta} = \Gamma^\phi{}_{r\phi} = \frac{1}{r}\,, \qquad \Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta\,, \qquad \Gamma^\phi{}_{\theta\phi} = \cot\theta\,, \tag{310}$$

where $f' = df/dr$ and $h' = dh/dr$.

Remark. No individual Christoffel symbol is difficult to compute—one could assign any one of them as a first-year calculus exercise. The real challenge is ensuring that the inevitable sign errors and missed factors are caught before they propagate. Computer algebra systems are an invaluable complement to hand calculation for this purpose.

16.4.3. *Ricci tensor.* The Ricci tensor is computed from $R_{\mu\nu} = \partial_\kappa\,\Gamma^\kappa{}_{\mu\nu} - \partial_\nu\,\Gamma^\kappa{}_{\mu\kappa} + \Gamma^\kappa{}_{\kappa\lambda}\,\Gamma^\lambda{}_{\mu\nu} - \Gamma^\kappa{}_{\nu\lambda}\,\Gamma^\lambda{}_{\mu\kappa}$. One first verifies that $R_{\mu\nu} = 0$ for $\mu \neq \nu$ (exercise—this considerably reduces the work). The diagonal components are:

$$R_{rr} = -\frac{f''}{2f} - \frac{1}{4}\frac{f'}{f}\Big(\frac{h'}{h} + \frac{f'}{f}\Big) - \frac{1}{r}\frac{h'}{h}\,, \tag{311}$$

$$R_{\theta\theta} = -\Big(-1 + \frac{r}{2h}\Big(-\frac{h'}{h} + \frac{f'}{f}\Big) + \frac{1}{h}\Big)\,, \tag{312}$$

$$R_{\phi\phi} = \sin^2\theta\; R_{\theta\theta}\,, \tag{313}$$

$$R_{tt} = -\Big(\frac{f''}{2h} - \frac{1}{4}\frac{f'}{f}\Big(\frac{h'}{h} + \frac{f'}{f}\Big) - \frac{1}{r}\frac{f'}{h}\Big)\,. \tag{314}$$

Exercise 16.1. Derive the Ricci tensor components (311)–(314) from the Christoffel symbols (308)–(310). Start by confirming that $R_{\mu\nu} = 0$ for $\mu \neq \nu$.

16.4.4. *Vacuum equations and the solution.* We model the gravitational field *outside* a spherically symmetric body (a star, say). In this exterior region the stress-energy tensor vanishes, and using the trace-reversed form of Einstein's equations (229) (recalled from §13.1.1), the vacuum field equations reduce to

$$R_{\mu\nu} = 0\,. \tag{315}$$

Since $R_{\phi\phi} = \sin^2\theta\; R_{\theta\theta}$, it suffices to solve the three equations $R_{tt} = R_{rr} = R_{\theta\theta} = 0$.

Step 1: Relate f and h. A key observation is that the combination

$$\frac{R_{tt}}{f} + \frac{R_{rr}}{h} = -\frac{1}{r\,h}\Big(\frac{h'}{h} + \frac{f'}{f}\Big) = 0 \tag{316}$$

yields the remarkably simple equation

$$\frac{f'}{f} + \frac{h'}{h} = 0\,, \qquad \text{i.e.} \quad \frac{d}{dr}\big(\ln(f\,h)\big) = 0\,,$$

so that $f\,h = K$ for some constant K. By rescaling the time coordinate $t \to \sqrt{K}\,t$, we may set $K = 1$:

$$f = h^{-1} \tag{317}$$

This immediately eliminates one of the two unknown functions.

Step 2: Solve for f. Substituting $h = f^{-1}$ into $R_{\theta\theta} = 0$ gives (exercise):

$$-f' + \frac{1-f}{r} = 0\,, \qquad \text{i.e.} \quad \frac{d}{dr}(r\,f) = 1\,. \tag{318}$$

Integrating:

$$f(r) = 1 + \frac{C}{r}\,, \tag{319}$$

where C is a constant of integration.

Step 3: Interpret the constant. As $r \to \infty$, the metric (307) with $f = 1 + C/r$ approaches the Minkowski metric: the C/r terms are small perturbations. Comparing with the Newtonian limit derived in §12.7—specifically the metric perturbation $\gamma_{00} = -2\phi$ where $\phi = -M/r$ is the Newtonian potential—we identify

$$C = -2M\,, \tag{320}$$

where M is the mass of the central body (in geometrised units $G = c = 1$).

Exercise 16.2. Carry out the comparison explicitly: write the Schwarzschild metric to leading order in M/r and compare with the linearised metric (223) from the Newtonian limit.

16.5. The Schwarzschild metric. Substituting $C = -2M$ and defining the **Schwarzschild radius** $r_s = 2M$:

Schwarzschild metric

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2 \tag{321}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ is the standard metric on S^2. This is the unique vacuum, static, spherically symmetric solution of Einstein's field equations, obtained by Karl Schwarzschild in 1916.

The hydrogen atom of general relativity

We selected, in this course, the two very best exact solutions of Einstein's equations: the FLRW cosmologies and the Schwarzschild metric. Both exploit symmetry to reduce the full complexity of the field equations to tractable problems, and both yield an extraordinary amount of physics. The Schwarzschild solution describes the spacetime outside any spherically symmetric, non-rotating mass—stars, planets, non-rotating black holes—and is the foundation for the classical tests of general relativity (perihelion precession, light

deflection, gravitational redshift) that we will study in the coming lectures. It went rather painlessly; unfortunately, this is not the general nature of things—solving Einstein's equations without high symmetry is an extraordinarily intricate task.

16.6. Singularities of the Schwarzschild metric. The metric (321) has apparent singularities at two values of r:

(1) $r = 0$: both g_{tt} and g_{rr} diverge.
(2) $r = r_s = 2M$: $g_{tt} = 0$ and $g_{rr} \to \infty$.

The singularity at $r = 2M$ is a **coordinate singularity**: the spacetime manifold is perfectly smooth there, and the apparent divergence is an artefact of the Schwarzschild coordinates (much as the apparent singularity of polar coordinates at the origin of $\mathbb{R}^2$). Alternative coordinate systems (Eddington–Finkelstein, Kruskal–Szekeres) are regular at $r = 2M$. We will not prove this here.

The singularity at $r = 0$, by contrast, is a genuine **curvature singularity**: the Kretschner scalar $R_{abcd}\, R^{abcd}$ diverges as $r \to 0$, indicating that the tidal forces become infinite.

Remark. For "usual" objects (stars, planets)—as opposed to black holes—the Schwarzschild radius $r_s = 2M$ lies deep within the interior of the body, where the vacuum solution does not apply. For the Sun, $r_s \approx 2.95\,\text{km}$, compared to the solar radius of $\approx 7 \times 10^5\,\text{km}$. In such cases, the vacuum Schwarzschild solution is joined to an *interior solution* (with non-zero stress-energy) at the surface of the body, and neither singularity is physically relevant. The singularities become significant only for sufficiently compact objects—**black holes**—which are the subject of more advanced courses.

16.7. Interior solution: a static spherically symmetric star. The Schwarzschild metric (321) describes the gravitational field in the *vacuum region* outside a spherically symmetric body. We now find the solution *inside* the body, where the stress-energy tensor is non-zero.

We model the star as a static, spherically symmetric perfect fluid—a "big hot boiling liquid"—with stress-energy tensor

$$T_{ab} = \rho\, u_a\, u_b + p\,(g_{ab} + u_a\, u_b)\,, \tag{322}$$

where $\rho(r)$ is the energy density, $p(r)$ the pressure, and $u^a = -\sqrt{f}\,(dt)^a$ points in the time direction (compatible with staticity). The metric retains the general static spherically symmetric form (307).

16.7.1. *Einstein's equations for the interior.* Applying $G_{ab} = 8\pi\, T_{ab}$ in an orthonormal basis (e_0, e_1, e_2, e_3) adapted to the metric (where $(e_0)_a = \sqrt{f}\,(dt)_a$, $(e_1)_a = \sqrt{h}\,(dr)_a$, $(e_2)_a = r\,(d\theta)_a$, $(e_3)_a = r\sin\theta\,(d\phi)_a$), one obtains three independent equations (exercise):

$$8\pi\,\rho = (r\,h^2)^{-1}\,h' + r^{-2}(1 - h^{-1})\,, \tag{1}$$

$$8\pi\,p = (r\,h)^{-1}\,f'/f - r^{-2}(1 - h^{-1})\,, \tag{2}$$

$$8\pi\,p = \tfrac{1}{2}(f\,h)^{-1/2}\,\frac{d}{dr}\big[(f\,h)^{-1/2}\,f'\big] + \tfrac{1}{2}(r\,f\,h)^{-1}\,f' - \tfrac{1}{2}(r\,h^2)^{-1}\,h'\,. \tag{3}$$

16.7.2. *Solving for h: the mass function.* Equation (1) involves h alone and can be rewritten as

$$\frac{1}{r^2}\,\frac{d}{dr}\big[r\,(1 - h^{-1})\big] = 8\pi\,\rho\,.$$

Integrating with the boundary condition $h(0) = 1$ (smoothness at the centre):

$$h(r) = \left(1 - \frac{2\,m(r)}{r}\right)^{-1}, \tag{323}$$

where the **mass function**

$$m(r) = 4\pi \int_0^r \rho(r')\, r'^2\, dr' \tag{324}$$

represents the total mass-energy enclosed within radius r. Since the spatial hypersurfaces Σ_t must be spacelike, we require $h > 0$, i.e. $r > 2\,m(r)$ throughout the interior.

At the surface $r = R$ of the star (where $\rho = 0$ for $r > R$), the interior solution must match the exterior Schwarzschild solution. This requires $M = m(R)$: the Schwarzschild mass parameter equals the integrated mass function evaluated at the stellar radius.

16.7.3. *Solving for f: the gravitational potential.* Writing $f = e^{2\phi}$ (so that ϕ will reduce to the Newtonian gravitational potential in the weak-field limit), equation (2) with (323) becomes

$$\frac{d\phi}{dr} = \frac{m(r) + 4\pi\, r^3\, p}{r\big(r - 2\,m(r)\big)}\,. \tag{325}$$

In the Newtonian limit ($r^3\, p \ll m(r)$ and $m(r) \ll r$), this reduces to $d\phi/dr \approx m(r)/r^2$—Poisson's equation for the gravitational potential, confirming the identification of ϕ with the Newtonian potential.

16.7.4. *The Tolman–Oppenheimer–Volkoff equation.* Substituting (323) and (325) into the remaining equation (3) yields, after considerable algebra, the **Tolman–Oppenheimer–Volkoff (TOV) equation** of hydrostatic equilibrium:

$$\frac{dp}{dr} = -(\rho + p)\,\frac{m(r) + 4\pi\, r^3\, p}{r\big(r - 2\,m(r)\big)} \tag{326}$$

This is the relativistic generalisation of the Newtonian equation of hydrostatic equilibrium $dp/dr = -\rho\, m(r)/r^2$. Every factor in (326) has a clear physical interpretation:

- $\rho + p$ replaces ρ: in relativity, pressure contributes to the gravitational mass.
- $m(r) + 4\pi\, r^3\, p$ replaces $m(r)$: the pressure itself gravitates.
- $r - 2\,m(r)$ replaces r: a relativistic correction from the spatial curvature.

All three corrections *strengthen* gravity compared to the Newtonian prediction, making relativistic stars more prone to gravitational collapse.

Interior Schwarzschild metric

The spacetime geometry inside a static, spherically symmetric, perfect-fluid star is

$$ds^2 = -e^{2\phi(r)}\, dt^2 + \left(1 - \frac{2\,m(r)}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2\,, \tag{327}$$

where $m(r)$ is given by (324) and $\phi(r)$ by (325), subject to the TOV equation (326) and an equation of state $p = p(\rho)$.

16.8. Geodesics of the Schwarzschild solution. We now turn to the motion of test bodies and light rays in the exterior Schwarzschild spacetime ($r > 2M$). These are the geodesics of the metric (321), and their study leads to some of the most celebrated predictions of general relativity.

16.8.1. *Killing vectors and constants of motion.* Before computing geodesics explicitly, we derive a powerful general result that will simplify the analysis considerably.

Proposition 16.1. *Let ξ^a be a Killing vector field and γ a geodesic with tangent u^a. Then $\xi_a\, u^a$ is constant along γ.*

Proof. Differentiate along the geodesic:

$$u^b\, \nabla_b(\xi_a\, u^a) = u^a\, u^b\, \nabla_b \xi_a + \xi_a\, u^b\, \nabla_b u^a\,.$$

The first term vanishes because $\nabla_{(a}\xi_{b)} = 0$ (Killing's equation) and $u^a u^b$ is symmetric. The second term vanishes by the geodesic equation $u^b\, \nabla_b u^a = 0$. □

The Schwarzschild metric has two manifest Killing vectors:

- $\xi^a = (\partial/\partial t)^a$ (time-translation invariance, stationarity),
- $\psi^a = (\partial/\partial\phi)^a$ (axial rotational invariance).

By Proposition 16.1, each yields a conserved quantity along any geodesic. We will exploit these in the next lecture to reduce the geodesic equation to a single ODE.

16.8.2. *Restriction to the equatorial plane.* The Schwarzschild metric is also invariant under the parity reflection $\theta \mapsto \pi - \theta$. This symmetry has an important consequence:

Geodesics are planar

If the initial position and initial tangent vector of a geodesic both lie in the equatorial plane $\theta = \pi/2$, then the entire geodesic lies in that plane.
Since any initial conditions can be rotated into the equatorial plane by an $SO(3)$ rotation (which is an isometry of the Schwarzschild metric), it follows that *every* geodesic in the Schwarzschild spacetime is planar. We may therefore set $\theta = \pi/2$ without loss of generality when analysing geodesic motion.

This reduces the effective dimension of the problem: instead of four coordinates (t, r, θ, ϕ), we work with (t, r, ϕ) at $\theta = \pi/2$. Combined with the two conserved quantities from the Killing vectors, the geodesic equation will reduce to a single first-order ODE for r—an "effective one-dimensional problem" that we will solve in the next lecture.

16.9. Constants of motion and the effective potential. We study timelike geodesics ($\kappa = 1$) in the exterior Schwarzschild spacetime ($r > 2M$), parameterised by proper time τ. Setting $\theta = \pi/2$ (no loss of generality by §16.8.2), the metric norm $g_{ab}\, u^a u^b = -1$ gives

$$-\Big(1 - \frac{2M}{r}\Big)\dot{t}^2 + \Big(1 - \frac{2M}{r}\Big)^{-1}\dot{r}^2 + r^2\,\dot{\phi}^2 = -\kappa\,, \tag{328}$$

where $\kappa = 1$ for timelike and $\kappa = 0$ for null geodesics, and dots denote $d/d\tau$.

The two Killing vectors $\xi^a = (\partial/\partial t)^a$ and $\psi^a = (\partial/\partial\phi)^a$ yield, via Proposition 16.1, the conserved quantities:

$$E = \Big(1 - \frac{2M}{r}\Big)\,\dot{t}\,, \tag{i}$$

$$L = r^2\,\dot{\phi}\,. \tag{ii}$$

For a timelike particle far from the source, E reduces to the special-relativistic energy per unit rest mass as measured by a static observer; L is the angular momentum per unit rest mass. The constancy of L is the relativistic analogue of Kepler's second law: equal areas are swept out in equal proper times.

Substituting (i) and (ii) into (328):

$$\tfrac{1}{2}\dot{r}^2 + V(r) = \tfrac{1}{2}E^2 \tag{329}$$

where the **effective potential** is

$$V(r) = \tfrac{1}{2}\kappa - \frac{\kappa M}{r} + \frac{L^2}{2r^2} - \frac{ML^2}{r^3} \tag{330}$$

One-dimensional mechanics

Equation (329) has the structure of a unit-mass particle with "kinetic energy" $\frac{1}{2}\dot{r}^2$ moving in the potential $V(r)$ with "total energy" $\frac{1}{2}E^2$. We can therefore import the full machinery of one-dimensional classical mechanics—turning points, potential wells, stability analysis—to understand the radial motion. The angular motion is then recovered from (ii).

The first three terms of (330) are familiar from Newtonian orbital mechanics ($\frac{1}{2}\kappa$ is a constant, $-\kappa M/r$ is the gravitational potential, $L^2/2r^2$ is the centrifugal barrier). The *new* term is

$$-\frac{ML^2}{r^3}, \tag{331}$$

which is the relativistic correction. It is negligible for large r, but *dominates over the centrifugal barrier for small* r, dragging the potential to $-\infty$ and qualitatively changing the orbital dynamics.

16.10. Circular orbits and their stability. Circular orbits correspond to extrema of $V(r)$ (where $\dot{r} = 0$). Setting $dV/dr = 0$ for $\kappa = 1$:

$$0 = \frac{dV}{dr} = \frac{1}{r^4}\left(Mr^2 - L^2 r + 3ML^2\right). \tag{332}$$

The roots are

$$r_\pm = \frac{L^2 \pm \sqrt{L^4 - 12\,L^2M^2}}{2M}. \tag{333}$$

16.10.1. *Small angular momentum: $L^2 < 12M^2$.* The discriminant is negative: **no extrema exist**. The potential decreases monotonically for small r (due to the $-ML^2/r^3$ term), so any particle spirals inexorably inward. There are no stable orbits.

16.10.2. *Large angular momentum: $L^2 > 12M^2$.* Two extrema exist: r_+ is a **minimum** (stable circular orbit) and r_- is a **maximum** (unstable circular orbit).

Innermost stable circular orbit

Since $r_+ > 6M$ always, there are **no stable circular orbits** at radii $r \leq 6M$. The **innermost stable circular orbit** (ISCO) is at $r = 6M$ (achieved in the limit $L^2 \to 12M^2$). The unstable circular orbits lie in the range $3M < r_- < 6M$.

In the Newtonian limit $L \gg M$, the stable radius $r_+ \approx L^2/M$ reproduces the Newtonian formula for the circular orbital radius.

Exercise 16.3. Show that the energy of a particle in a circular orbit at radius R is

$$E(R) = \frac{R - 2M}{\sqrt{R}\,\sqrt{R - 3M}}\,.$$

Verify that $E > 1$ for $R < 4M$ (particles in unstable circular orbits between $3M$ and $4M$ can escape to infinity if perturbed outward).

16.11. Perihelion precession. Consider a particle in a slightly perturbed stable circular orbit at radius r_+. For sufficiently small displacements, the potential is effectively quadratic and the particle executes simple harmonic motion in r with radial frequency (exercise):

$$\omega_r^2 = \left.\frac{d^2V}{dr^2}\right|_{r_+} = \frac{M(r_+ - 6M)}{r_+^3\,(r_+ - 3M)}\,. \tag{334}$$

(Note: the "time" here is the proper time τ of the orbiting particle.)

The angular frequency of the circular orbit is found from $\omega_\phi = \dot{\phi} = L/r_+^2$; eliminating L using the circular orbit condition:

$$\omega_\phi^2 = \frac{M}{r_+^2\,(r_+ - 3M)}\,. \tag{335}$$

In the Newtonian limit ($r_+ \gg M$), both frequencies approach $\omega_r^2 \approx \omega_\phi^2 \approx M/r_+^3$: the oscillations are **commensurate** and the orbit *closes* (as it must for a $1/r$ potential, by Bertrand's theorem).

In general relativity, however, $\omega_r \neq \omega_\phi$. The orbit does *not* close: the perihelion (point of closest approach) shifts from one orbit to the next.

Precession of the perihelion

The precession rate per orbit is

$$\omega_p = \omega_\phi - \omega_r = -\left(\sqrt{1 - \frac{6M}{r_+}} - 1\right)\omega_\phi\,. \tag{336}$$

For $r_+ \gg M$:

$$\boxed{\omega_p \approx \frac{3\,M^{3/2}}{r_+^{5/2}}} \tag{337}$$

(radians per orbit in proper time).

Mercury's perihelion: a triumph of general relativity

The precession of Mercury's perihelion had been a long-standing puzzle in celestial mechanics. After accounting for the gravitational perturbations of the other planets, an anomalous precession of approximately 43″ (seconds of arc) per century remained unexplained by Newtonian gravity. When Einstein completed his general theory of relativity in November 1915, one of his first calculations was the perihelion precession of Mercury. The result—43″ per century—matched the observed anomaly precisely, without any adjustable parameters. Einstein later wrote that this calculation gave him "palpitations of the heart." It remains one of the most celebrated confirmations of general relativity.

In the final lecture, we will study **null geodesics** in the Schwarzschild spacetime—the trajectories of light rays—and derive the deflection of light by a massive body, another celebrated prediction of Einstein's theory.

16.12. Null geodesics and the deflection of light. For the final lecture of this course we study the trajectories of light rays—null geodesics ($\kappa = 0$)—in the Schwarzschild spacetime. For me, this is one of the most spectacular predictions of general relativity: that light itself is curved by the gravitational field, and that this curvature can be detected with terrestrial observations.

16.12.1. *The null effective potential.* Setting $\kappa = 0$ in the effective potential (330):

$$V_{\text{null}}(r) = \frac{L^2}{2r^2}\left(1 - \frac{2M}{r}\right) = \frac{L^2}{2r^3}(r - 2M)\,. \tag{338}$$

The energy equation (329) becomes

$$\tfrac{1}{2}\dot{r}^2 + V_{\text{null}}(r) = \tfrac{1}{2}E^2\,. \tag{339}$$

The potential has a single maximum at $r = 3M$ (exercise: set $dV_{\text{null}}/dr = 0$) with height $V_{\max} = L^2M/(2(3M)^3) = L^2/(54M^2)$.

Scattering vs. capture

A photon approaching from infinity either has enough energy to crest the potential barrier and plunge to $r = 0$ (**capture**), or it reflects off the barrier and escapes back to infinity (**scattering**). The critical case is the unstable circular photon orbit at $r = 3M$—the **photon sphere**.

16.12.2. *The impact parameter and capture cross section.* The ratio $b = L/E$ has a natural interpretation. In flat spacetime ($M = 0$), a null geodesic travelling along a straight line at distance r_0 from the origin satisfies $b = r_0$: the impact parameter equals the distance of closest approach. For the Schwarzschild geometry, which is **asymptotically flat**, $b = L/E$ represents the **apparent impact parameter** as seen by a distant observer.

A photon is captured if its energy exceeds the potential maximum: $\frac{1}{2}E^2 > V_{\max}$, i.e. $L^2/E^2 < 27M^2$. The **capture radius** is therefore

$$b_c = 3\sqrt{3}\,M\,, \tag{340}$$

and the **capture cross section** is

$$\sigma = \pi\, b_c^2 = 27\pi\, M^2\,. \tag{341}$$

Any photon with impact parameter $b < b_c$ is captured; those with $b > b_c$ are deflected and escape.

16.12.3. *Total angular deflection.* For a scattering trajectory ($b > b_c$), the photon approaches from $r = \infty$, reaches a turning point at $r = R_0$ (where $\dot{r} = 0$, i.e. $E^2 = L^2(R_0 - 2M)/R_0^3$), and recedes to $r = \infty$. The total change in the angular coordinate is

$$\Delta\phi = 2\int_{R_0}^{\infty} \frac{L\,dr}{r^2\sqrt{E^2 - \frac{L^2}{r^3}(r - 2M)}} \tag{342}$$

(the factor of 2 accounts for the symmetry of the trajectory about the turning point).

Substituting $u = 1/r$ and eliminating b via the turning-point condition $R_0^3 = b^2(R_0 - 2M)$:

$$\Delta\phi = 2\int_0^{1/R_0} \frac{du}{\sqrt{R_0^{-2} - 2MR_0^{-3} - u^2 + 2Mu^3}}\,. \tag{343}$$

Sanity check. For $M = 0$ (flat spacetime), $\Delta\phi = \pi$ (exercise)—the photon travels in a straight line and the total angular change from $\phi(-\infty)$ to $\phi(+\infty)$ is exactly π, as expected.

16.12.4. *Perturbative expansion for small M.* For realistic celestial bodies ($M/R_0 \ll 1$), we expand $\Delta\phi$ to first order in M. The expansion is subtle because the turning point R_0 itself depends on M; one promotes M and R_0 to independent variables and Taylor-expands. Using the turning-point condition to relate R_0 and b to zeroth order ($R_0 = b + O(M)$), one finds after a careful calculation:

$$\Delta\phi \approx \pi + \frac{4M}{b} \tag{344}$$

The **deflection angle** (the deviation from the flat-spacetime value of π) is therefore

$$\delta\phi = \Delta\phi - \pi = \frac{4M}{b}\,. \tag{345}$$

Restoring factors of G and c: $\delta\phi = 4GM/(c^2 b)$.

Deflection of light by a massive body

A light ray passing a spherically symmetric body of mass M at impact parameter b is deflected by an angle

$$\delta\phi = \frac{4M}{b}$$

(in geometrised units). For a ray grazing the surface of the Sun ($M_\odot \approx 1.5\,\text{km}$, $R_\odot \approx 7 \times 10^5\,\text{km}$):

$$\delta\phi \approx \frac{4 \times 1.5}{7 \times 10^5} \approx 8.5 \times 10^{-6}\ \text{rad} \approx 1.75''\,.$$

The 1919 eclipse expedition

The deflection of light by the Sun was confirmed by the Dyson–Eddington–Davidson expedition during the total solar eclipse of 29 May 1919. Photographs of stars near the eclipsed Sun showed apparent displacements consistent with Einstein's prediction of $1.75''$, rather than the $0.87''$ predicted by a naive Newtonian calculation (which treats light as a particle subject to Newtonian gravity). The result was announced on 6 November 1919 and made Einstein an international celebrity overnight. Modern measurements using radio waves from quasars bent by the Sun achieve far better precision and have confirmed the prediction to within 0.01%.

This concludes our introduction to general relativity. We have travelled from the equivalence principle and the mathematics of manifolds, through the differential geometry of connections, curvature, and geodesics, to Einstein's field equations and their consequences: gravitational waves, the expanding universe, and the geometry of spacetime around stars and black holes. We have only scratched the surface of an extraordinary theory—one that has been the subject of active research for over a century and whose deepest mysteries, from the nature of dark energy to the quantum structure of spacetime, remain open.

An extraordinary theory

General relativity is a theory of unparalleled beauty in physics. From a handful of physical principles—the equivalence of gravitational and inertial mass, the requirement that the laws of physics be independent of the coordinate system—Einstein constructed a framework that describes the large-scale structure of the universe, the dynamics of black holes, and the propagation of gravitational waves, all as consequences of the curvature of spacetime. That the same theory which explains the perihelion of Mercury also predicts the Big Bang and gravitational waves from merging black holes is, by any measure, breathtaking.

17. Retrospective and outlook

We close these lecture notes by stepping back from the details to survey the logical architecture of general relativity, to appreciate what we have built, and to glimpse what lies beyond.

17.1. The logical structure of the theory. General relativity rests on a remarkably small number of physical principles. The entire theory can be summarised in four lines:

The axioms of general relativity

(i) Spacetime is a four-dimensional Lorentzian manifold $(\mathcal{M}, g_{ab})$.
(ii) Freely falling test bodies move on geodesics of g_{ab}.
(iii) The laws of physics, expressed as tensor equations, take the same form in all coordinate systems (*general covariance*).
(iv) The metric is determined by the matter content via Einstein's field equation: $G_{ab} + \Lambda\, g_{ab} = 8\pi\, T_{ab}$.

From these four statements, the entire course follows. Axiom (i) required us to develop the theory of manifolds, tangent spaces, and tensor fields (Lectures 3–6). Axiom (ii) led us to connections, parallel transport, and the geodesic equation (Lectures 7–10). The requirement that the connection be metric-compatible and torsion-free singled out the Levi-Civita connection and its curvature, the Riemann tensor (Lectures 10–11). Axiom (iii) was our guiding principle throughout, ensuring that every equation we wrote was intrinsic. And axiom (iv)—arrived at by demanding consistency with the Bianchi identity and the Newtonian limit—gave us the field equations whose solutions occupied the second half of the course (Lectures 12–22).

The miracle of self-consistency

The most striking feature of this logical structure is its *self-consistency*. The geodesic hypothesis (ii) was an *input* to the theory—we assumed it as a physical postulate to motivate the field equations. But Einstein's field equations (iv), via the Bianchi identity $\nabla^a G_{ab} = 0$, automatically imply stress-energy conservation $\nabla^a T_{ab} = 0$, which in turn implies the geodesic motion of pressureless matter (§13.2.3). The postulate is recovered as a *theorem*. This kind of bootstrapping is extraordinarily rare in physics and is one of the strongest pieces of internal evidence that the theory is on the right track.

17.2. What we solved. The general Einstein equations are ten coupled, nonlinear, second-order partial differential equations—a system of formidable complexity. We attacked them through three strategies:

(1) **Perturbation theory** (Lectures 12–14). Linearising around flat spacetime reduced the field equations to a wave equation, yielding the Newtonian limit ($\phi = -M/r$), gravitational radiation (two polarizations propagating at c), and the gauge structure of the theory.
(2) **Maximal symmetry** (Lectures 14–18). Imposing homogeneity and isotropy reduced the ten PDEs to two ODEs—the Friedmann equations—for a single function $a(\tau)$. The solutions describe the Big Bang, the expansion of the universe, and (for $k = +1$) the Big Crunch.
(3) **Spherical symmetry** (Lectures 18–22). Imposing staticity and spherical symmetry reduced the equations to two functions of one variable, $f(r)$ and $h(r)$, which we solved *exactly*: the Schwarzschild metric. Its geodesics gave us the precession of Mercury's perihelion (43″/century) and the deflection of starlight by the Sun (1.75″).

In each case, symmetry was the key that unlocked the equations. Without it, we would have been helpless.

17.3. The experimental pillars. General relativity is not merely a mathematical edifice; it is a physical theory, and its predictions have been confirmed with extraordinary precision:

Prediction	**Confirmation**	**Precision**
Perihelion precession of Mercury	Le Verrier (1859), Einstein (1915)	$\sim 0.1\%$
Deflection of starlight	Eddington (1919), radio VLBI	$\sim 0.01\%$
Gravitational redshift	Pound–Rebka (1960), GPS	$\sim 10^{-4}$
Shapiro time delay	Cassini spacecraft (2003)	$\sim 10^{-5}$
Gravitational waves	LIGO/Virgo (2015–)	waveform match
Expansion of the universe	Hubble (1929), Planck (2018)	$\sim 1\%$
Black hole shadow	Event Horizon Telescope (2019)	image

Every prediction of general relativity that has been tested has been confirmed. No other classical theory of gravity comes close to this record.

17.4. What we did not cover. A one-semester course can only scratch the surface. Major topics that await the reader in more advanced treatments include:

- **Black holes.** The Schwarzschild coordinate singularity at $r = 2M$ conceals the rich causal structure of black holes: the event horizon, the Penrose diagram, the Kerr solution for rotating black holes, the laws of black hole mechanics, and Hawking radiation.
- **Gravitational wave physics.** Beyond the linearised theory: post-Newtonian approximations, numerical relativity, the two-body problem, and the extraction of astrophysical parameters from LIGO/Virgo signals.
- **Singularity theorems.** The Penrose–Hawking theorems show that singularities are *generic* in general relativity, not artefacts of high symmetry. Their proof requires the theory of causal structure and the Raychaudhuri equation.
- **Initial value formulation.** General relativity admits a well-posed initial value problem (the ADM formalism), which is the foundation for numerical relativity and the mathematical study of existence and uniqueness of solutions.
- **Cosmological perturbation theory.** The seeds of structure in the universe—galaxies, clusters, the CMB anisotropies—arise from perturbations of the FLRW background.

Their study connects general relativity to inflationary cosmology and observational astrophysics.

- **Quantum gravity.** General relativity and quantum mechanics are individually successful but mutually incompatible. Approaches to their unification—string theory, loop quantum gravity, causal set theory, and others—remain among the deepest open problems in physics.

17.5. The enduring mystery. We began this course (§1) by noting that general relativity is, by wide consensus, one of the most beautiful theories in physics—and also that it is *wrong*, or at least incomplete. It describes the large-scale structure of the universe with breathtaking accuracy, yet it breaks down at singularities and cannot be reconciled with quantum mechanics.

The deepest questions remain open. What is the quantum structure of spacetime? What happens at the singularity inside a black hole, or at the Big Bang? What is the nature of dark energy, which drives the accelerating expansion of the universe? Is spacetime fundamental, or does it emerge from something deeper?

These are questions for the next generation of physicists. The tools developed in this course—manifolds, tensors, curvature, geodesics, Einstein's equations—are the language in which these questions are posed and in which their answers will be expressed.

A final thought

Einstein's general relativity is a theory of unparalleled beauty and scope. From a handful of physical principles—the equivalence of gravitational and inertial mass, the requirement that physics be independent of coordinates—it constructs a framework that describes the expansion of the universe, the dynamics of black holes, and the propagation of gravitational waves, all as consequences of the curvature of spacetime. That the same equation, $G_{ab} = 8\pi\, T_{ab}$, predicts both the $43''$-per-century precession of Mercury and the existence of gravitational waves from merging black holes detected a century later is, by any measure, extraordinary.

We have only scratched the surface. There has been active research in general relativity for over a century, and the largest mysteries that remain open in physics—the nature of dark energy, the quantum structure of spacetime, the origin of the universe—are those that arise at the intersection of gravity and quantum theory. The journey continues.

18. Addendum: the energy-momentum tensor from microscopics

In Lecture 6 (§6.5.2) we defined the energy-momentum tensor of a system of point particles via the microscopic expressions (79) and (81), and left the proof that these two forms are equivalent—and that the result is a Lorentz tensor—as an exercise (Exercise 6.2). The standard argument, due to Weinberg [Wei72], consists of inserting a coordinate-time integral, observing that "the differentials dt' cancel," and replacing dt' with the invariant proper-time element $d\tau$. While physically convincing, this argument glosses over the distributional nature of the objects involved and leaves the equivalence of the two forms essentially unproved.

In this addendum we supply the missing rigour. For the equivalence theorem itself (§18.3), the key step is the classical **change-of-variables (substitution) formula** for a C^1 diffeomorphism—an elementary result that needs no heavy machinery. However, the natural *generalisation* to slicing by arbitrary Cauchy surfaces (§18.6) requires the **coarea formula** of geometric measure theory [Fed69], which provides the framework for decomposing integrals over level sets of Lipschitz maps. We therefore state the coarea formula in its general form and explain how the one-dimensional case used in the equivalence proof reduces to the classical substitution rule, while the full coarea/slicing machinery becomes essential for the general slicing theorem.

18.1. Distributions supported on worldlines. Throughout this addendum, $(\mathbb{R}^4, \eta_{\alpha\beta})$ denotes Minkowski spacetime with signature $(-,+,+,+)$, and $\mathcal{D}(\mathbb{R}^4) = C_c^\infty(\mathbb{R}^4)$ is the space of compactly supported smooth test functions. We write $\mathcal{D}'(\mathbb{R}^4)$ for the topological dual: the space of distributions (continuous linear functionals on $\mathcal{D}$).

Definition 18.1 (Timelike worldline)**.** A **timelike worldline** is a smooth, future-directed embedding $\gamma\colon I \to \mathbb{R}^4$ (where $I \subseteq \mathbb{R}$ is an open interval) parametrised by proper time τ, so that

$$\eta_{\alpha\beta}\,\dot\gamma^\alpha\dot\gamma^\beta = -1 \qquad \text{and} \qquad \frac{d\gamma^0}{d\tau} > 0 \tag{346}$$

everywhere on I. Here $\dot\gamma^\alpha = d\gamma^\alpha/d\tau$. Throughout this addendum we assume $I = \mathbb{R}$, i.e. the worldline is inextendible.

For a massive particle of rest mass m, the **four-momentum** along γ is $p^\alpha(\tau) = m\,\dot\gamma^\alpha(\tau)$.

The microscopic energy-momentum tensor is built from delta functions localised on worldlines. To make this precise, we define it directly as a distribution.

Definition 18.2 (Distributional energy-momentum tensor)**.** Given N timelike worldlines $\gamma_j\colon \mathbb{R} \to \mathbb{R}^4$ carrying four-momenta $p_j^\alpha(\tau)$, the **(distributional) energy-momentum tensor** $T^{\alpha\beta} \in \mathcal{D}'(\mathbb{R}^4)$ is defined by its action on test functions:

$$\langle T^{\alpha\beta}, \phi\rangle = \sum_{j=1}^{N} \int_{\mathbb{R}} p_j^\alpha(\tau)\,\dot\gamma_j^\beta(\tau)\,\phi\big(\gamma_j(\tau)\big)\,\mathrm{d}\tau \tag{347}$$

for every $\phi \in \mathcal{D}(\mathbb{R}^4)$.

Proposition 18.1 (Well-definedness)**.** *The functional* (347) *defines a distribution of order zero, that is, a tensor-valued Radon measure on* $\mathbb{R}^4$.

Proof. It suffices to treat a single worldline γ; the sum over j follows by linearity.

Step 1: γ^0 is a diffeomorphism $\mathbb{R} \to \mathbb{R}$. By Definition 18.1, the worldline is future-directed: $\dot\gamma^0 > 0$. The normalisation $\eta_{\alpha\beta}\,\dot\gamma^\alpha\dot\gamma^\beta = -1$ with signature $(-,+,+,+)$ gives $(\dot\gamma^0)^2 = 1 + \sum_{i=1}^3(\dot\gamma^i)^2 \geq 1$, so $\dot\gamma^0 \geq 1$. Hence $\gamma^0(\tau) \geq \gamma^0(0) + \tau$ for $\tau > 0$ and $\gamma^0(\tau) \leq \gamma^0(0) + \tau$ for $\tau < 0$, which forces $\gamma^0 \to \pm\infty$ as $\tau \to \pm\infty$. Thus $f\colon \mathbb{R} \to \mathbb{R}$, $f(\tau) = \gamma^0(\tau)$, is a strictly increasing smooth surjection and therefore a diffeomorphism $\mathbb{R} \to \mathbb{R}$.

Step 2: γ is proper. We must show that $\gamma^{-1}(K)$ is compact for every compact $K \subset \mathbb{R}^4$.

(i) *Bounded τ-range.* Since K is compact, its time projection $\pi_0(K) \subset \mathbb{R}$ is a compact interval. The diffeomorphism f maps $\gamma^{-1}(K)$ into $f^{-1}(\pi_0(K))$, which is a compact interval (as the preimage of a compact interval under a diffeomorphism $\mathbb{R} \to \mathbb{R}$). Hence $\gamma^{-1}(K)$ is contained in a bounded τ-interval.

(ii) *Bounded spatial coordinates.* The normalisation $\eta_{\alpha\beta}\,\dot\gamma^\alpha\dot\gamma^\beta = -1$ with signature $(-,+,+,+)$ gives $\sum_{i=1}^3(\dot\gamma^i)^2 = (\dot\gamma^0)^2 - 1 < (\dot\gamma^0)^2$, so $|\dot\gamma^i| < \dot\gamma^0$ for each spatial component. In particular, the spatial components of γ grow no faster than γ^0, so they remain bounded on the bounded τ-interval from (i).

(iii) *Compactness.* Since γ is an embedding (hence a homeomorphism onto its image), $\gamma^{-1}(K)$ is closed in $\mathbb{R}$ (as the preimage of a compact, hence closed, set under a continuous map). By (i) and (ii), it is also bounded. Hence $\gamma^{-1}(K)$ is compact by the Heine–Borel theorem.

Step 3: The integral converges and yields order-zero continuity. Let $\phi \in \mathcal{D}(\mathbb{R}^4)$ have support in a compact set K. The function $\tau \mapsto p^\alpha(\tau)\,\dot\gamma^\beta(\tau)\,\phi(\gamma(\tau))$ is smooth with support in

the compact set $\gamma^{-1}(K)$, so the integral converges absolutely. Linearity of the functional in ϕ is immediate. Moreover,

$$|\langle T^{\alpha\beta}, \phi\rangle| \;\le\; \sum_j \int_{\gamma_j^{-1}(K)} |p_j^\alpha|\,|\dot\gamma_j^\beta|\,\mathrm{d}\tau \cdot \|\phi\|_\infty \;\le\; C_K\,\|\phi\|_\infty\,,$$

where, for fixed α, β, $C_K = \sum_j \int_{\gamma_j^{-1}(K)} |p_j^\alpha|\,|\dot\gamma_j^\beta|\,\mathrm{d}\tau < \infty$ depends only on K (not on ϕ). Thus $T^{\alpha\beta}$ is continuous in the C^0-topology and hence is a distribution of order zero. (By the Riesz representation theorem, every distribution of order zero corresponds to a Radon measure, so $T^{\alpha\beta}$ is indeed a tensor-valued Radon measure on $\mathbb{R}^4$.) □

Remark. Definition 18.2 is the *pushforward* of a measure on the worldline to a distribution on spacetime. In the language of de Rham's theory of currents, the functional $\phi \mapsto \int \phi(\gamma(\tau))\,\mathrm{d}\tau$ defines a 0-current on $\mathbb{R}^4$ (it acts on 0-forms, i.e. test functions). The *integration current* of the worldline, which acts on 1-forms via $\omega \mapsto \int \omega_\alpha(\gamma(\tau))\,\dot\gamma^\alpha(\tau)\,\mathrm{d}\tau$, is a 1-current. Each component $T^{\alpha\beta}$ (for fixed α, β) is obtained from the 0-current by weighting the integrand with $p^\alpha\,\dot\gamma^\beta$. See Federer [Fed69, §4.1].

Informally, Definition 18.2 says that $T^{\alpha\beta}$ is the distribution

$$T^{\alpha\beta}(x) = \sum_{j=1}^{N} \int \mathrm{d}\tau\;\, p_j^\alpha(\tau)\,\dot\gamma_j^\beta(\tau)\,\delta^{(4)}\big(x - \gamma_j(\tau)\big)\,, \tag{348}$$

which is precisely equation (81). The distributional definition makes it clear that this expression is well defined and coordinate-independent; what remains is to show that it reduces to the 3+1 form (79) upon slicing by a constant-time hyperplane.

18.2. The coarea formula and the substitution rule. The bridge between the covariant form (348) and the 3+1 splitting (79) is ultimately the classical *change-of-variables formula* for a diffeomorphism. To place this in its proper context—and to prepare for the general slicing theorem of §18.6—we state the coarea formula of Federer and then discuss the one-dimensional case separately.

Theorem 18.2 (Coarea formula [Fed69, Theorem 3.2.12])**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}^m$ be a Lipschitz map with $m < n$ (strict inequality). Then for any nonnegative Borel function $g\colon \mathbb{R}^n \to [0,\infty]$,*

$$\int_{\mathbb{R}^n} g(x)\,J_m f(x)\,\mathrm{d}\mathcal{H}^n(x) = \int_{\mathbb{R}^m}\Big(\int_{f^{-1}(y)} g\;\mathrm{d}\mathcal{H}^{n-m}\Big)\mathrm{d}\mathcal{H}^m(y)\,, \tag{349}$$

where $J_m f$ denotes the m-dimensional Jacobian of f and $\mathcal{H}^k$ the k-dimensional Hausdorff measure (which coincides with the Lebesgue measure on $\mathbb{R}^k$; see [Fed69, 2.10.35]*).*

The one-dimensional case ($n = m = 1$). When domain and codomain have the same dimension, the strict inequality $m < n$ is *not* satisfied, so Theorem 18.2 does not apply. The relevant result is instead the **area formula** [Fed69, Theorem 3.2.3]: for $f\colon \mathbb{R} \to \mathbb{R}$ Lipschitz and $g\colon \mathbb{R} \to [0,\infty]$ Borel,

$$\boxed{\int_{\mathbb{R}} g(\tau)\,|f'(\tau)|\,\mathrm{d}\tau = \int_{\mathbb{R}} \sum_{\tau\in f^{-1}(t)} g(\tau)\,\mathrm{d}t}\,. \tag{350}$$

When f is injective (in particular, a diffeomorphism), each level set $f^{-1}(t)$ is at most a singleton and the sum is trivial: one recovers the classical **substitution (change-of-variables) formula**

$$\int_{\mathbb{R}} g(\tau)\,|f'(\tau)|\,\mathrm{d}\tau = \int_{\mathbb{R}} g\big(\tau_*(t)\big)\,\mathrm{d}t\,, \tag{351}$$

where $\tau_*(t) = f^{-1}(t)$ is the unique preimage. This is the identity that drives the proof of Theorem 18.3; the full coarea machinery is not needed for that argument but becomes essential in §18.6.

Remark (Delta-function composition rule). The substitution formula (351) is equivalent to the distributional identity

$$\delta\big(f(\tau) - t\big) = \frac{\delta(\tau - \tau_*)}{|f'(\tau_*)|}\,. \tag{352}$$

To see this, note that (351) states that the pushforward of the measure $|f'(\tau)|\,\mathrm{d}\tau$ under f equals Lebesgue measure $\mathrm{d}t$ (by the Riesz representation theorem, since the identity holds for all non-neg. Borel g). Testing against a Dirac mass at t then yields (352). More generally, when f has finitely many roots $\{\tau_k\}$ of $f(\tau) = t$, the area formula (350) gives the standard multi-root identity $\delta(f(\tau) - t) = \sum_k \delta(\tau - \tau_k)/|f'(\tau_k)|$. The substitution formula thus provides the rigorous foundation for the "delta-function composition rule" that is often invoked without proof in physics texts.

18.3. From the covariant form to the spatial-slice form. We now prove the main result: the covariant definition (348) reproduces the 3+1 form (79) of Lecture 6 upon slicing by a constant-time hyperplane.

Theorem 18.3 (Equivalence of forms)**.** *Let $T^{\alpha\beta}$ be the distribution defined in* (347)*. Then, in the sense of distributions on $\mathbb{R}^4$,*

$$T^{\alpha\beta}(x) = \sum_{j=1}^{N} p_j^\alpha(t)\,\frac{d\gamma_j^\beta}{dt}(t)\,\delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{\gamma}_j(t)\big)\,, \tag{353}$$

where $t = x^0$ and $\boldsymbol{\gamma}_j(t) = \big(\gamma_j^1(t), \gamma_j^2(t), \gamma_j^3(t)\big)$ is the spatial position of the j-th particle at coordinate time t.

Proof. It suffices to prove the identity for a single worldline; the sum over j follows by linearity. Drop the subscript j and let $\gamma\colon \mathbb{R} \to \mathbb{R}^4$ be a timelike worldline carrying momentum p^α.

Step 1: Factor out the time velocity. Define the *time function along the worldline* $f\colon \mathbb{R} \to \mathbb{R}$ by $f(\tau) = \gamma^0(\tau)$. Since γ is future-directed and timelike, $f'(\tau) = d\gamma^0/d\tau > 0$, so f is a smooth diffeomorphism from $\mathbb{R}$ onto $\mathbb{R}$.

Decompose the four-velocity:

$$\dot\gamma^\beta(\tau) = \frac{d\gamma^\beta}{d\tau} = \frac{d\gamma^\beta}{dt}\bigg|_{t=f(\tau)} f'(\tau)\,.$$

Substituting into Definition 18.2:

$$\langle T^{\alpha\beta}, \phi\rangle = \int_{\mathbb{R}} p^\alpha(\tau)\,\frac{d\gamma^\beta}{dt}\bigg|_{t=f(\tau)} \phi\big(\gamma(\tau)\big)\,f'(\tau)\,\mathrm{d}\tau\,. \tag{354}$$

Step 2: Change of variables. Since f is a smooth diffeomorphism with $f' > 0$, the level set $f^{-1}(t)$ is a singleton $\{\tau_*(t)\}$ for each t, and the classical change of variables (substitution) gives

$$\int_{\mathbb{R}} g(\tau)\,f'(\tau)\,\mathrm{d}\tau = \int_{\mathbb{R}} g\big(f^{-1}(t)\big)\,\mathrm{d}t$$

for any integrable g, without sign restriction on the integrand. (This is the diffeomorphism case of the one-dimensional area formula; see Remark 18.2.) Applying this to (354):

$$\langle T^{\alpha\beta}, \phi\rangle = \int_{\mathbb{R}} p^\alpha\big(\tau_*(t)\big)\,\frac{d\gamma^\beta}{dt}\bigg|_t \phi\big(\gamma(\tau_*(t))\big)\,\mathrm{d}t\,. \tag{355}$$

Step 3: Identify the spatial delta function. By definition of $\tau_*(t)$, we have $\gamma^0(\tau_*(t)) = t$, so $\gamma(\tau_*(t)) = \big(t,\, \boldsymbol{\gamma}(t)\big)$. Writing $p^\alpha(\tau_*(t)) = p^\alpha(t)$ and $d\gamma^\beta/dt\,\big|_{\tau_*(t)} = d\gamma^\beta/dt\,(t)$ as functions of coordinate time, equation (355) becomes

$$\begin{aligned}\langle T^{\alpha\beta}, \phi\rangle &= \int_{\mathbb{R}} p^\alpha(t)\,\frac{d\gamma^\beta}{dt}(t)\,\phi\big(t, \boldsymbol{\gamma}(t)\big)\,\mathrm{d}t \\ &= \int_{\mathbb{R}} \mathrm{d}t \int_{\mathbb{R}^3} \mathrm{d}^3\boldsymbol{x}\;\; p^\alpha(t)\,\frac{d\gamma^\beta}{dt}(t)\,\delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{\gamma}(t)\big)\,\phi(t, \boldsymbol{x})\,. \end{aligned} \tag{356}$$

where the second line uses the sifting property of $\delta^{(3)}$ in the reverse direction, and we apply Fubini's theorem to pass to the iterated integral (the integrand is in $L^1(\mathbb{R}\times\mathbb{R}^3)$ since ϕ has compact support). Since (356) holds for all $\phi \in \mathcal{D}(\mathbb{R}^4)$, we conclude that (353) holds in the distributional sense. □

Slicing worldlines by time

The theorem says that the natural (covariant) definition of $T^{\alpha\beta}$—as a measure spread along the worldline—can be "sliced" by a constant-time hyperplane $\Sigma_t = \{x^0 = t\}$ to recover the 3+1 expression. The change of variables provides exactly the right Jacobian factor $(1/f'(\tau_*))$ to convert the proper-time line element $\mathrm{d}\tau$ along the worldline into the transverse coordinate $\mathrm{d}t$. This factor cancels the $d\gamma^\beta/d\tau$ in the covariant integrand, producing $d\gamma^\beta/dt$ in the sliced expression.
Weinberg's observation that "the dt' cancel" is a compact (if informal) summary of exactly this mechanism.

18.4. Lorentz covariance. The covariant form (347) immediately yields the transformation law, resolving Exercise 6.2:

Proposition 18.4 (Lorentz covariance)**.** *Under a Lorentz transformation $x'^\mu = \Lambda^\mu{}_\nu\, x^\nu$, the distributional energy-momentum tensor transforms as*

$$T'^{\alpha\beta}(x') = \Lambda^\alpha{}_\gamma\,\Lambda^\beta{}_\delta\, T^{\gamma\delta}(x)\,. \tag{357}$$

Proof. Under $x' = \Lambda x$, the transformed worldline is $\gamma'(\tau) = \Lambda\gamma(\tau)$ and the transformed four-momentum is $p'^\alpha(\tau) = \Lambda^\alpha{}_\gamma\, p^\gamma(\tau)$. (Since τ is Lorentz-invariant and γ^α transforms as a four-vector, $\dot\gamma^\alpha = d\gamma^\alpha/d\tau$ transforms as a four-vector, and hence $p^\alpha = m\dot\gamma^\alpha$ does as well.)

We define the transformed distribution $T'^{\alpha\beta}$ by its action on test functions: for any $\phi \in \mathcal{D}(\mathbb{R}^4)$,

$$\begin{aligned}\langle T'^{\alpha\beta},\, \phi\rangle &= \sum_j \int_{\mathbb{R}} p'^\alpha_j(\tau)\,\dot\gamma'^\beta_j(\tau)\,\phi\big(\gamma'_j(\tau)\big)\,\mathrm{d}\tau \\ &= \sum_j \int_{\mathbb{R}} \Lambda^\alpha{}_\gamma\, p^\gamma_j(\tau)\;\cdot\;\Lambda^\beta{}_\delta\,\dot\gamma^\delta_j(\tau)\;\cdot\;\phi\big(\Lambda\gamma_j(\tau)\big)\,\mathrm{d}\tau \\ &= \Lambda^\alpha{}_\gamma\,\Lambda^\beta{}_\delta \sum_j \int_{\mathbb{R}} p^\gamma_j(\tau)\,\dot\gamma^\delta_j(\tau)\,(\phi\circ\Lambda)\big(\gamma_j(\tau)\big)\,\mathrm{d}\tau \\ &= \Lambda^\alpha{}_\gamma\,\Lambda^\beta{}_\delta\,\big\langle T^{\gamma\delta},\; \phi\circ\Lambda\big\rangle. \end{aligned} \tag{358}$$

This is the standard distributional transformation law $T'^{\alpha\beta} = \Lambda^\alpha{}_\gamma\,\Lambda^\beta{}_\delta\,(\Lambda^{-1})_* T^{\gamma\delta}$, which is the distributional restatement of (357). □

Remark. The 3+1 form (79) is *not* manifestly covariant: neither $\delta^{(3)}$ nor $d\gamma^\beta/dt$ is Lorentz-invariant in isolation. It is most naturally through Theorem 18.3—whose proof rests on the

substitution formula—that we can deduce that the combination $p^\alpha\,(d\gamma^\beta/dt)\,\delta^{(3)}$ transforms as a tensor. The miracle is that the Jacobian factors conspire to produce a Lorentz-invariant measure along the worldline; the change of variables explains *why* they conspire.

18.5. Distributional conservation law. We now prove the conservation law entirely within the distributional framework, avoiding the somewhat formal manipulation of derivatives of delta functions that appears in Weinberg [Wei72, §2.8].

Theorem 18.5 (Conservation of $T^{\alpha\beta}$)**.** *If each particle is free* ($\mathrm{d}p_j^\alpha/\mathrm{d}\tau = 0$)*, then*

$$\partial_\beta\, T^{\alpha\beta} = 0 \tag{359}$$

in the distributional sense. More generally, in the presence of external forces,

$$\partial_\beta\, T^{\alpha\beta}(x) = G^\alpha(x)\,, \tag{360}$$

where the ***force density***

$$G^\alpha(x) = \sum_j \int \mathrm{d}\tau\, \frac{\mathrm{d}p_j^\alpha}{\mathrm{d}\tau}(\tau)\,\delta^{(4)}\big(x - \gamma_j(\tau)\big)$$

is the distributional pushforward of the equations of motion.

Proof. We prove (359); equation (360) follows by the same argument, now retaining the nonzero right-hand side of (362) when $\mathrm{d}p_j^\alpha/\mathrm{d}\tau \neq 0$. For any $\phi \in \mathcal{D}(\mathbb{R}^4)$, the standard definition of the distributional derivative (integration by parts with a sign) gives:

$$\begin{aligned}
\langle \partial_\beta T^{\alpha\beta},\, \phi\rangle &= -\langle T^{\alpha\beta},\, \partial_\beta\phi\rangle \\
&= -\sum_j \int_{\mathbb{R}} p_j^\alpha(\tau)\,\dot\gamma_j^\beta(\tau)\,\partial_\beta\phi\big(\gamma_j(\tau)\big)\,\mathrm{d}\tau \\
&= -\sum_j \int_{\mathbb{R}} p_j^\alpha(\tau)\,\frac{\mathrm{d}}{\mathrm{d}\tau}\big[\phi(\gamma_j(\tau))\big]\,\mathrm{d}\tau\,,
\end{aligned} \tag{361}$$

where the last step uses the chain rule $\frac{\mathrm{d}}{\mathrm{d}\tau}\phi(\gamma(\tau)) = \dot\gamma^\beta(\tau)\,\partial_\beta\phi(\gamma(\tau))$. Since $p_j^\alpha = m_j\dot\gamma_j^\alpha$ and each γ_j is smooth (Definition 18.1), the four-momentum $p_j^\alpha(\tau)$ is smooth, so integration by parts is justified. Since γ_j is proper (Proposition 18.1), the set $\gamma_j^{-1}(\operatorname{spt}\phi)$ is compact, so the function $\tau \mapsto \phi(\gamma_j(\tau))$ has compact support in $\mathbb{R}$. The boundary terms therefore vanish, giving:

$$\langle \partial_\beta T^{\alpha\beta},\, \phi\rangle = \sum_j \int_{\mathbb{R}} \frac{\mathrm{d}p_j^\alpha}{\mathrm{d}\tau}(\tau)\,\phi\big(\gamma_j(\tau)\big)\,\mathrm{d}\tau\,. \tag{362}$$

For free particles, $\mathrm{d}p_j^\alpha/\mathrm{d}\tau = 0$, and the right-hand side vanishes for all ϕ. □

Remark. The elegance of the distributional proof is worth emphasising. In the 3+1 approach, the conservation law requires computing $\partial_i\big[p^\alpha\,(\mathrm{d}\gamma^\beta/\mathrm{d}t)\,\delta^{(3)}(\boldsymbol{x}-\boldsymbol{\gamma}(t))\big] + \partial_0\big[p^\alpha\,(\mathrm{d}\gamma^0/\mathrm{d}t)\,\delta^{(3)}(\boldsymbol{x}-\boldsymbol{\gamma}(t))\big]$, which involves the multivariable chain rule

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{\gamma}(t)\big) = -\frac{\mathrm{d}\gamma^i}{\mathrm{d}t}\,\frac{\partial}{\partial x^i}\,\delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{\gamma}(t)\big)$$

to convert time derivatives into spatial derivatives. The distributional proof (361)–(362) bypasses this entirely: the chain rule on test functions and integration by parts do all the work.

Exercise 18.1. Extend Theorem 18.5 to particles that undergo strictly localised collisions (so that worldlines begin and end at discrete spacetime events). Show that $\partial_\beta T^{\alpha\beta} = 0$ provided that four-momentum is conserved at each collision vertex: $\sum_{n\in c} p_n^\alpha = 0$ for each collision c. *Hint:* integrate by parts on each worldline segment; the boundary terms at each collision vertex sum to zero by momentum conservation.

18.6. Slicing by an arbitrary spacelike hypersurface. The equivalence of the covariant and 3+1 forms does not depend on the particular choice of time coordinate x^0. We now state the general result, which applies to arbitrary spacelike hypersurfaces and extends naturally to curved spacetimes.

Theorem 18.6 (General slicing theorem)**.** *Let $\Sigma = f^{-1}(0)$ be a smooth spacelike Cauchy surface in $(\mathbb{R}^4, \eta)$, defined as a level set of a smooth function $f\colon \mathbb{R}^4 \to \mathbb{R}$ with ∇f everywhere timelike (that is, $\eta^{\alpha\beta}\partial_\alpha f\, \partial_\beta f < 0$). Then each worldline γ_j intersects Σ transversally at a unique point $\gamma_j(\tau_j^*)$, and the restriction ("slice") of $T^{\alpha\beta}$ to Σ satisfies*

$$T^{\alpha\beta}\big|_\Sigma(\boldsymbol{y}) = \sum_{j=1}^{N} \frac{p_j^\alpha(\tau_j^*)\, \dot\gamma_j^\beta(\tau_j^*)}{\left|\partial_\gamma f\, \dot\gamma_j^\gamma(\tau_j^*)\right|}\, \delta_\Sigma(\boldsymbol{y} - \boldsymbol{y}_j)\,, \tag{363}$$

where $\boldsymbol{y}$ denotes coordinates on Σ, $\boldsymbol{y}_j$ is the location of the j-th crossing on Σ, and δ_Σ is the delta function with respect to the Riemannian measure on Σ induced by the ambient Minkowski metric; since Σ is spacelike, this induced metric is positive definite. (This coincides with the 3-dimensional Hausdorff measure $\mathcal{H}^3$ restricted to Σ, up to identification of Σ with $\mathbb{R}^3$ via coordinates.)

Sketch of proof. Transversality. The covector field $\partial_\alpha f$ satisfies $\eta^{\alpha\beta}\partial_\alpha f\, \partial_\beta f < 0$ (timelike), and the tangent vector $\dot\gamma^\alpha$ is also timelike. By the reverse Cauchy–Schwarz inequality for Lorentzian inner products (if u, v are both timelike and in the same time-cone, then $|g(u,v)| \geq |u|\,|v| > 0$), the contraction $\partial_\gamma f \cdot \dot\gamma^\gamma \neq 0$.

Uniqueness of intersection. Consider the composition $h = f \circ \gamma\colon \mathbb{R} \to \mathbb{R}$. Its derivative $h'(\tau) = \partial_\gamma f\big(\gamma(\tau)\big)\, \dot\gamma^\gamma(\tau)$ is nonzero everywhere by transversality, and has constant sign by continuity (since $\mathbb{R}$ is connected). Hence h is strictly monotone, so $h^{-1}(0)$ contains at most one point. Existence of the crossing follows from the Cauchy-surface hypothesis: by definition, every inextendible timelike curve meets a Cauchy surface, so each complete worldline γ_j intersects $\Sigma = f^{-1}(0)$ at least—hence exactly—once.

Slicing and the substitution formula. The proof proceeds in two stages. First, the classical change-of-variables formula applied to $h = f \circ \gamma\colon \mathbb{R} \to \mathbb{R}$ converts the proper-time integral into an integral with respect to the level-set parameter of f, producing the Jacobian factor $|\partial_\gamma f \cdot \dot\gamma^\gamma|^{-1}$. Second, the identification of the resulting object as a delta function on Σ uses the theory of slicing of normal 1-currents (Federer [Fed69, Theorem 4.3.2]), which provides the natural framework when Σ is not a coordinate hyperplane. □

Remark. For the standard time function $f(x) = x^0 - t_0$, we have $\partial_\gamma f = \delta^0_\gamma$, so $|\partial_\gamma f \cdot \dot\gamma^\gamma| = |\dot\gamma^0|$. Since γ is future-directed (Definition 18.1), $|\dot\gamma^0| = d\gamma^0/d\tau$, and the Jacobian factor combines with $\dot\gamma^\beta$ to give $d\gamma^\beta/dt$—recovering Theorem 18.3. In curved spacetime one replaces f with a Cauchy time function and δ_Σ with the delta function defined by the induced Riemannian metric on the Cauchy surface.

18.7. The four-current revisited. For completeness, we note that the entire development applies *mutatis mutandis* to the electromagnetic four-current J^α of §6.5.1. Given particles with charges q_j, define the distributional four-current by

$$\langle J^\alpha, \phi\rangle = \sum_{j=1}^{N} \int_{\mathbb{R}} q_j\, \dot\gamma_j^\alpha(\tau)\, \phi\big(\gamma_j(\tau)\big)\, \mathrm{d}\tau\,. \tag{364}$$

The change-of-variables argument of Theorem 18.3 shows that

$$J^\alpha(x) = \sum_{j=1}^{N} q_j\, \frac{d\gamma_j^\alpha}{dt}(t)\, \delta^{(3)}\big(\boldsymbol{x} - \boldsymbol{\gamma}_j(t)\big)\,, \tag{365}$$

which unifies the charge density (74) and current density (75). The conservation law $\partial_\alpha J^\alpha = 0$ follows from the same integration-by-parts argument as Theorem 18.5, using the constancy of charge along each worldline ($\mathrm{d}q_j/\mathrm{d}\tau = 0$).

18.8. Discussion.

Summary

(i) The covariant form (81) is the fundamental definition of the microscopic energy-momentum tensor: it is a well-defined distribution (Proposition 18.1) and a Lorentz tensor by construction (Proposition 18.4).
(ii) The 3+1 form (79) is derived from the covariant form by *change of variables along the worldline* (Theorem 18.3).
(iii) Conservation $\partial_\beta T^{\alpha\beta} = 0$ follows from the equations of motion by a distributional integration-by-parts argument (Theorem 18.5), without manipulating derivatives of delta functions.
(iv) The construction generalises immediately to arbitrary spacelike hypersurfaces via the slicing theorem for currents (Theorem 18.6).

Thus Weinberg's beautiful microscopic picture of the energy-momentum tensor—as the density and flux of four-momentum carried by point particles—rests on a rigorous distributional foundation. The change-of-variables formula along the worldline is the elementary mechanism that makes the equivalence work; the full coarea/slicing framework of geometric measure theory becomes essential for the generalisation to arbitrary Cauchy surfaces. Recognising this clarifies what would otherwise appear to be a sleight of hand in the passage from coordinate-time expressions to manifestly covariant ones.

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